

# SCIENTIFIC JOURNAL PUBLICATION TREND TRACKING SYSTEM

Group 6

# I. INTRODUCTION

## 1 Overview

### 1.1 Project Information

- Project Name: Scientific Journal Publication Trend Tracking System
- Group Name: Group 6
- Software Type: Web Application
- GitHub Repository:  
<https://github.com/NguyenNgocBaoTram10/Scientific-Journal-Publication-Trend-Tracking-System>

### 1.2 Project team

| Full Name              | Role   | Student ID   | Email                  |
|------------------------|--------|--------------|------------------------|
| Lê Thanh Tú            | Leader | 049206004778 | tult4778@ut.edu.vn     |
| Nguyễn Ngọc Bảo Trâm   | Member | 082306008015 | tramnnb8015@ut.edu.vn  |
| Lê Thị Huyền Trân      | Member | 077306003635 | tranlth3635@ut.edu.vn  |
| Trần Trung Hiếu        | Member | 089206008473 | hieutt8473@ut.edu.vn   |
| Trần Văn Học           | Member | 052206005618 | hoctv5618@ut.edu.vn    |
| Nguyễn Đoàn Nhật Quang | Member | 079206010585 | quangndn0585@ut.edu.vn |

Bảng 1: Project Team Information

## 2 Project Scope and Limitations

### 2.1 Major Features

#### 2.1.1 Major Features for User

- Register and log in to the system.
- Search scientific publications using keywords, authors, journals, or research fields.
- View detailed publication information.
- Save publications to bookmarks.

- Follow research topics of interest.
- Receive notifications about new publications and research trends.
- View publication trend analysis and statistical dashboards.
- Generate and export analytical reports (PDF/CSV).

### **2.1.2 Major Features for Administrator**

- Log in and log out.
- Manage user accounts.
- Configure academic API sources (Semantic Scholar, OpenAlex, and Crossref).
- Monitor the data synchronization process.
- Manage system notifications.
- Generate system reports.

## **2.2 Project Limitations**

### **2.2.1 Limited Use of Full-Text Data**

- The system only uses publication metadata such as title, abstract, keywords, authors, and publication year.
- It does not process full-text content, which limits the depth of analysis and understanding of research papers.

### **2.2.2 Dependence on External APIs**

- The system relies on external APIs such as OpenAlex, and Crossref.
- Any API downtime, structural changes, or request limitations may affect system stability and data collection performance.

### **2.2.3 Simplified Trend Analysis and Update Frequency**

- Trend analysis is mainly based on publication frequency and growth rate.
- It does not apply advanced Machine Learning or AI models for prediction.
- Data is synchronized periodically (daily or weekly), so real-time trends may not be immediately reflected.

## II. PROJECT MANAGEMENT PLAN

### 1 Overview

#### 1.1 Scope and estimation

| #                                                  | WBS Item                                                             | Complexity | Duration (Days) |
|----------------------------------------------------|----------------------------------------------------------------------|------------|-----------------|
| <b>1. Project Initiation</b>                       |                                                                      |            |                 |
| 1.1                                                | Define project scope                                                 | Medium     | 1               |
| 1.2                                                | Collect and analyze requirements                                     | Complex    | 2               |
| <b>2. Project Management and Environment Setup</b> |                                                                      |            |                 |
| 2.1                                                | Set up development environment (GitHub, Database, Backend, Frontend) | Simple     | 1               |
| 2.2                                                | Design database (ERD, Database Schema)                               | Complex    | 2               |
| 2.3                                                | Configure security (JWT, RBAC, Password Hashing)                     | Complex    | 2               |
| <b>3. System Design</b>                            |                                                                      |            |                 |
| 3.1                                                | Design Use Case Diagram and Activity Diagram                         | Medium     | 1               |
| 3.2                                                | Design Sequence Diagram                                              | Medium     | 1               |
| 3.3                                                | Design DFD and ERD                                                   | Medium     | 1               |
| <b>4. System Development</b>                       |                                                                      |            |                 |
| 4.1                                                | User Registration and Login                                          | Medium     | 2               |
| 4.2                                                | Search Scientific Publications                                       | Complex    | 2               |
| 4.3                                                | View Publication Details                                             | Medium     | 1               |
| 4.4                                                | Bookmark Management                                                  | Medium     | 1               |
| 4.5                                                | Research Topic Following                                             | Medium     | 1               |
| 4.6                                                | Notification Management                                              | Medium     | 1               |
| 4.7                                                | Publication Trend Analysis                                           | Complex    | 3               |
| 4.8                                                | Develop Statistical Dashboard                                        | Complex    | 2               |
| 4.9                                                | Display Trending Topics                                              | Medium     | 1               |
| 4.10                                               | Generate Statistical Reports (PDF/CSV)                               | Complex    | 2               |

| #                                  | WBS Item                                                       | Complexity | Duration (Days)        |
|------------------------------------|----------------------------------------------------------------|------------|------------------------|
| <b>5. Administration Functions</b> |                                                                |            |                        |
| 5.1                                | User Management                                                | Complex    | 2                      |
| 5.2                                | Configure Academic API Sources                                 | Complex    | 2                      |
| 5.3                                | Synchronize Data from Semantic Scholar, OpenAlex, and Crossref | Complex    | 3                      |
| <b>6. Testing</b>                  |                                                                |            |                        |
| 6.1                                | Prepare Test Plan                                              | Medium     | 1                      |
| 6.2                                | Design Test Cases                                              | Medium     | 2                      |
| 6.3                                | Execute Testing                                                | Complex    | 2                      |
| 6.4                                | Fix Defects After Testing                                      | Complex    | 2                      |
| <b>7. Project Finalization</b>     |                                                                |            |                        |
| 7.1                                | Complete Project Documentation                                 | Medium     | 2                      |
| 7.2                                | Deploy the System                                              | Medium     | 1                      |
| <b>Total Duration</b>              |                                                                |            | <b>45 Working Days</b> |

## 1.2 Project Objective

- **Timeline:** The project must be completed before **July 3, 2026**.
- **Allocated Effort:** 45 man-days.

## 1.3 Project Risks

| # | Risk Description                                                                                            | Impact                                                                                                        | Possibility | Response Plan                                                                                                           |
|---|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------|
| 1 | External academic APIs (Semantic Scholar, OpenAlex, Crossref) become unavailable or change their endpoints. | The system cannot retrieve publication data, causing incomplete search results and inaccurate trend analysis. | High        | Implement API fallback mechanisms, monitor API status regularly, and cache previously retrieved data whenever possible. |

| # | Risk Description                                                                            | Impact                                                                                  | Possibility | Response Plan                                                                                                              |
|---|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------|
| 2 | Large volumes of publication data lead to slow system performance.                          | Users experience long response times when searching publications or viewing dashboards. | Medium      | Optimize database indexing, implement pagination, caching, and asynchronous background synchronization.                    |
| 3 | Security vulnerabilities expose user accounts or stored data.                               | Unauthorized access, data leakage, and loss of user trust.                              | Medium      | Use JWT authentication, RBAC authorization, password hashing, HTTPS, and regular security testing.                         |
| 4 | Trend analysis results become inaccurate because of outdated or incomplete datasets.        | Generated reports and statistics may mislead users' research decisions.                 | Medium      | Schedule automatic data synchronization and validate collected data before analysis.                                       |
| 5 | System integration between frontend, backend, and database encounters compatibility issues. | Development delays and unexpected system failures during deployment.                    | Medium      | Perform integration testing continuously and maintain version compatibility among all components.                          |
| 6 | Insufficient testing before deployment leaves critical defects unresolved.                  | System crashes, incorrect functionality, and poor user experience.                      | High        | Prepare comprehensive Test Plans, Test Cases, conduct functional, integration, and user acceptance testing before release. |
| 7 | Project schedule delays due to underestimated implementation effort.                        | Failure to complete the project within the planned timeline.                            | Medium      | Monitor project progress weekly, prioritize critical features, and adjust the implementation schedule when necessary.      |

## 2 Project Deliverables

| Sprint          | Week   | Duration           | Deliverables                                                                                                                                            | Note |
|-----------------|--------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Sprint 1</b> | Week 1 | 14/05 – 20/05/2026 | <ul style="list-style-type: none"> <li>• Define project scope</li> <li>• Analysis requirement</li> </ul>                                                |      |
| <b>Sprint 2</b> | Week 2 | 21/05 – 27/05/2026 | <ul style="list-style-type: none"> <li>• Finish Product Overview</li> <li>• Finish User Requirement</li> <li>• Finish Functional Requirement</li> </ul> |      |
| <b>Sprint 3</b> | Week 3 | 28/05 – 03/06/2026 | <ul style="list-style-type: none"> <li>• Finish Database Design</li> <li>• Finish Software design description</li> </ul>                                |      |
| <b>Sprint 4</b> | Week 4 | 04/06 – 10/06/2026 | <ul style="list-style-type: none"> <li>• Finish Software requirement specification</li> </ul>                                                           |      |
| <b>Sprint 5</b> | Week 5 | 11/06 – 17/06/2026 | <ul style="list-style-type: none"> <li>• Finish Software design description (continued)</li> </ul>                                                      |      |
| <b>Sprint 6</b> | Week 6 | 18/06 – 24/06/2026 | <ul style="list-style-type: none"> <li>• Finish Software testing document</li> <li>• Finish Release package and user guide</li> </ul>                   |      |
| <b>Sprint 7</b> | Week 7 | 25/06 – 01/07/2026 | <ul style="list-style-type: none"> <li>• Testing</li> <li>• Finish document</li> </ul>                                                                  |      |

### 3. Responsibility Assignment

*Do ~D; Review ~R; Support ~S*

| Responsibility                        | Thanh Tú | Bảo Trâm | Huyền Trân | Trung Hiếu | Văn Học | Nhật Quang |
|---------------------------------------|----------|----------|------------|------------|---------|------------|
| Project Planning & Tracking           | S        | D        | S          | S          | S       | S          |
| Prepare Project Introduction Document | R        | D        | R          | R          | R       | R          |
| Prepare Software Requirement Document | S        | D        | S          | S          | S       | S          |
| Prepare Software Design Document      | D        | D        | D          | D          | D       | D          |
| Prepare Test Document                 | D        | D        | D          | D          | D       | D          |
| Prepare Software User Guides          | D        | D        | D          | D          | D       | D          |
| Prepare Final Report                  | D        | D        | D          | D          | D       | D          |

#### 4. Project Communications

| Communication Item  | Who    | Purpose                                                                                                                                    | When           | Type                   |
|---------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------|
| Team Weekly Meeting | Group  | <ul style="list-style-type: none"> <li>Report work project</li> <li>Assign work for next week</li> <li>Control project deadline</li> </ul> | Every Sunday   | Meeting on Google Meet |
| Weekly Report       | Member | <ul style="list-style-type: none"> <li>Report work process</li> </ul>                                                                      | 1 day / week   | Report via Zalo        |
| Group Meeting       | Group  | <ul style="list-style-type: none"> <li>Report work process</li> </ul>                                                                      | Every Thursday | Offline in class       |

## 5. Configuration Management

### 5.1 Document Management

- Document tool: Confluence
- Task management tool: Jira

## 5.2 Source Code Management

- Source code management tool: GitHub

## 5.3 Tools & Infrastructure

| Category                | Tools / Infrastructure          |
|-------------------------|---------------------------------|
| Programming Languages   | PHP, HTML5, CSS3, JavaScript    |
| Database                | MySQL                           |
| Database Administration | phpMyAdmin                      |
| Academic APIs           | OpenAlex API, Crossref REST API |
| API Testing             | Postman                         |
| Version Control         | GitHub                          |
| Project Management      | Jira, Confluence                |
| Diagramming             | Draw.io, PlantUML               |
| IDE / Code Editor       | Visual Studio Code              |
| Local Server            | WAMP Server                     |

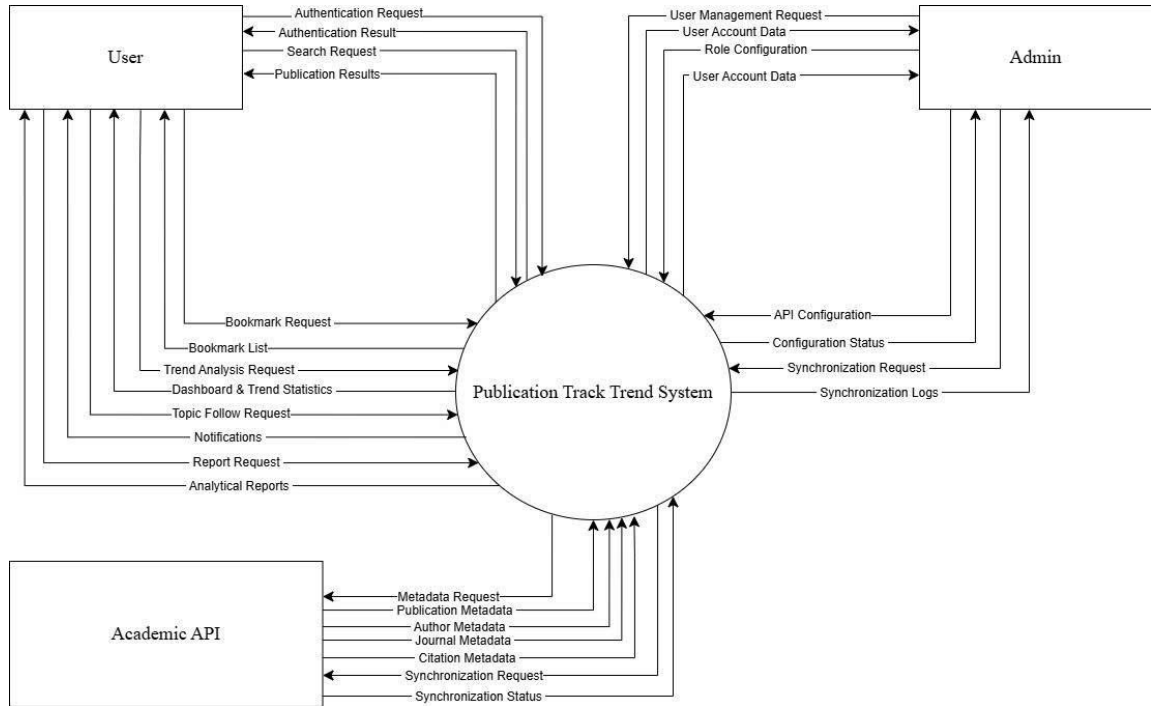
# III. SOFTWARE REQUIREMENT SPECIFICATION

## 1 Product Overview

The Scientific Journal Publication Trend Tracking System (SJPTTS) is a web-based application developed to help researchers, students, lecturers, and academic institutions search, monitor, and analyze scientific publication trends. The system integrates academic data from OpenAlex, and Crossref to provide comprehensive and up-to-date publication information.

The system allows users to search scientific publications using keywords, authors, journals, research fields. Users can view detailed publication information, bookmark publications, follow research topics, receive notifications about newly published papers, analyze publication trends through interactive dashboards, and export reports in PDF or CSV format.

Administrators can manage user accounts, configure academic API sources, synchronize publication data, monitor system operations, and maintain the overall performance of the platform. The system aims to provide an efficient solution for discovering scientific literature and supporting research through publication trend analysis.



Hình 1: Publication Trend Tracking System

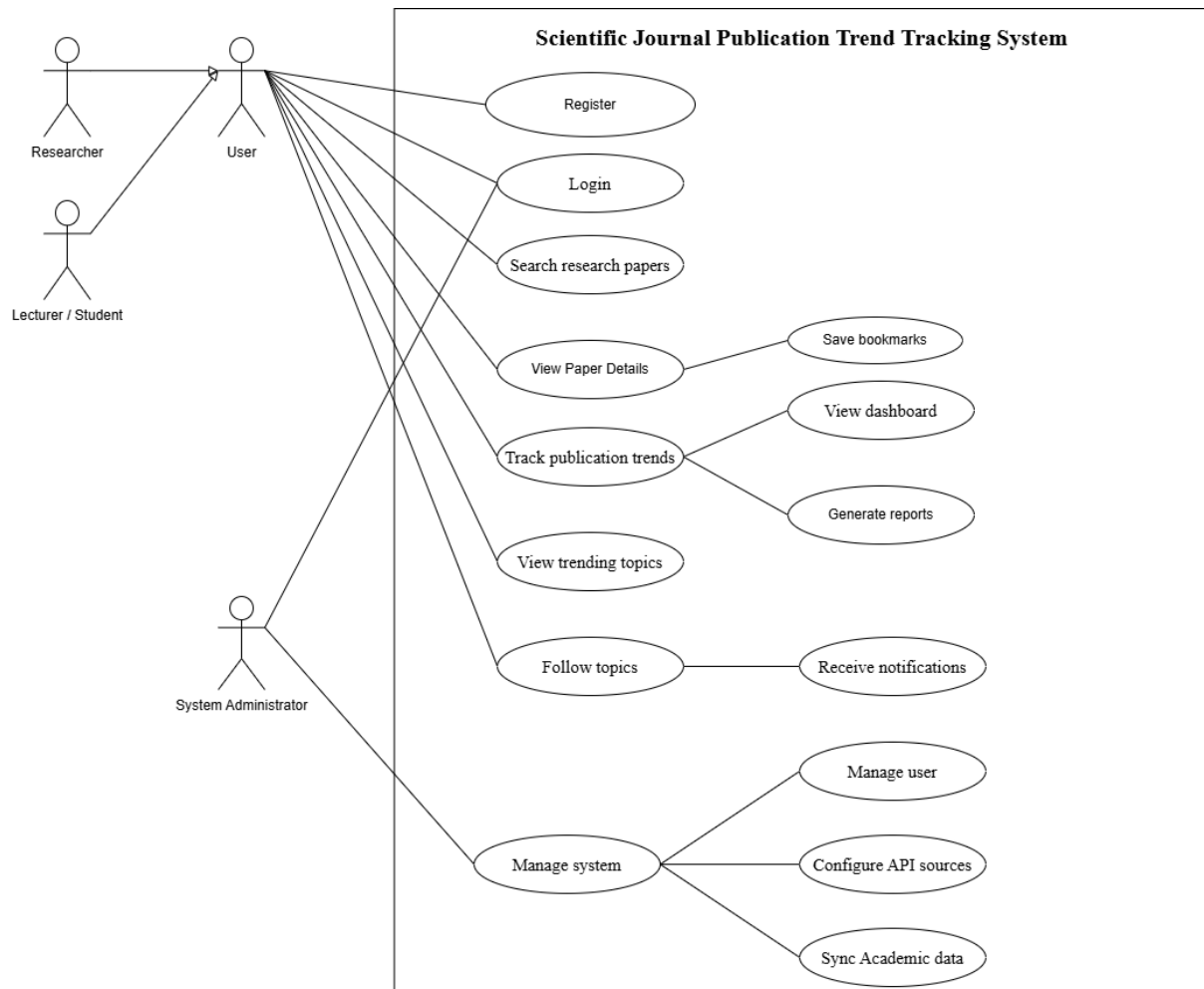
## 2 User Requirements

### 2.1 Actors

| # | Actor                             | Description                                                                                                                                                                                               |
|---|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | User                              | System user who can search scientific publications, view publication details, save bookmarks, follow research topics, receive notifications, and analyze publication trends.                              |
| 2 | Administrator                     | System administrator responsible for managing users, configuring the system, setting up and validating academic data sources (OpenAlex, Crossref), synchronizing data, and monitoring system operations.  |
| 3 | Publication Trend Tracking System | The core system that handles all functionalities such as paper search, trend analysis, dashboard generation, trending topic detection, bookmark management, notification delivery, and report generation. |
| 4 | Academic API                      | External services such as OpenAlex and Crossref that provide scientific publication metadata (authors, journals, citations) for system synchronization and analysis.                                      |

## 2.2 Use Case Diagrams

### 2.2.1 Diagram



Hình 2: Use Case Diagram

### 2.2.2 Use Case Descriptions

| Main UC Group  | UC | Main Agent (Actor) | Sub-UCs         | Brief Description of the Process                                                                             | Output                                                                    |
|----------------|----|--------------------|-----------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Manage Account |    | User               | Register, Login | Users create an account and authenticate to access the Scientific Journal Publication Trend Tracking System. | A registered account is created and a secure user session is established. |

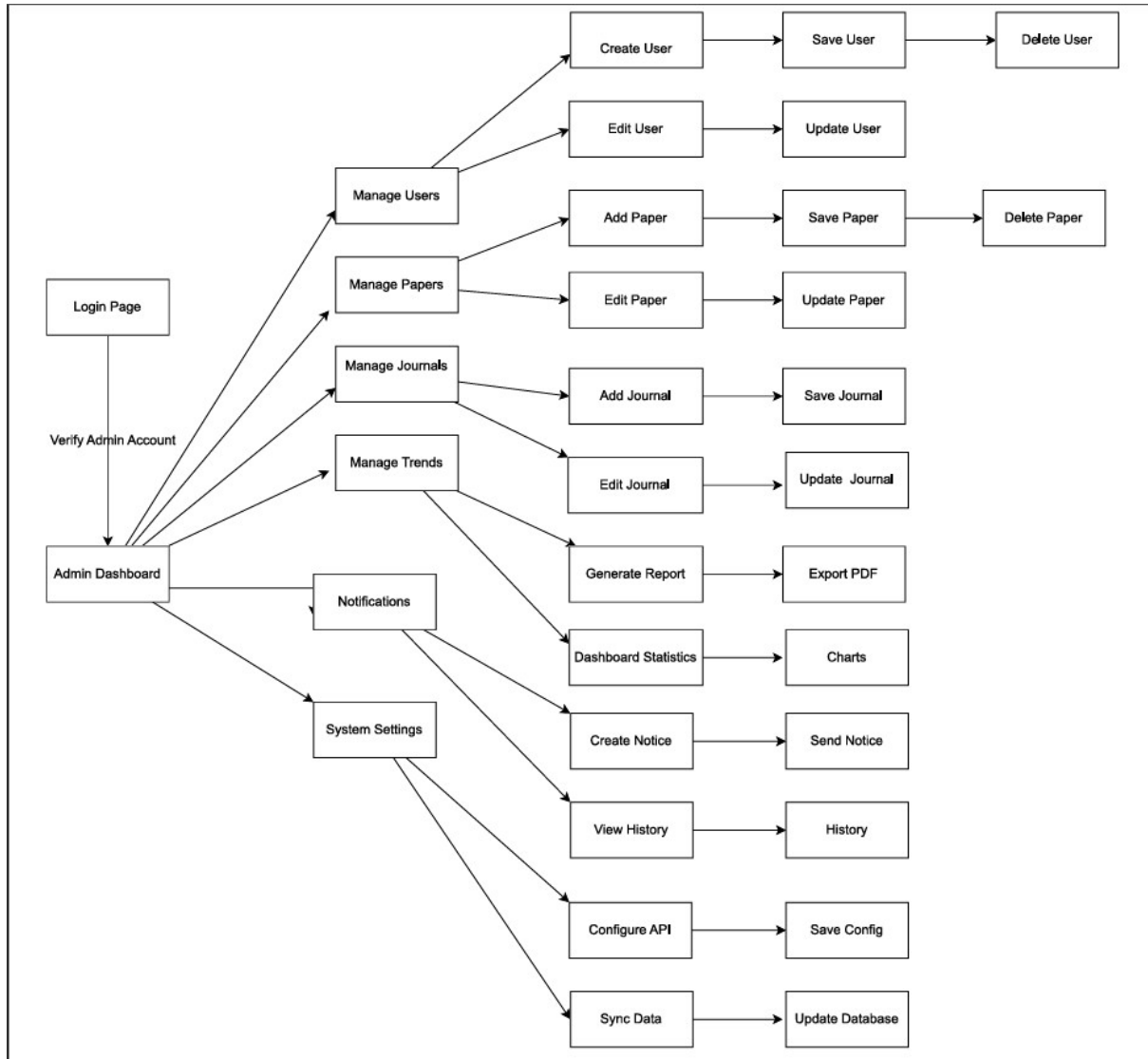
| Main UC Group            | Main Agent (Actor) | Sub-UCs                                                         | Brief Description of the Process                                                                                                                                      | Output                                                                                            |
|--------------------------|--------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Search Publications      | User               | Search Research Papers, View Paper Details                      | Users search scientific publications using keywords, author names, or journal names and view detailed publication meta-data.                                          | Matching publications and detailed publication information are displayed.                         |
| Manage Bookmarks         | User               | Save Bookmarks                                                  | Users save selected publications into their personal bookmark collection for future reference.                                                                        | The selected publication is stored in the user's bookmark list.                                   |
| Track Publication Trends | User               | Track Publication Trends, View Dashboard, View Trending Topics  | Users analyze publication trends by keyword, topic, author, journal, and time range. The system generates statistical charts and identifies trending research topics. | Trend statistics, dashboards and trending topics are displayed.                                   |
| Follow Topics            | User               | Follow Topics, Receive Notifications                            | Users follow research topics to receive notifications about tracked activities                                                                                        | The selected topic is added to the user's followed-topic list and notifications become available. |
| Generate Reports         | User               | Generate Reports                                                | Users generate analytical reports based on publication trend analysis and selected report parameters.                                                                 | An analytical report summarizing publication trends is generated.                                 |
| Manage System            | Administrator      | Manage System, Configure API Sources, Synchronize Academic Data | Administrators manage system settings, configure academic APIs, synchronize publication metadata and maintain system operations.                                      | System settings and publication metadata are successfully updated.                                |
| Manage Users             | Administrator      | Manage Users                                                    | Administrators manage user accounts, roles, permissions and account status.                                                                                           | User information and access permissions are updated.                                              |
| Configure API Sources    | Administrator      | Configure API Sources                                           | Administrators configure, update and test external academic APIs such as OpenAlex, Crossref and Semantic Scholar.                                                     | API configuration settings are successfully validated and stored.                                 |

| Main UC Group             | Main Agent (Actor) | Sub-UCs                   | Brief Description of the Process                                                                                                      | Output                                                                             |
|---------------------------|--------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Synchronize Academic Data | Administrator      | Synchronize Academic Data | Administrators synchronize publication metadata from external academic APIs into the local database for searching and trend analysis. | The publication metadata database is updated with the latest synchronized records. |

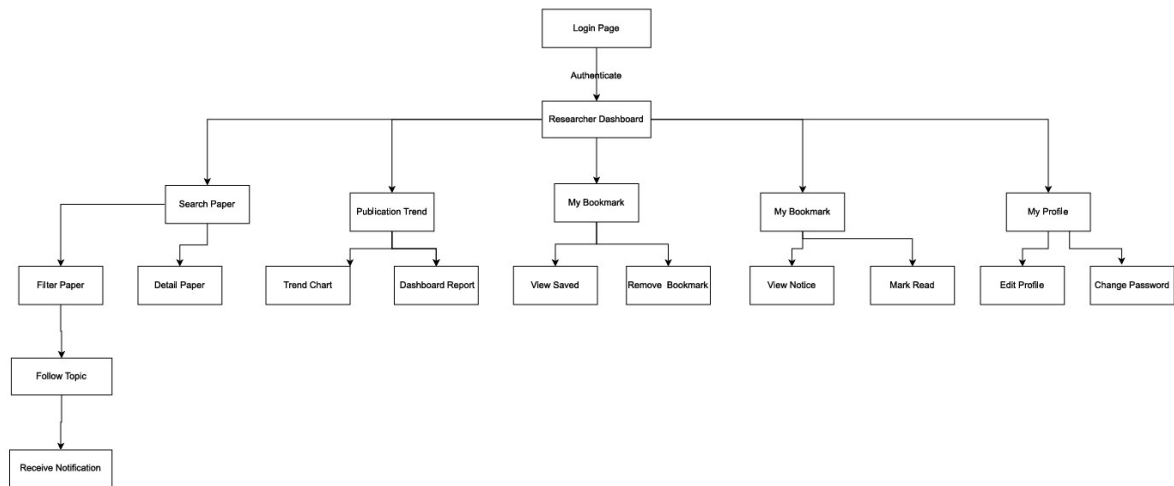
## 3 Functional Requirement

### 3.1 System Functional Overview

#### 3.1.1 Screens Flow



Hình 3: Screens Flow



Hình 4: Screen Description

| Functional Group    | Screen Name            | Detailed Description                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User Authentication | Login                  | Purpose: Allows users to log into the system. Components: Username, Password, Login button, and Register link. Flow: The user enters credentials → The system authenticates the user → The main system interface is displayed. Rule: Username and password are required; invalid credentials will generate an error message.                                                       |
| User Authentication | Register               | Purpose: Allows new users to create an account. Components: Full Name, Email, Username, Password, and Register button. Flow: The user enters the required information → The system validates the email and username → A new account is created → The user is redirected to the Login page. Rules: Email and username must be unique; passwords are securely hashed before storage. |
| Paper Management    | Search Research Papers | Purpose: Allows users to search scientific publications by keyword, author, or journal. Components: Search box, filters, Search button, and result list. Flow: The user enters search criteria → The system searches publication metadata → Matching publications are displayed.                                                                                                   |

| Functional Group           | Screen Name              | Detailed Description                                                                                                                                                                                                                                                                                        |
|----------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Paper Management           | View Paper Details       | Purpose: Displays detailed metadata of a selected research paper. Components: Title, Authors, Abstract, Keywords, Publication Year, Journal Name, Citation Information, and Save Bookmark button. Flow: The user selects a paper → The system retrieves publication metadata → Paper details are displayed. |
| Paper Management           | Bookmarks                | Purpose: Manages the user's bookmarked papers. Components: Bookmark list, Delete Bookmark button. Flow: The user views or removes bookmarked papers. Rule: Duplicate bookmarks are not allowed.                                                                                                             |
| Publication Trend Analysis | Track Publication Trends | Purpose: Analyzes publication trends based on keywords, research topics, authors, or journals. Components: Search filters, time range selector, and Analyze button. Flow: The user enters analysis criteria → The system analyzes publication metadata → Statistical results and charts are displayed.      |
| Publication Trend Analysis | Dashboard                | Purpose: Displays an overview of publication statistics. Components: Publications by year chart, Topic distribution chart, Journal distribution chart, and Growth trend chart. Flow: The system generates dashboard visualizations after trend analysis is completed.                                       |
| Publication Trend Analysis | Trending Topics          | Purpose: Displays the most popular research topics. Flow: The system analyzes publication data → Ranks research topics → Displays trending topics.                                                                                                                                                          |
| Topic Management           | Follow Topics            | Purpose: Allows users to follow research topics of interest. Components: Topic list and Follow button. Flow: The user selects a topic → The system stores it in the followed-topic list.                                                                                                                    |
| Notifications              | Notifications            | Purpose: Displays system notifications or notifications when following a new topic. Flow: The system generates notifications → The user views notification details and marks them as read.                                                                                                                  |

| Functional Group      | Screen Name               | Detailed Description                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reporting             | Generate Reports          | Purpose: Generates reports based on publication trend analysis. Flow: The user clicks "Generate Report" → The system displays options: Preview & Print / Save as PDF, Direct PDF Download, and Download Report (HTML) → The system executes the request based on the selected option and completes the generation, display, or export of the analytical report. |
| System Administration | Manage System             | Purpose: Provides administrative functions for system management. Components: User Management, API Configuration, and Academic Data Synchronization. Flow: The administrator selects a management function → The system performs the requested operation.                                                                                                       |
| System Administration | Manage Users              | Purpose: Manages user accounts. Components: User list, Change User Role, Activate Account, and Deactivate Account. Flow: The administrator selects a user → Updates account information → Saves the changes.                                                                                                                                                    |
| System Administration | Configure API Sources     | Purpose: To define and manage external academic data sources. Flow: The administrator enters source details (Name, URL, Priority, Status, Description). The administrator clicks "Add API Source" to save it to the system. The system displays the list, allowing the administrator to track, edit, or delete entries via the "Action" column.                 |
| System Administration | Synchronize Academic Data | Purpose: Synchronize publication data from external APIs into the system. Components: API List, Synchronize button, Synchronization Log. Workflow: Click Synchronize → The system retrieves data from the APIs → Updates the database → Displays the synchronization result.                                                                                    |

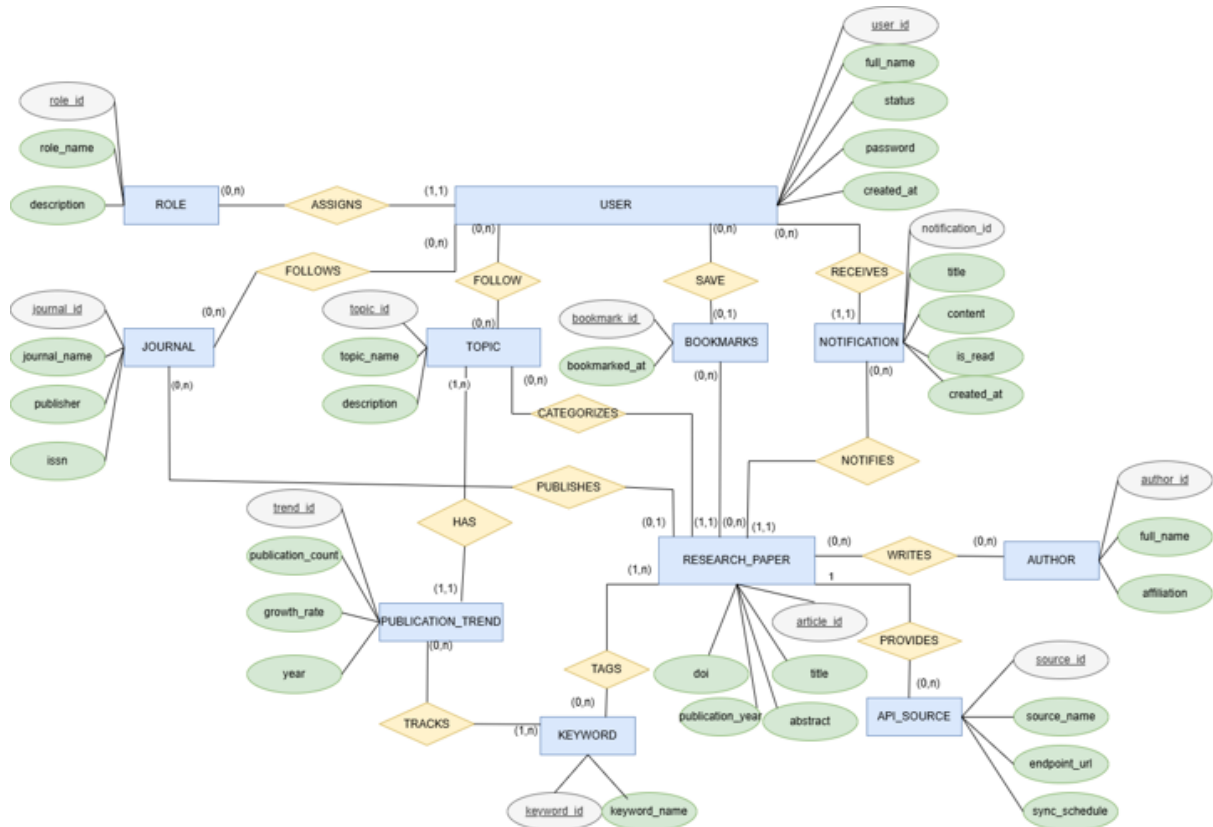
### 3.1.2 Screen Authorization

| Screen                   | Researcher | Lecturer/Student | System Administrator |
|--------------------------|------------|------------------|----------------------|
| Login                    | x          | x                | x                    |
| Logout                   | x          | x                | x                    |
| Register                 | x          | x                |                      |
| Search Research Papers   | x          | x                |                      |
| View Paper Details       | x          | x                |                      |
| Save Bookmarks           | x          | x                |                      |
| Track Publication Trends | x          | x                |                      |
| View Dashboard           | x          | x                |                      |
| Generate Reports         | x          | x                |                      |
| View Trending Topics     | x          | x                |                      |
| Follow Topics            | x          | x                |                      |
| Receive Notifications    | x          | x                |                      |
| View Profile             | x          | x                |                      |
| Edit Profile             | x          | x                |                      |
| View Notifications       | x          | x                |                      |
| Create Report            | x          | x                |                      |
| Manage System            |            |                  | x                    |
| Manage Users             |            |                  | x                    |
| Configure API Sources    |            |                  | x                    |
| Sync Academic Data       |            |                  | x                    |

### 3.1.3 Non-Screen Functions

| # | Feature                                 | System Function | Description                                                                                                                                                                                                                                                                                                         |
|---|-----------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Automated Academic Data Synchronization | Scheduled Job   | Scheduled job that automatically retrieves publication metadata directly from external academic APIs (OpenAlex and Crossref). The retrieved data is used for publication search, statistical analysis, and publication trend tracking.                                                                              |
| 2 | Automated Trend Calculation             | Scheduled Job   | Scheduled job that automatically recalculates publication trend statistics after each data synchronization from external academic APIs. The system analyzes total publications, total citations, Open Access rate, and research topic trends, then updates the data displayed on the dashboard and Trending Topics. |

## 3.2 Entity Relationship Diagram



Hình 5: Entity Relationship Diagram (ERD)

## 3.3 Authentication

### 3.3.1 Login

**Role:** User

**Function trigger:**

- The user enters a username and password to log in.

**Function description:**

- **Purpose:** Allow users to access the system.
- **Data processing:** Authenticate the entered username and password against the system database.

**Function Details:**

- If the login is successful, redirect the user to the Home page.
- If the login fails, display the error message: “Tên đăng nhập hoặc mật khẩu không đúng.”



## Chào mừng trở lại

Đăng nhập vào ScholarTrend

Tên đăng nhập

username

Mật khẩu

Password

**Đăng nhập**

Chưa có tài khoản? Đăng ký

Hình 6:

ScholarTrend



**Chào mừng trở lại**  
Đăng nhập vào ScholarTrend

Tên đăng nhập

Mật khẩu

Tên đăng nhập hoặc mật khẩu không đúng.

Đăng nhập

Hình 7:

DỮ LIỆU HỌC THUẬT THỜI GIAN THỰC

# Giải mã xu hướng Khoa học toàn cầu

Theo dõi hàng triệu ấn phẩm, phân tích tốc độ phát triển của các lĩnh vực và đón đầu những đột phá nghiên cứu quan trọng nhất.



Tìm kiếm bài báo nghiên cứu, tác giả hoặc từ khóa...

Khám phá ngay →

## Chủ đề đang nổi bật

Xem tất cả xu hướng ↗



+21% so với cùng kỳ

### Năng lượng Tái tạo

Hiệu suất pin mặt trời thế hệ mới và lưu trữ năng lượng quy mô lớn.

4.758+

Bài báo mới



+17% so với cùng kỳ

### Thu giữ Carbon

Công nghệ thu giữ carbon trực tiếp từ không khí và lưu trữ địa chất.

14.186+

Bài báo mới



+7% so với cùng kỳ

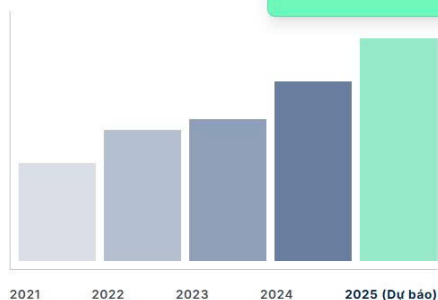
### Chỉnh sửa Gen CRISPR

Ứng dụng CRISPR trong điều trị bệnh di truyền và nông nghiệp.

3.977+

Bài báo mới

## Tần suất xuất bản theo năm



TỈ LỆ TĂNG TRƯỞNG  
21,6%  
Trung bình năm

## Phân tích xu hướng chuyên sâu

ScholarTrend hỗ trợ thu thập và phân tích metadata bài báo khoa học từ các nguồn học thuật uy tín. Hệ thống giúp người dùng theo dõi xu hướng nghiên cứu, khám phá các chủ đề nổi bật và hỗ trợ quá trình tìm kiếm, phân tích tài liệu khoa học.



Theo dõi số lượng bài báo theo từng năm và xu hướng xuất bản.



Phân tích các chủ đề nghiên cứu nổi bật dựa trên dữ liệu học thuật.



Hiển thị Dashboard trực quan với biểu đồ thống kê và kết quả phân tích.



## Đừng bỏ lỡ bất kỳ đột phá nào

Theo dõi các chủ đề nghiên cứu để nhận thông báo khi có bài báo mới hoặc khi xu hướng xuất bản của chủ đề thay đổi.

### 3.3.2 Register

**Role:** User

**Function trigger:** The user clicks Đăng kí.

**Function description:**

- Purpose: Create a new user account.
- Data processing: Validate user information and store it in the database.

**Function Details:**

- Whether the username already exists
- Validate password requirements.
- Create a new account.
- Display a success message.
- Sign up and check your email.

## Tạo tài khoản ScholarTrend

Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.

Họ và tên

 Nguyễn Văn A

Email

 example@email.com

Tên đăng nhập

 Username

Mật khẩu

 Password

Đăng ký

Đã có tài khoản? [Đăng nhập](#)

Hình 9:

Hình 10:

### 3.3.3 Profile Management

#### Update Profile

**Role:** User

**Function trigger:** The user opens the Profile page and clicks *Chỉnh sửa thông tin*.

**Function description:**

- Purpose: Allow users to update their personal information.
- Data processing: Validate and save the updated information.

**Function Details:**

- Update name, Email and other personal information.
- Display a success message after saving.

Thông tin tài khoản

Quản lý hồ sơ cá nhân và cài đặt bảo mật của bạn.

TH

HỌ VÀ TÊN

Trần Trung Hiếu

EMAIL

eldenring.674@gamil.com

TÊN ĐĂNG NHẬP

Hizeu

@

MÃ NGƯỜI DÙNG

31

#

TRẠNG THÁI EMAIL

✓

Đã xác minh

✎

Chỉnh sửa thông tin

↗

Đăng xuất

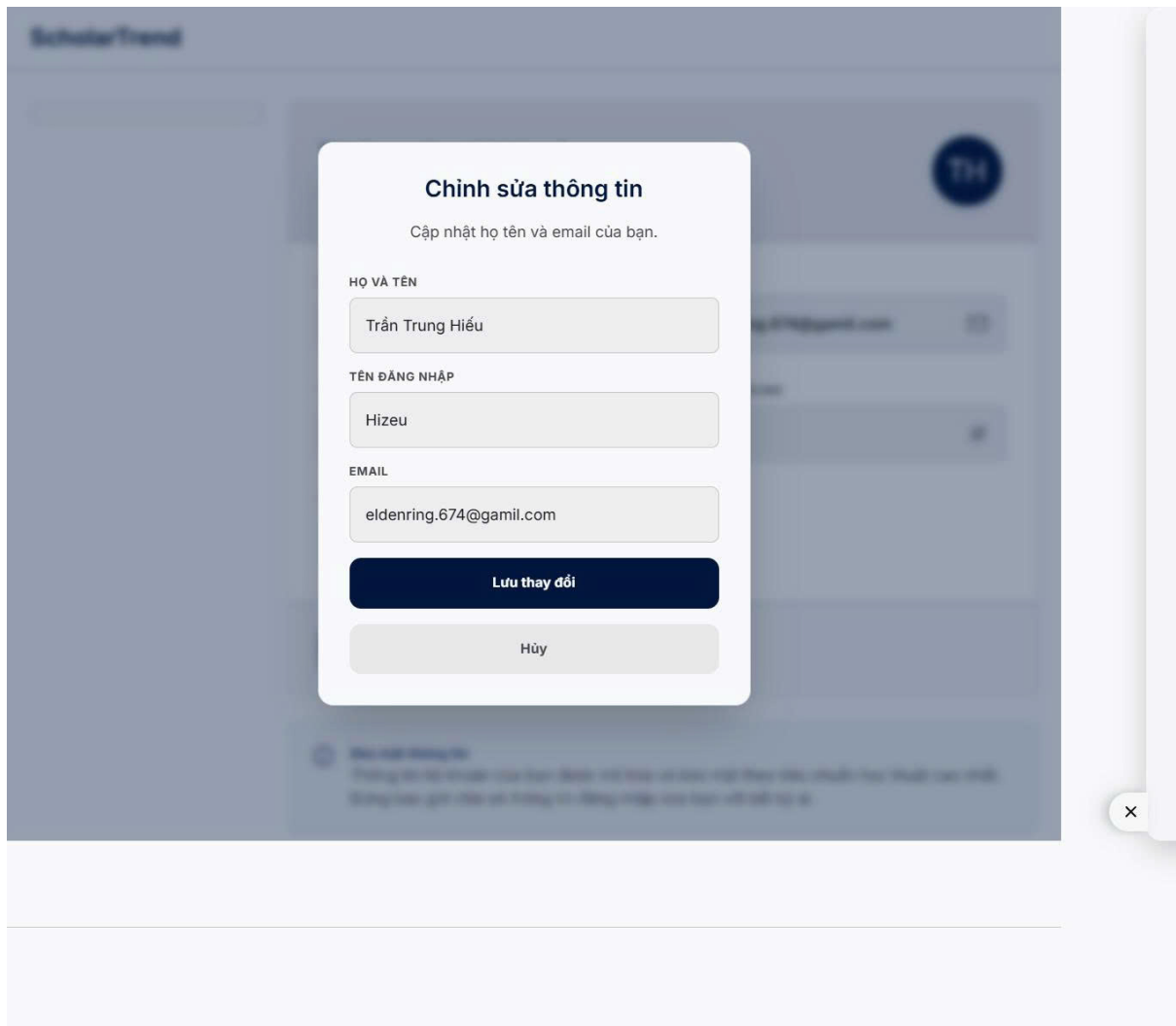
?

Bảo mật thông tin

Thông tin tài khoản của bạn được mã hóa và bảo mật theo tiêu chuẩn học thuật cao nhất. Đừng bao giờ chia sẻ thông tin đăng nhập của bạn với bất kỳ ai.

X

Hình 11:



Hình 12:

### 3.3.4 Logout

**Role:** User

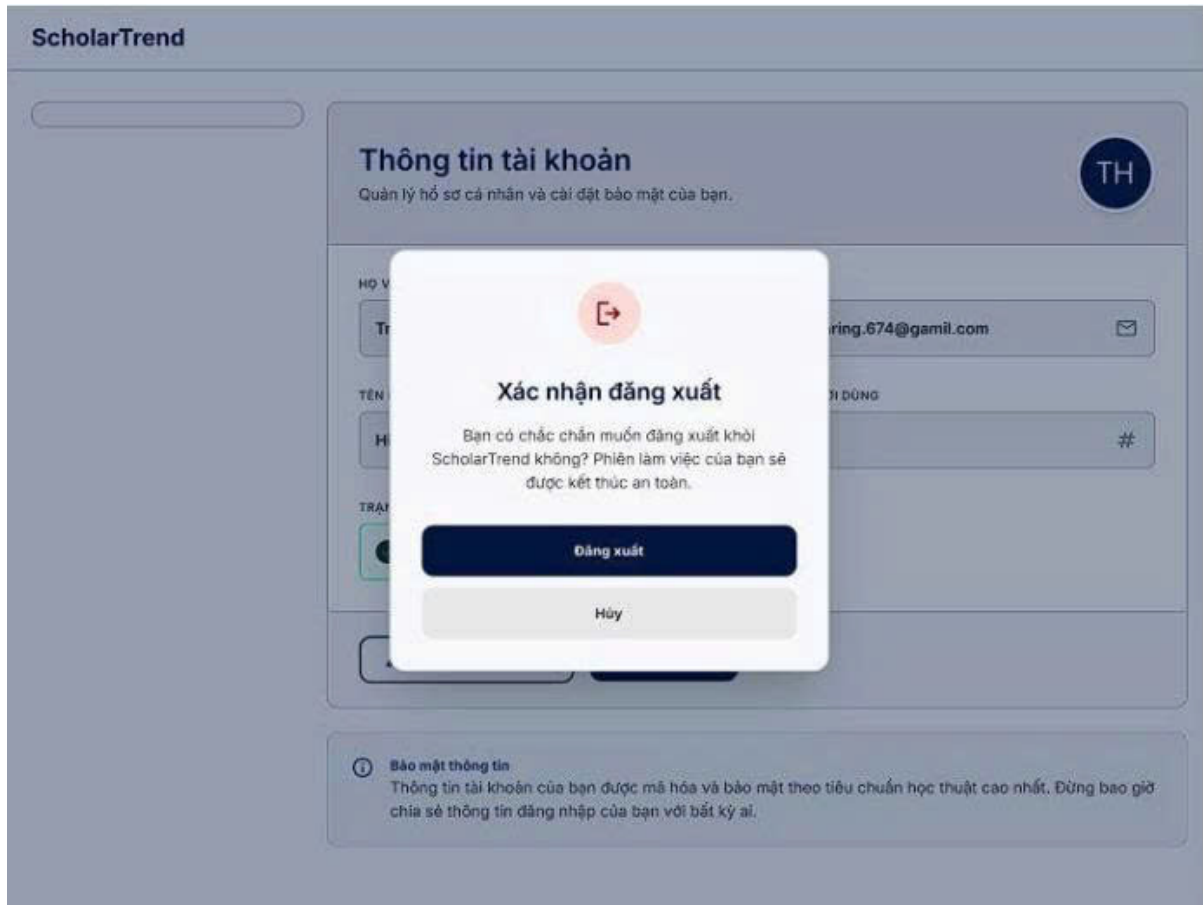
**Function trigger:** The user opens the Profile page and clicks Đăng xuất.

**Function description:**

- Purpose: Allow the user to securely log out of the system.
- Data processing: End the current user session.

**Function Details:**

- Display a confirmation dialog asking whether the user wants to Đăng xuất.
- If the user selects Đăng xuất, the system logs the user out and redirects them to the Đăng nhập page.
- If the user selects Hủy, the dialog is closed and the user remains on the current page.



Hình 13:

### 3.3.5 Search Papers

**Role:** User

#### **Function Trigger**

User opens “*Tìm kiếm*” → Enters a keyword → Selects filters (Publication Year, Open Access, Paper Type) → Clicks **Search**.

#### **Function Description**

- **Purpose:**

Allow users to search scientific papers from academic databases based on keywords and filtering conditions.

- **Data Processing:**

Validate the search input, send the request to academic databases (e.g., Crossref API), retrieve matching publications, apply selected filters, and display the search results.

#### **Function Details**

- **Validation:**

- Keyword cannot be empty.

- Start year must not be greater than End year.
- **Business Rules:**
  - Search supports Title, Keyword and DOI.
  - Users may filter by publication year.
  - Users may filter by paper type.
  - Users may choose to display only Open Access publications.
  - Results are displayed with pagination (10 items per page).
  - Each result includes Title, Authors, Year, DOI and Citation Count (if available), with the option to save the paper to Favorites.
- **Normal Case:**
  - User enters a keyword and selects filters.
  - User clicks the **Search** button.
  - System queries the academic database.
  - Matching publications are returned and displayed.
  - User can view paper details, open the DOI link, or save the paper to Favorites.
- **Abnormal Case:**
  - No papers found → Display “*No matching papers found.*”
  - Invalid year range → Request the user to correct the input.
  - API unavailable → Display “*Unable to retrieve publication data. Please try again later.*”

## Bộ lọc

## NĂM XUẤT BẢN

2020

2025

2020-2025

2015-2019

2010-2014

## TRUY CẬP



Chỉ Open Access

## LOẠI BÀI BÁO

- ☒ Tất cả
- ☐ Bài báo (Article)
- ☐ Tổng quan (Review)
- ☐ Chương sách
- ☐ Preprint
- ☐ Dữ liệu (Dataset)

Xoá bộ lọc

## Kết quả tìm kiếm

"Next-generation" —

2020 – 2025 ×

Next-generation



Đang tìm kiếm...

Hình 14:

## Bộ lọc

## NĂM XUẤT BẢN

2020

2025

2020–2025

2015–2019

2010–2014

## TRUY CẬP



Chỉ Open Access

## LOẠI BÀI BÁO

☒ Tất cả☐ Bài báo (Article)☐ Tổng quan (Review)☐ Chương sách☐ Preprint☐ Dữ liệu (Dataset)

Xoá bộ lọc

## Kết quả tìm kiếm

"Next-generation" — 261.869 bài báo (nguồn: Crossref)

Next-generation



2020 – 2025

1 Chương sách Next-Generation Sequencing Data Analysis 2023

## What's Next for Next-Generation Sequencing (NGS)?

Xinkun Wang

10.1201/9780429329180-20

DOI

Yêu thích

2 Bài báo Foods 2025 Open Access

## Apple Waste/By-Products and Microbial Resources to Promote the Design of Added-Value Foods: A Review

Hiba Selmi, Ester Presutto, Martina Totaro, Giuseppe Spano *et al.*

Apple fruit is among the most consumed fruits in the world, both in fresh and processed forms (e.g., ready-to-eat fresh slices, juice, jam, cider, and dried slices). During apple consumption/processing, a significant amount of apple residue is discarded. These residues can also be interesting materi...

8 trích dẫn 10.3390/foods14111850

DOI

Yêu thích

3 Chương sách Next-Generation Sequencing Data Analysis 2023

## Early-Stage Next-Generation Sequencing (NGS) Data Analysis

Xinkun Wang

10.1201/9780429329180-7

DOI

Yêu thích

4 Chương sách Next Generation Adaptation 2021

## INTRODUCTION:

ALLEN H. REDMON

10.3307/1-cty1455m6-3

DOI

Yêu thích

### Bộ lọc

#### NĂM XUẤT BẢN

2015

2019

2020-2025

2015-2019

2010-2014

#### TRUY CẬP



Chỉ Open Access

#### LOẠI BÀI BÁO

☒ Tất cả☐ Bài báo (Article)☐ Tổng quan (Review)☐ Chương sách☐ Preprint☐ Dữ liệu (Dataset) Xóa bộ lọc

## Kết quả tìm kiếm

**"Néch-gen-ra-ti-on"** — 0 bài báo

2015 - 2019 X

Néch-gen-ra-ti-on



Không tìm thấy kết quả với bộ lọc này.

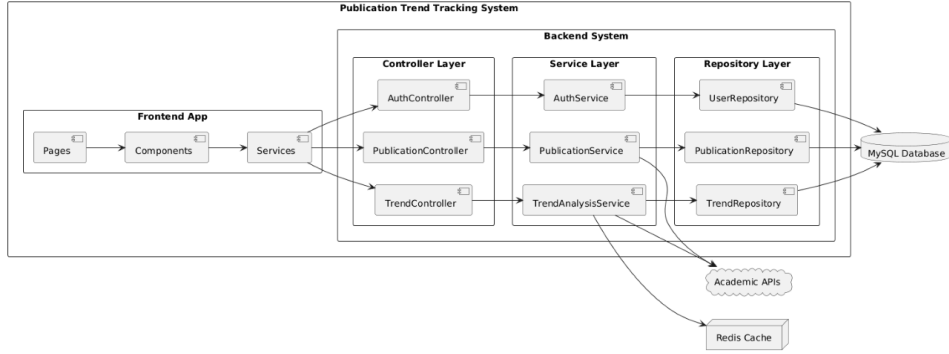
Xóa bộ lọc

Hình 16:

# IV. Software Design Description

## 1 System Design

### 1.1 System Architecture



#### 1.1.1 Frontend App

**TypeScript (TS):** Core technology used for frontend development, enabling the implementation of scalable, maintainable, and type-safe user interfaces.

**Pages:** Responsible for rendering the main user interfaces, including Login Page, Dashboard Page, Publication Search Page, and Trend Analysis Page.

**Components:** Reusable user interface elements used to construct application pages, such as search bars, filter panels, tables, charts, and publication cards.

**Services:** Responsible for handling communication between the frontend and backend through HTTP requests and responses.

#### 1.1.2 Backend System

The backend system follows a layered architecture consisting of Controller Layer, Service Layer, and Repository Layer.

##### Controller Layer

**AuthController:** Handles authentication requests such as login, logout, and user authorization.

**PublicationController:** Processes publication-related requests, including publication searching, filtering, and detailed information retrieval.

**TrendController:** Handles requests related to publication trend analysis, dashboard visualization, and trend monitoring.

##### Service Layer

**AuthService:** Implements authentication logic, including user verification and access control.

**PublicationService:** Handles publication-related business logic such as processing search queries, filtering publications, and managing publication metadata.

**TrendAnalysisService:** Responsible for trend analysis, topic identification, dashboard analytics, and report generation.

#### **Repository Layer**

**UserRepository:** Manages data access operations related to user accounts and authentication data.

**PublicationRepository:** Handles database operations related to publication metadata, authors, and research fields.

**TrendRepository:** Responsible for storing and retrieving trend-related analytical data.

### **1.1.3 Database**

**MySQL Database:** Relational database management system used to store system data, including user accounts, publication metadata, trend records, and system-generated reports.

### **1.1.4 External Services**

**Academic APIs:** External academic data sources used to retrieve scientific publication metadata, including publication titles, authors, research fields, publication years, and citation-related information.

Examples include:

- OpenAlex
- Crossref
- Google Scholar

These APIs support publication synchronization and provide up-to-date research data for the system.

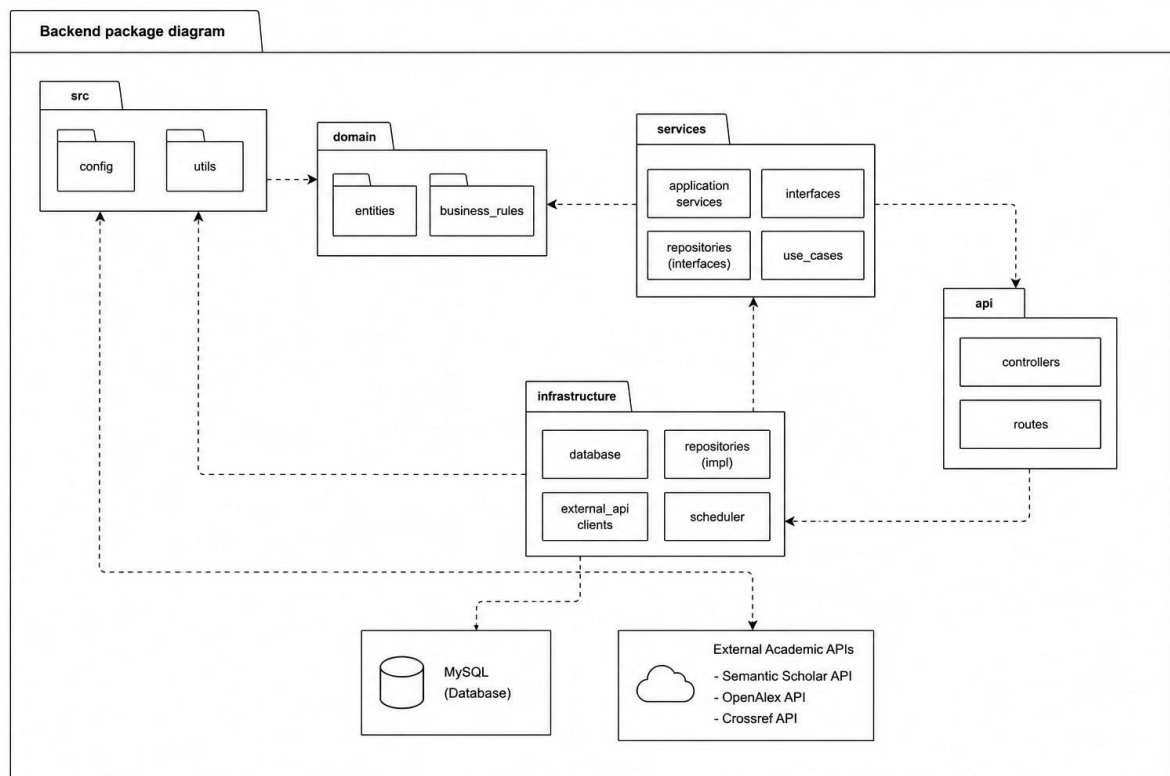
### **1.1.5 Cache Management**

**Redis Cache:** In-memory caching system used to store frequently accessed data such as dashboard statistics, publication trends, and popular search results to improve system performance and reduce database workload.

## 1.2 Package Diagram

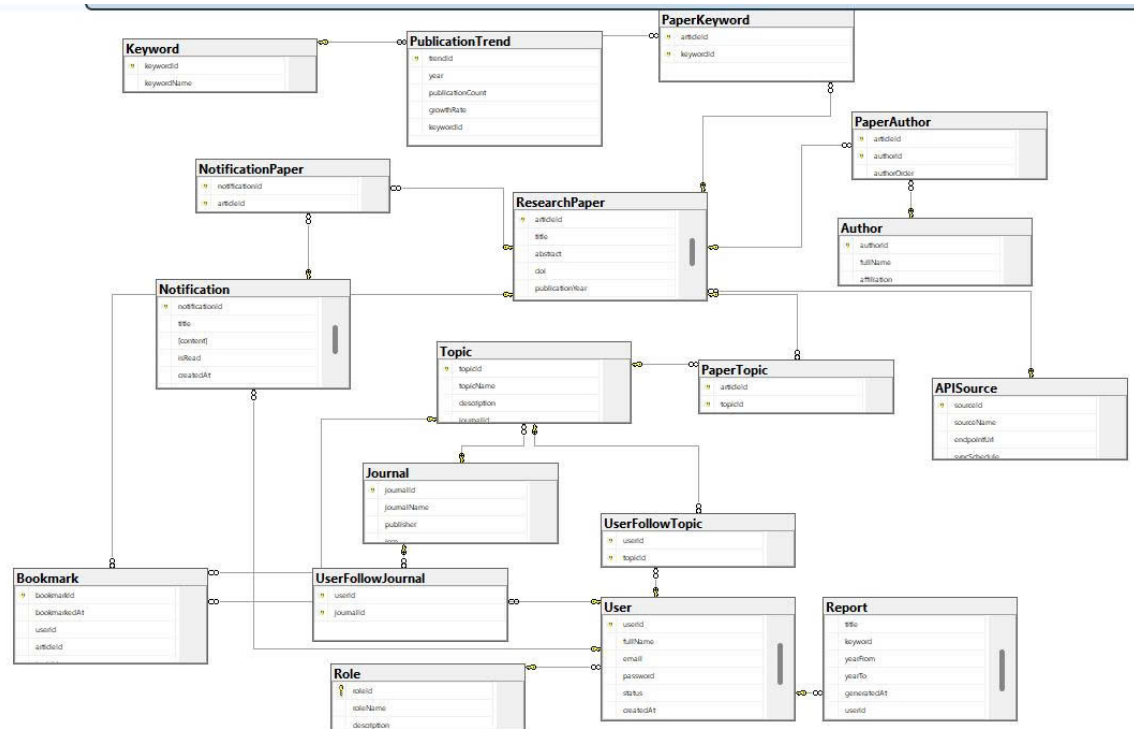
### 1.2.1 Front End

### 1.2.2 Back End



Hình 17: Backend package diagram

## 2 Database Design



Hình 18: Database Design

### Entities Description

## 2.1 Role Table

| Field  | Type | Description | PK  | FK |
|--------|------|-------------|-----|----|
| roleId | INT  | Role ID     | Yes | No |

|             |               |             |    |    |
|-------------|---------------|-------------|----|----|
| roleName    | NVARCHAR(100) | Role name   | No | No |
| description | NVARCHAR(255) | Description | No | No |

## 2.2 User Table

| Field     | Type          | Description    | PK  | FK         |
|-----------|---------------|----------------|-----|------------|
| userId    | INT           | User ID        | Yes | No         |
| fullName  | NVARCHAR(150) | Full name      | No  | No         |
| email     | NVARCHAR(150) | Email          | No  | No         |
| password  | NVARCHAR(255) | Password       | No  | No         |
| status    | NVARCHAR(50)  | Status         | No  | No         |
| createdAt | DATETIME2     | Created time   | No  | No         |
| roleId    | INT           | Role reference | No  | Yes → Role |

## 2.3 Journal Table

| Field       | Type          | Description  | PK  | FK |
|-------------|---------------|--------------|-----|----|
| journalId   | INT           | Journal ID   | Yes | No |
| journalName | NVARCHAR(200) | Journal name | No  | No |
| publisher   | NVARCHAR(150) | Publisher    | No  | No |
| issn        | NVARCHAR(20)  | ISSN         | No  | No |

## 2.4 UserFollowJournal (M:N)

| Field  | Type | Description | PK                 | FK         |
|--------|------|-------------|--------------------|------------|
| userId | INT  | User        | Yes<br>(Composite) | Yes → User |

|           |     |         |                    |                  |
|-----------|-----|---------|--------------------|------------------|
| journalId | INT | Journal | Yes<br>(Composite) | Yes →<br>Journal |
|-----------|-----|---------|--------------------|------------------|

## 2.5 Topic Table

| Field       | Type          | Description       | PK  | FK               |
|-------------|---------------|-------------------|-----|------------------|
| topicId     | INT           | Topic ID          | Yes | No               |
| topicName   | NVARCHAR(150) | Topic name        | No  | No               |
| description | NVARCHAR(MAX) | Description       | No  | No               |
| journalId   | INT           | Journal reference | No  | Yes →<br>Journal |

## 2.6 UserFollowTopic (M:N)

| Field   | Type | Description | PK                 | FK             |
|---------|------|-------------|--------------------|----------------|
| userId  | INT  | User        | Yes<br>(Composite) | Yes →<br>User  |
| topicId | INT  | Topic       | Yes<br>(Composite) | Yes →<br>Topic |

## 2.7 APISource Table

| Field        | Type          | Description   | PK  | FK |
|--------------|---------------|---------------|-----|----|
| sourceId     | INT           | Source ID     | Yes | No |
| sourceName   | NVARCHAR(150) | Source name   | No  | No |
| endpointUrl  | NVARCHAR(500) | API URL       | No  | No |
| syncSchedule | NVARCHAR(100) | Sync schedule | No  | No |

## 2.8 Author Table

| Field       | Type          | Description | PK  | FK |
|-------------|---------------|-------------|-----|----|
| authorId    | INT           | Author ID   | Yes | No |
| fullName    | NVARCHAR(150) | Full name   | No  | No |
| affiliation | NVARCHAR(255) | Affiliation | No  | No |

## 2.9 Keyword Table

| Field       | Type          | Description | PK  | FK |
|-------------|---------------|-------------|-----|----|
| keywordId   | INT           | Keyword ID  | Yes | No |
| keywordName | NVARCHAR(100) | Keyword     | No  | No |

## 2.10 Research Paper Table

| Field           | Type          | Description | PK  | FK              |
|-----------------|---------------|-------------|-----|-----------------|
| articleId       | INT           | Paper ID    | Yes | No              |
| title           | NVARCHAR(500) | Title       | No  | No              |
| abstract        | NVARCHAR(MAX) | Abstract    | No  | No              |
| doi             | NVARCHAR(200) | DOI         | No  | No              |
| publicationYear | INT           | Year        | No  | No              |
| sourceId        | INT           | API Source  | No  | Yes → APISource |

## 2.11 PaperTopic (M:N)

| Field | Type | Description | PK | FK |
|-------|------|-------------|----|----|
|-------|------|-------------|----|----|

|           |     |       |                         |                                  |
|-----------|-----|-------|-------------------------|----------------------------------|
| articleId | INT | Paper | Yes<br>(Com-<br>posite) | Yes<br>→ Re-<br>search-<br>Paper |
| topicId   | INT | Topic | Yes<br>(Com-<br>posite) | Yes →<br>Topic                   |

## 2.12 PaperKeyword (M:N)

| Field     | Type | Description | PK                      | FK                               |
|-----------|------|-------------|-------------------------|----------------------------------|
| articleId | INT  | Paper       | Yes<br>(Com-<br>posite) | Yes<br>→ Re-<br>search-<br>Paper |
| keywordId | INT  | Keyword     | Yes<br>(Com-<br>posite) | Yes →<br>Keyword                 |

## 2.13 PaperAuthor (M:N)

| Field       | Type | Description  | PK                      | FK                               |
|-------------|------|--------------|-------------------------|----------------------------------|
| articleId   | INT  | Paper        | Yes<br>(Com-<br>posite) | Yes<br>→ Re-<br>search-<br>Paper |
| authorId    | INT  | Author       | Yes<br>(Com-<br>posite) | Yes →<br>Author                  |
| authorOrder | INT  | Author order | No                      | No                               |

## 2.14 PublicationTrend Table

| Field            | Type  | Description | PK  | FK               |
|------------------|-------|-------------|-----|------------------|
| trendId          | INT   | Trend ID    | Yes | No               |
| year             | INT   | Year        | No  | No               |
| publicationCount | INT   | Count       | No  | No               |
| growthRate       | FLOAT | Growth rate | No  | No               |
| keywordId        | INT   | Keyword     | No  | Yes →<br>Keyword |

## 2.15 Report Table

| Field       | Type          | Description  | PK  | FK            |
|-------------|---------------|--------------|-----|---------------|
| reportId    | INT           | Report ID    | Yes | No            |
| title       | NVARCHAR(300) | Title        | No  | No            |
| keyword     | NVARCHAR(200) | Keyword      | No  | No            |
| yearFrom    | INT           | From year    | No  | No            |
| yearTo      | INT           | To year      | No  | No            |
| generatedAt | DATETIME2     | Created time | No  | No            |
| userId      | INT           | User         | No  | Yes →<br>User |

## 2.16 Notification Table

| Field          | Type          | Description     | PK  | FK |
|----------------|---------------|-----------------|-----|----|
| notificationId | INT           | Notification ID | Yes | No |
| title          | NVARCHAR(300) | Title           | No  | No |
| content        | NVARCHAR(MAX) | Content         | No  | No |
| isRead         | BIT           | Read status     | No  | No |
| createdAt      | DATETIME2     | Created time    | No  | No |

|        |     |      |    |            |
|--------|-----|------|----|------------|
| userId | INT | User | No | Yes → User |
|--------|-----|------|----|------------|

## 2.17 NotificationPaper (M:N)

| Field          | Type | Description  | PK                 | FK                   |
|----------------|------|--------------|--------------------|----------------------|
| notificationId | INT  | Notification | Yes<br>(Composite) | Yes → Notification   |
| articleId      | INT  | Paper        | Yes<br>(Composite) | Yes → Research-Paper |

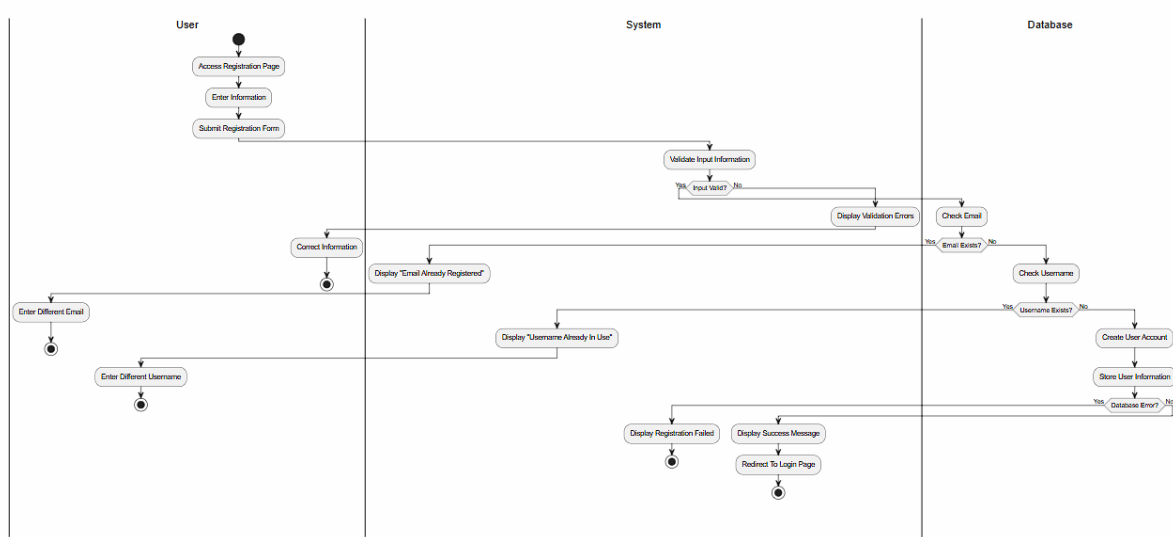
## 2.18 Bookmark Table

| Field        | Type      | Description | PK  | FK                   |
|--------------|-----------|-------------|-----|----------------------|
| bookmarkId   | INT       | Bookmark ID | Yes | No                   |
| bookmarkedAt | DATETIME2 | Time saved  | No  | No                   |
| userId       | INT       | User        | No  | Yes → User           |
| articleId    | INT       | Paper       | No  | Yes → Research-Paper |
| topicId      | INT       | Topic       | No  | Yes → Topic          |

## 3 Detail Design

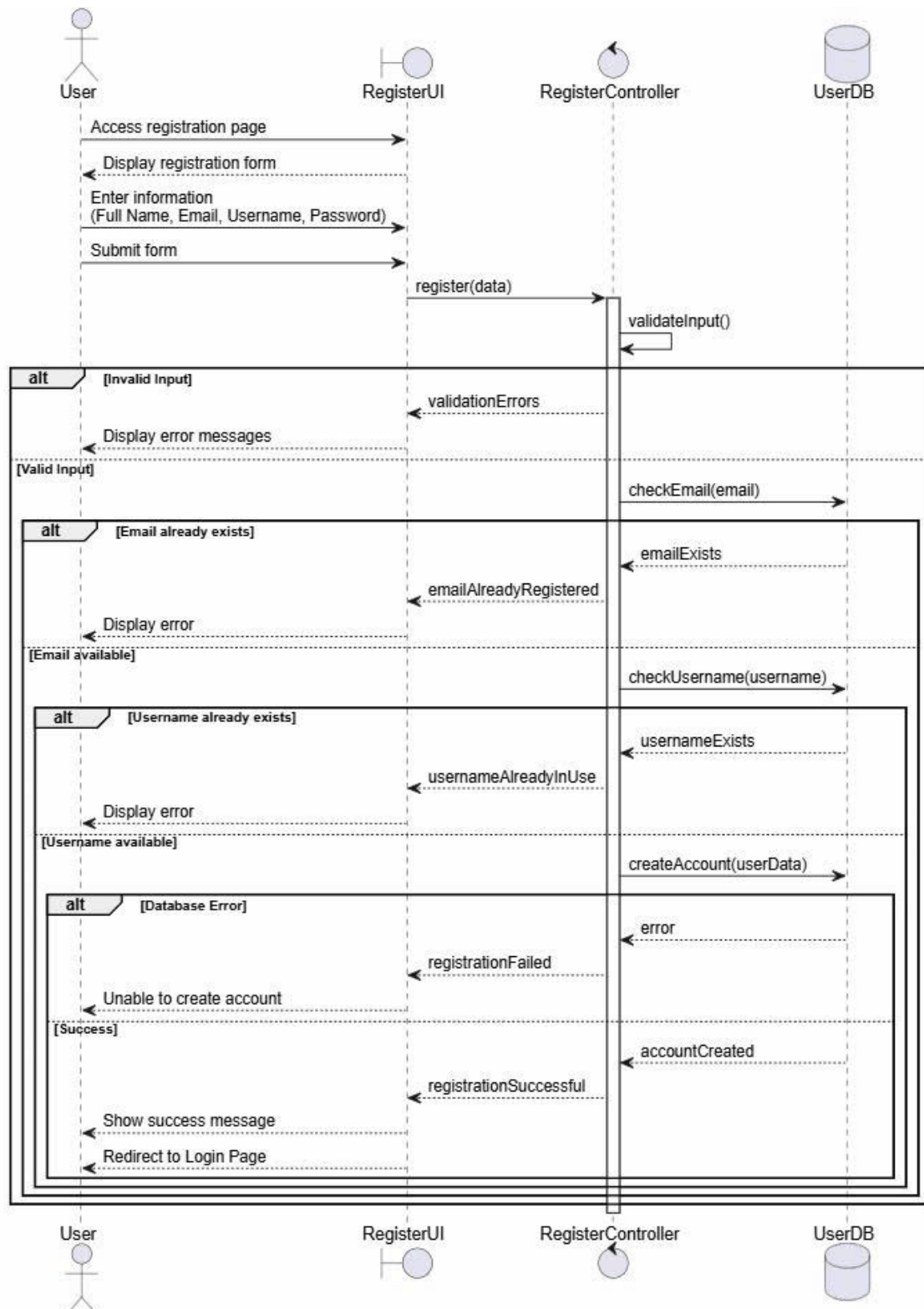
### 3.1 Register

#### 3.1.1 Activity diagram



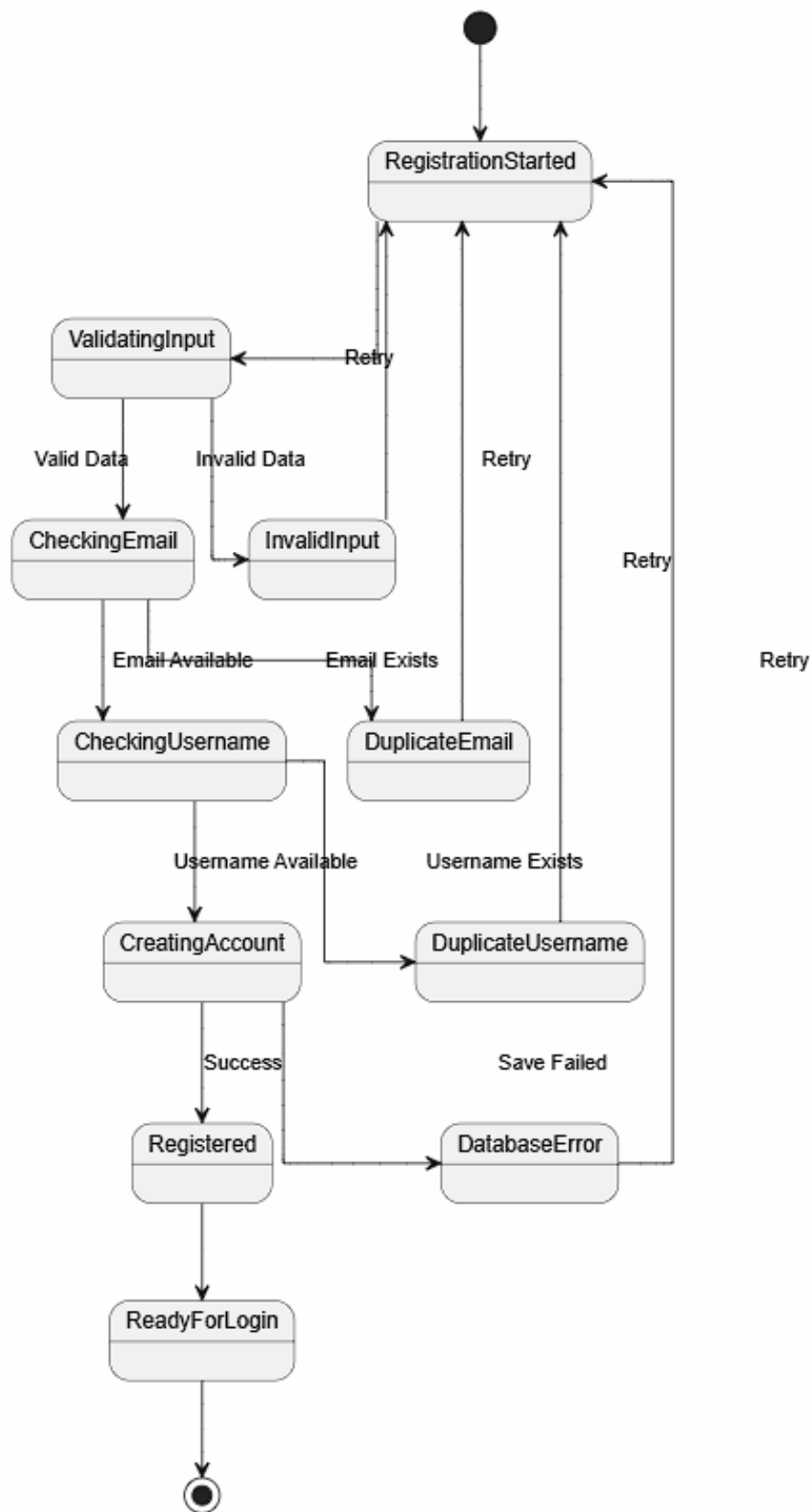
Hình 19: Activity diagram

### 3.1.2 Sequence Diagram



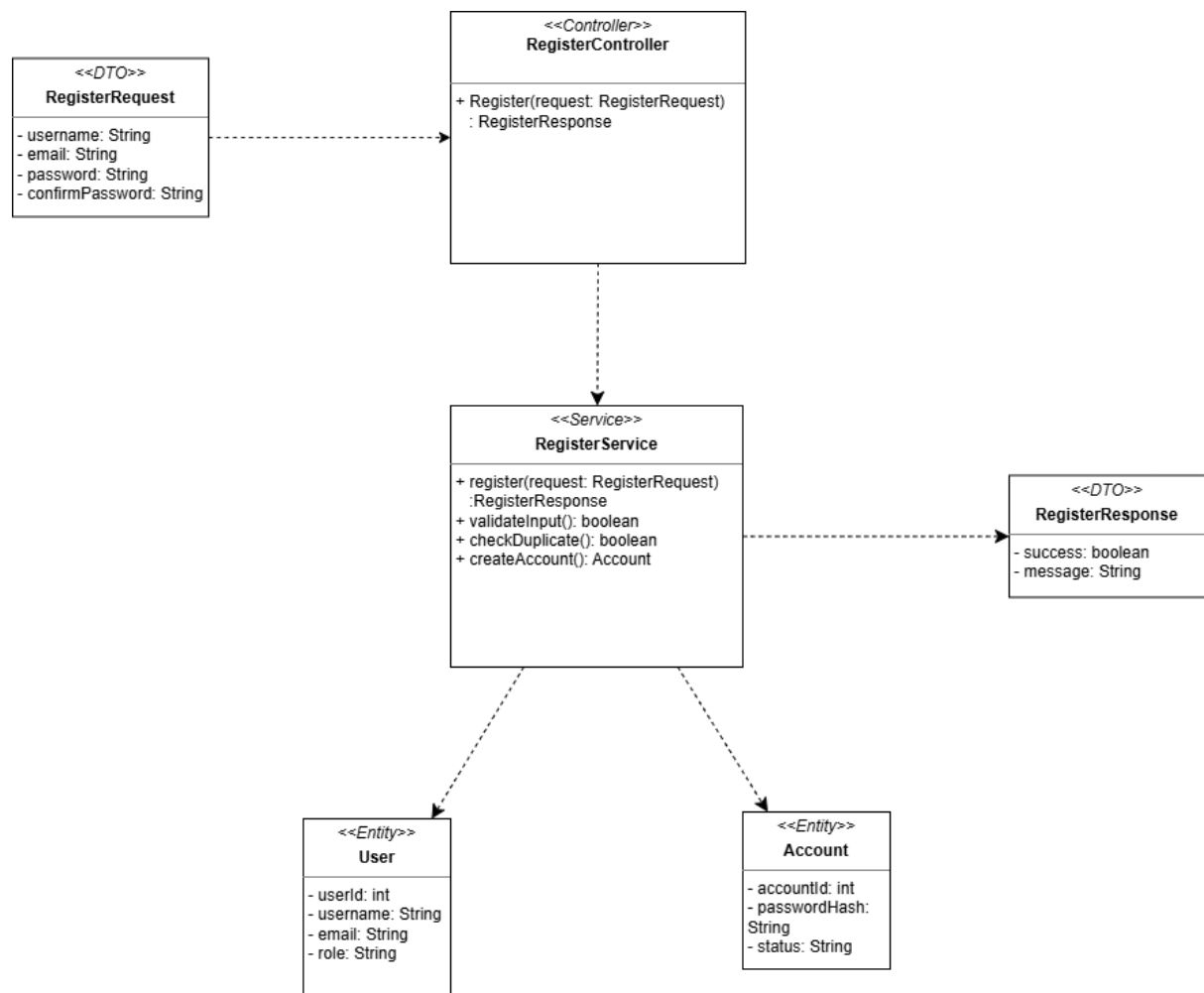
Hình 20: Sequence Diagram

### 3.1.3 State Diagram



Hình 21: State diagram

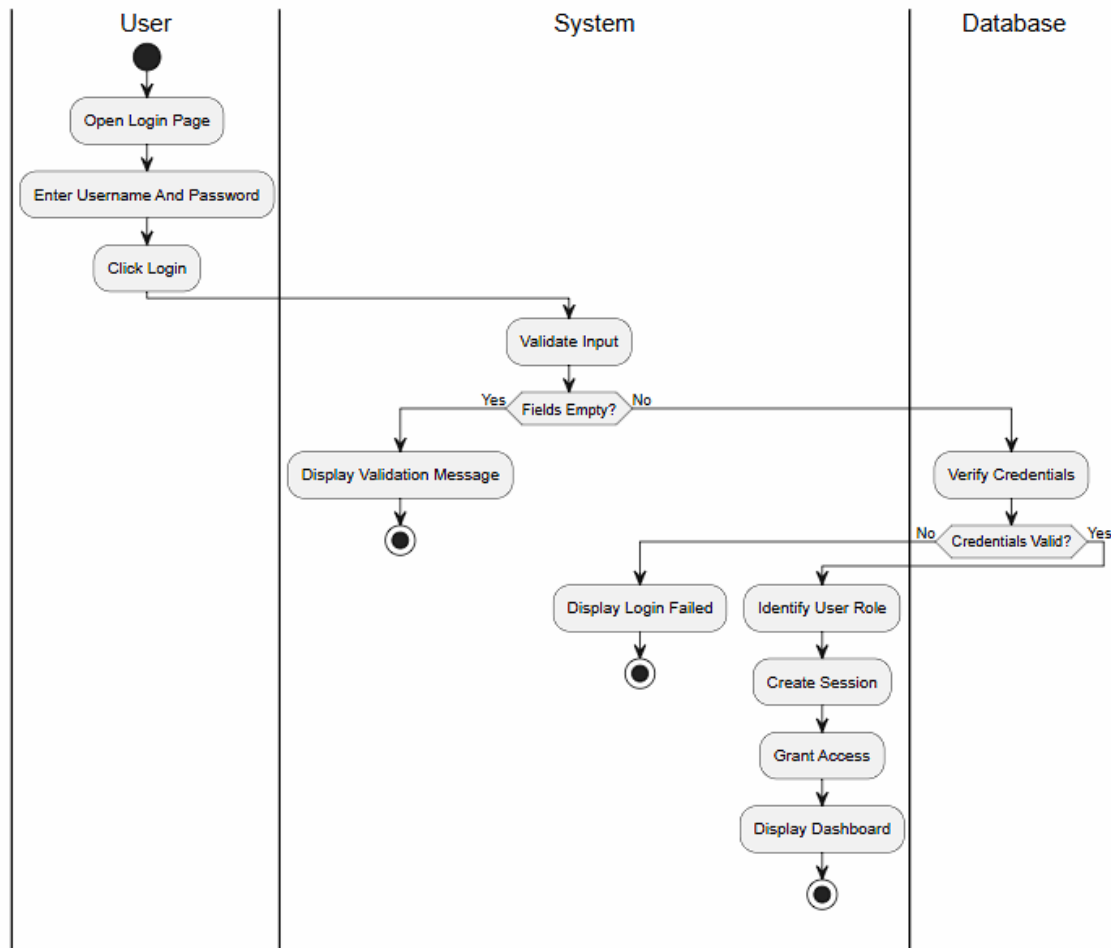
### 3.1.4 Class Diagram



Hình 22: Class Diagram: Register

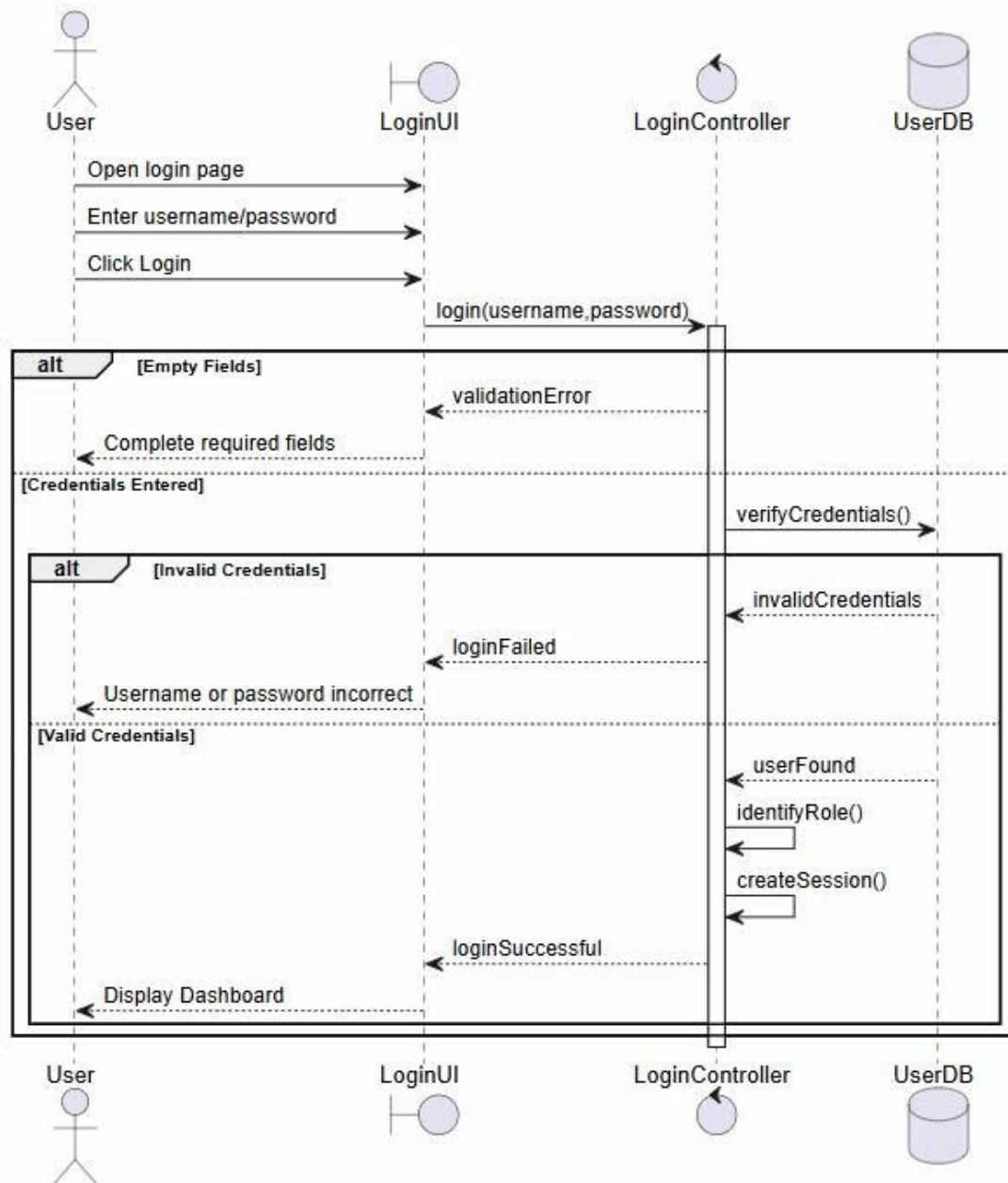
## 3.2 Login

### 3.2.1 Activity Diagram



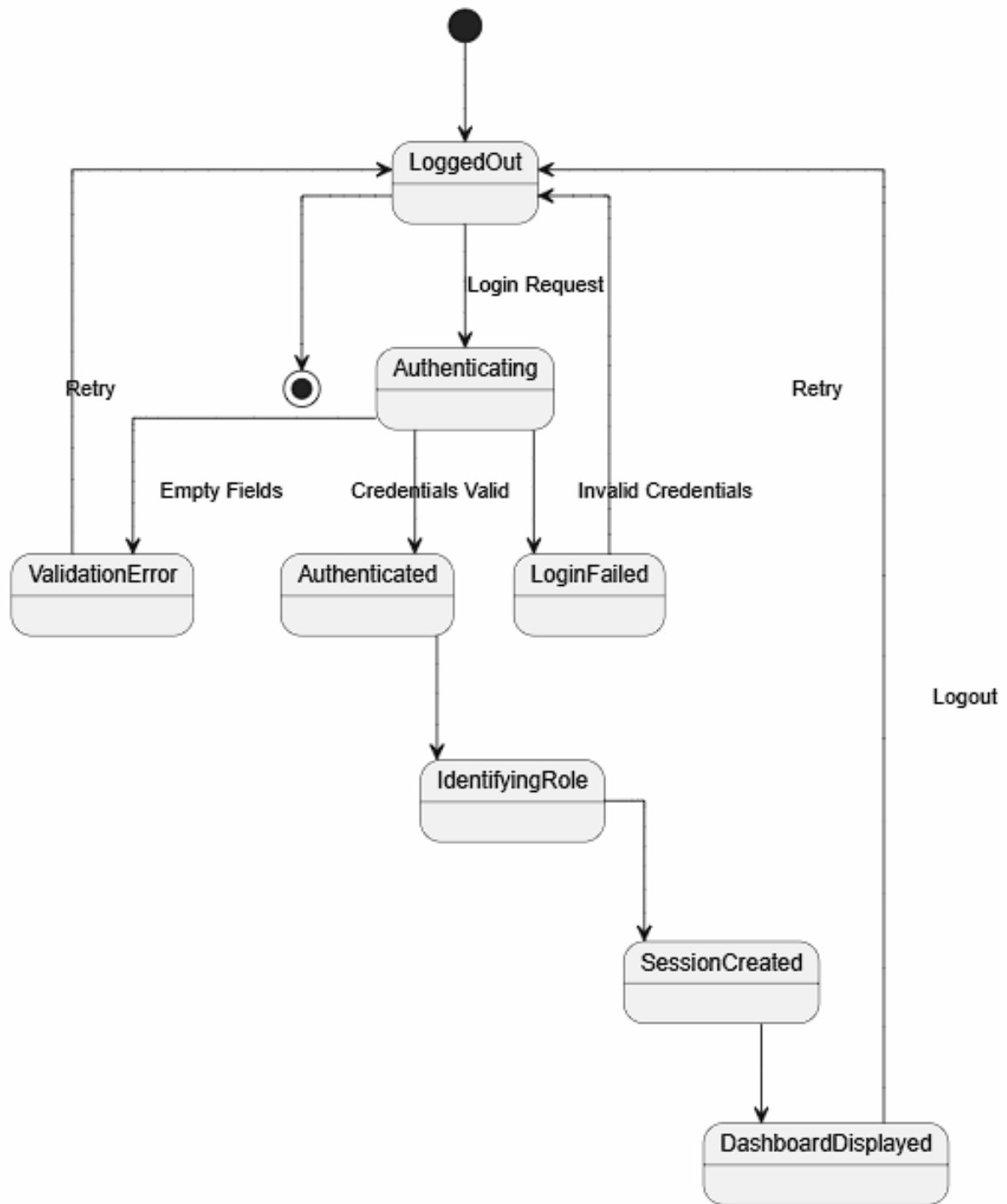
Hình 23: Activity diagram

### 3.2.2 Sequence Diagram



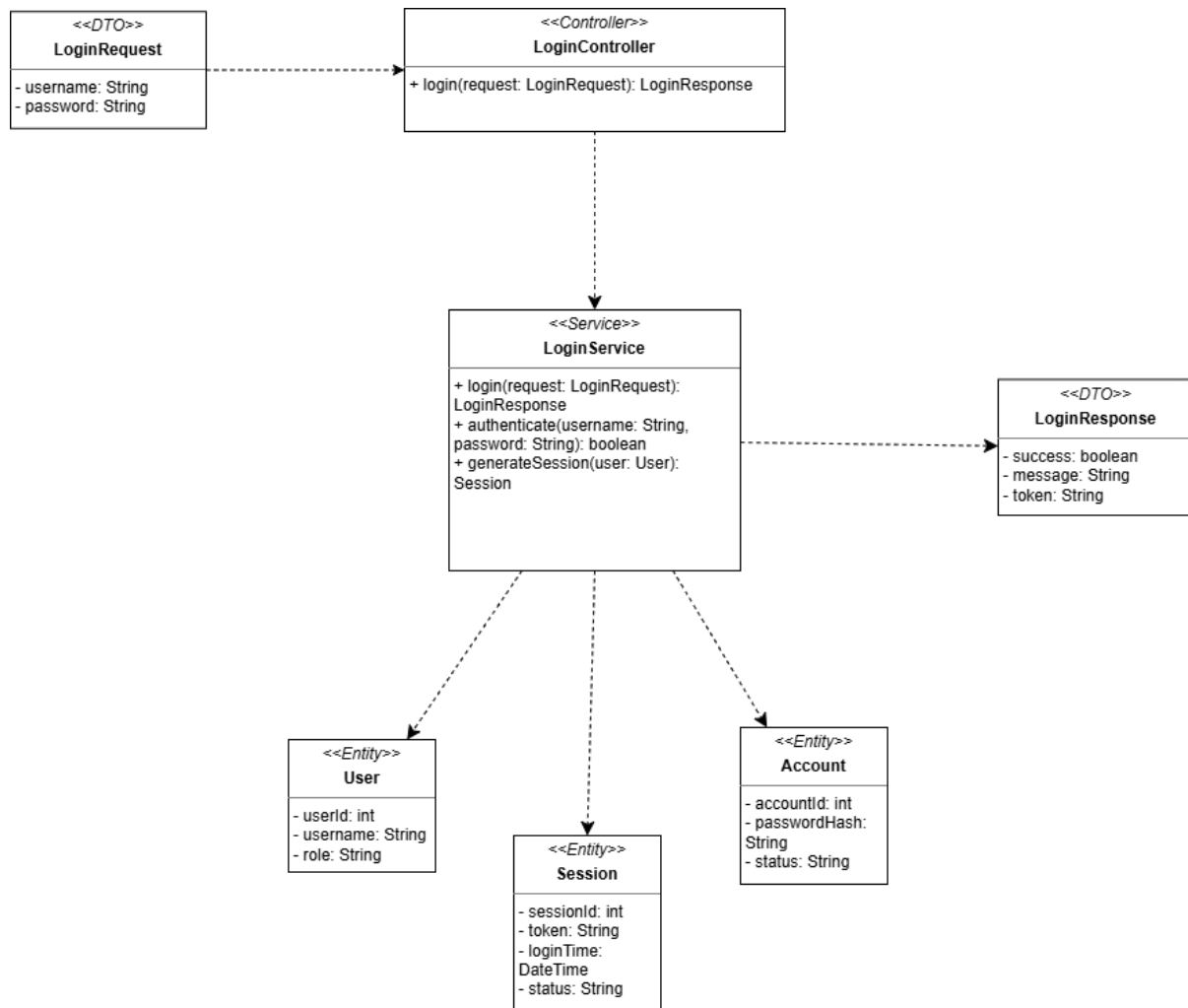
Hình 24: Sequence Diagram

### 3.2.3 State Diagram



Hình 25: State diagram

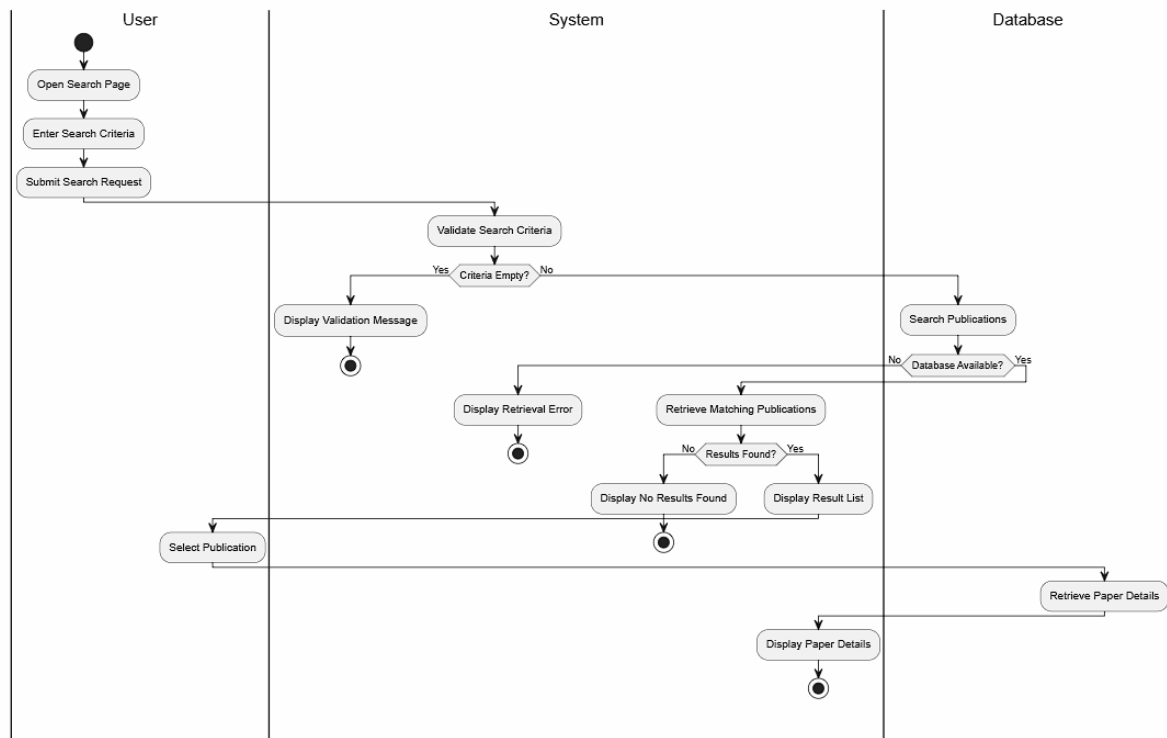
### 3.2.4 Class Diagram



Hình 26: Class Diagram: Login

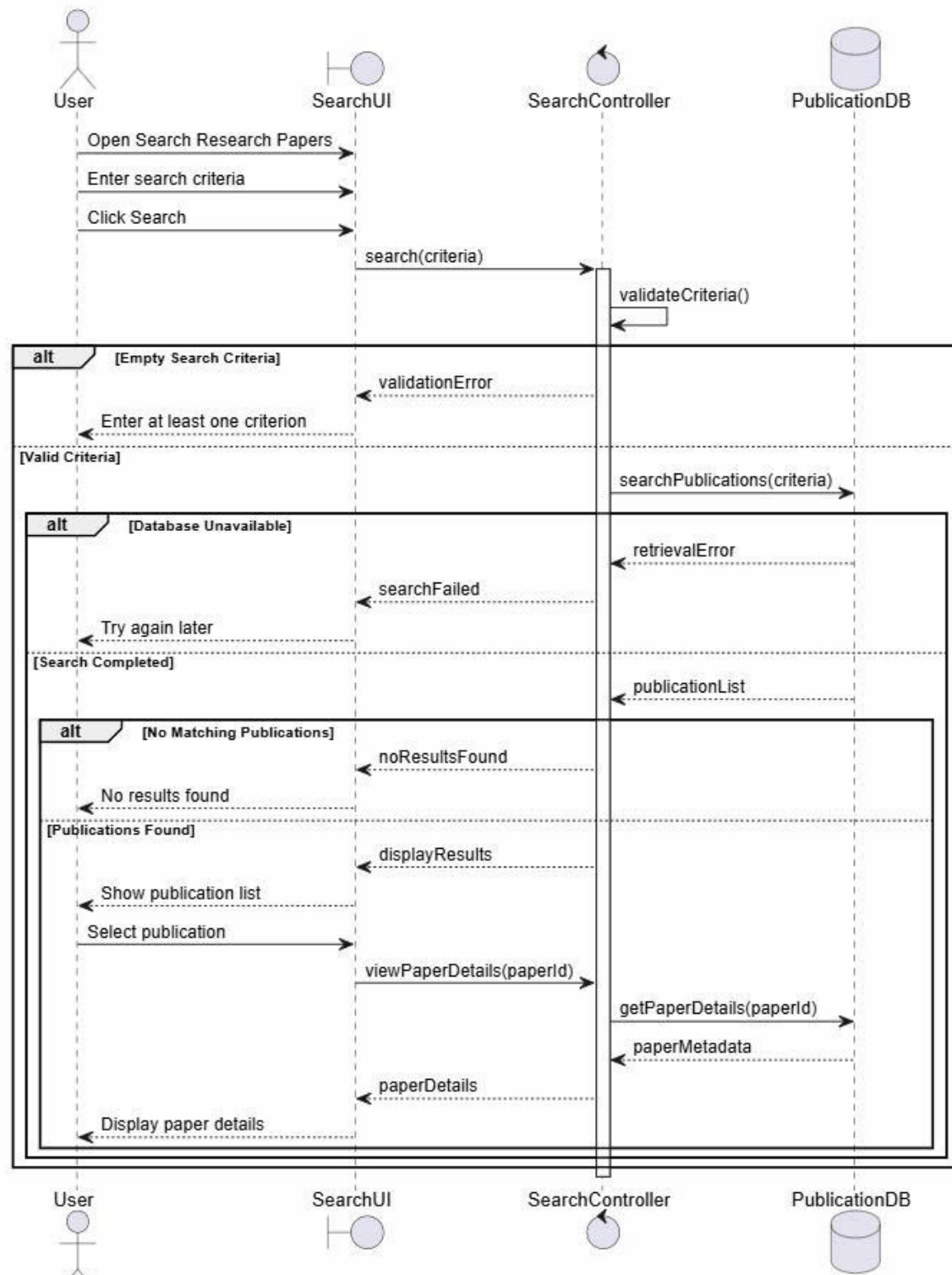
## 3.3 Search Research Papers

### 3.3.1 Activity Diagram



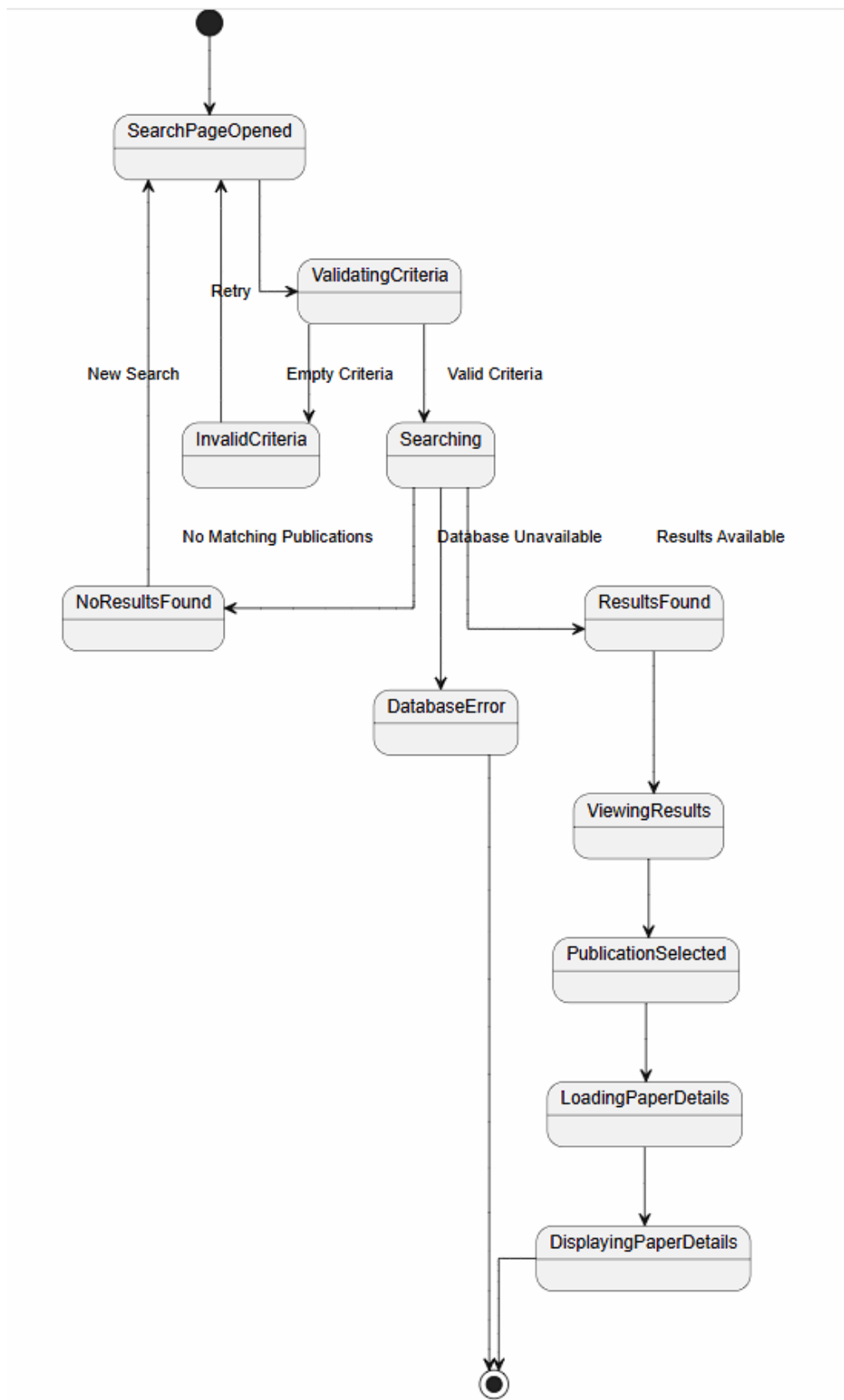
Hình 27: Activity diagram

### 3.3.2 Sequence Diagram



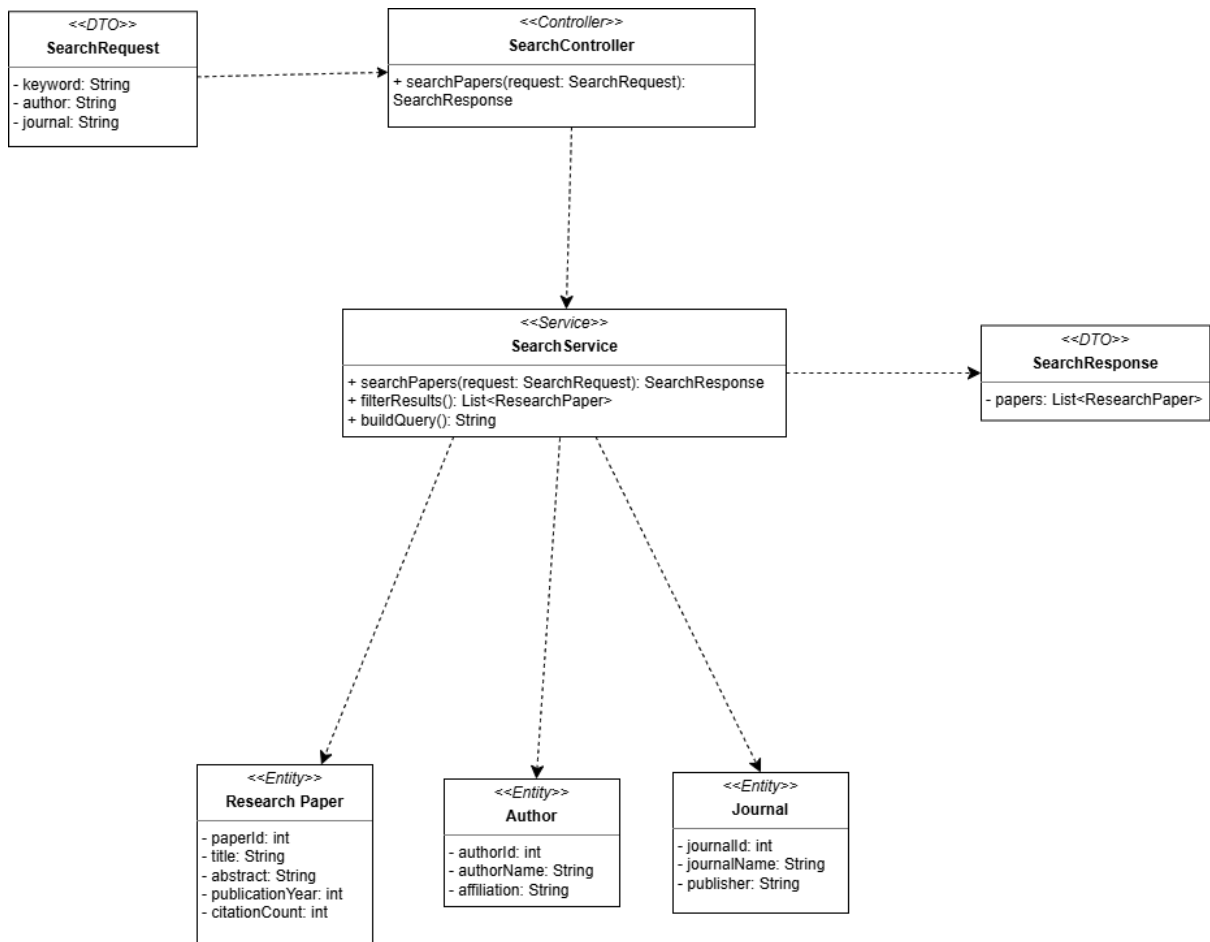
Hình 28: Sequence Diagram

### 3.3.3 State Diagram



Hình 29: State diagram

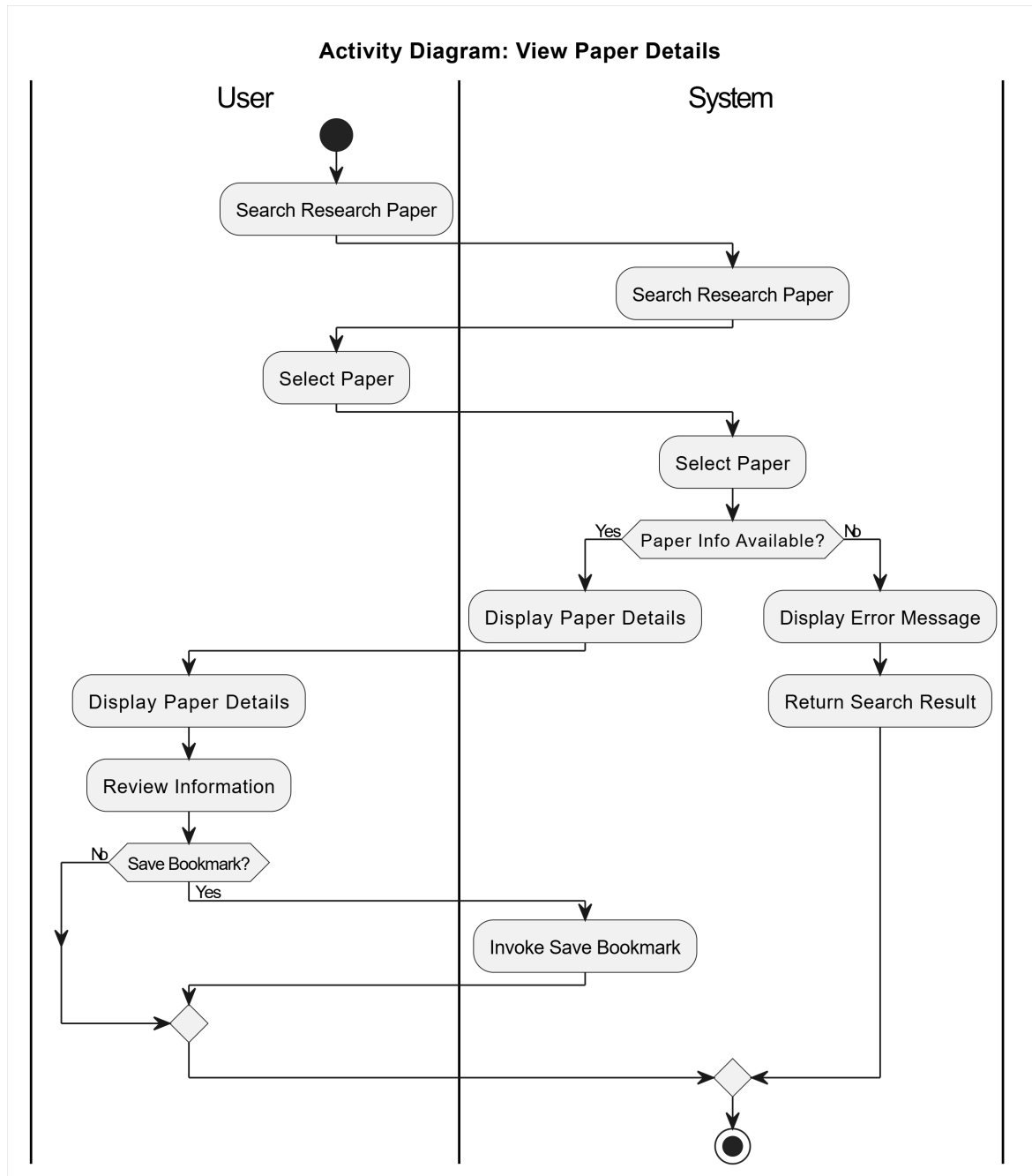
### 3.3.4 Class Diagram



Hình 30: Class Diagram: Search Research Papers

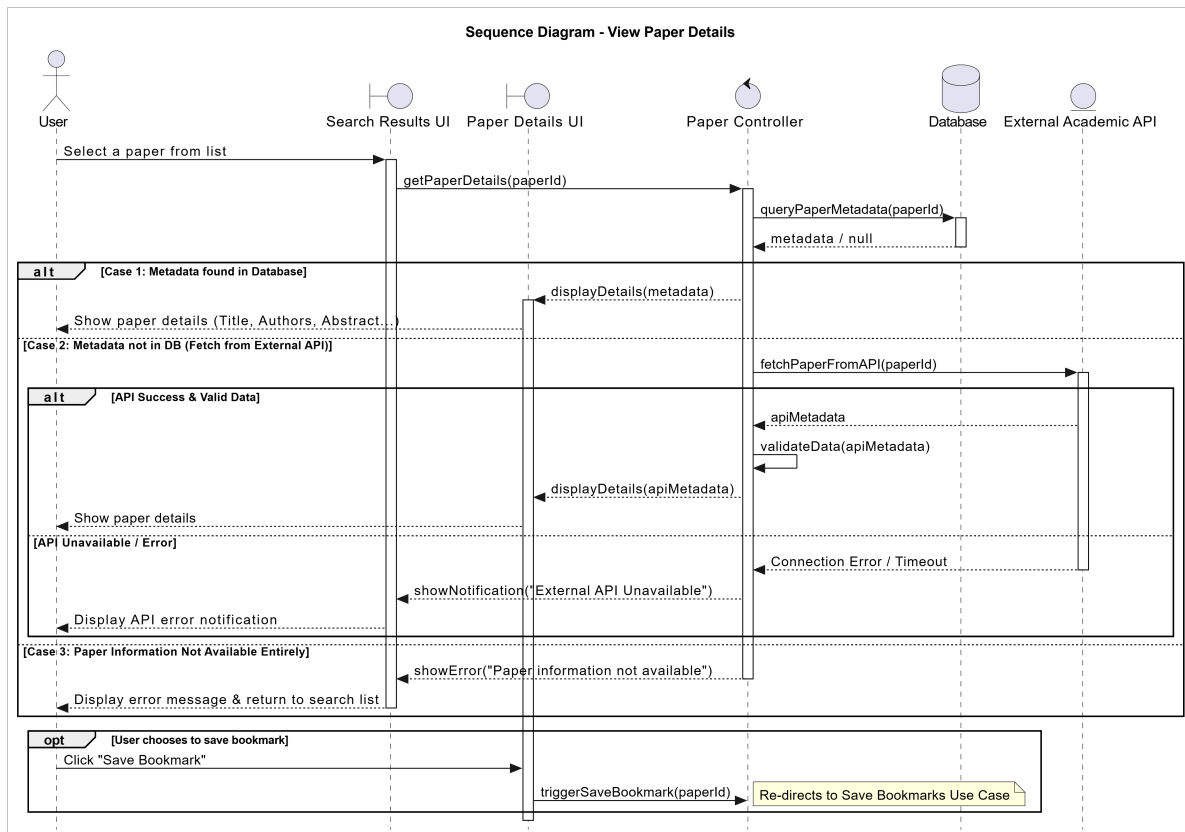
## 3.4 View Paper Details

### 3.4.1 Activity Diagram



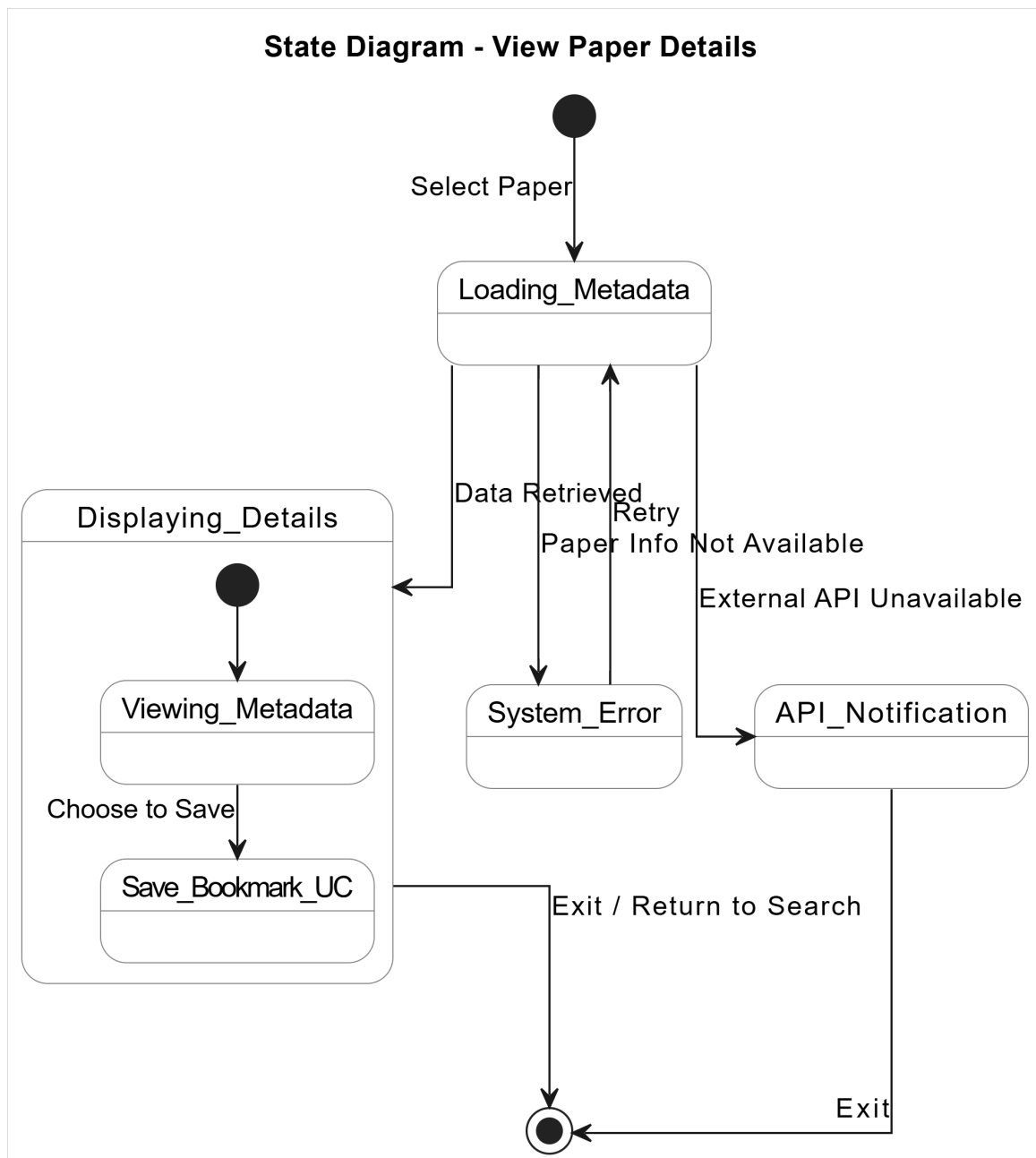
Hình 31: Activity Diagram: View Paper Details

### 3.4.2 Sequence Diagram



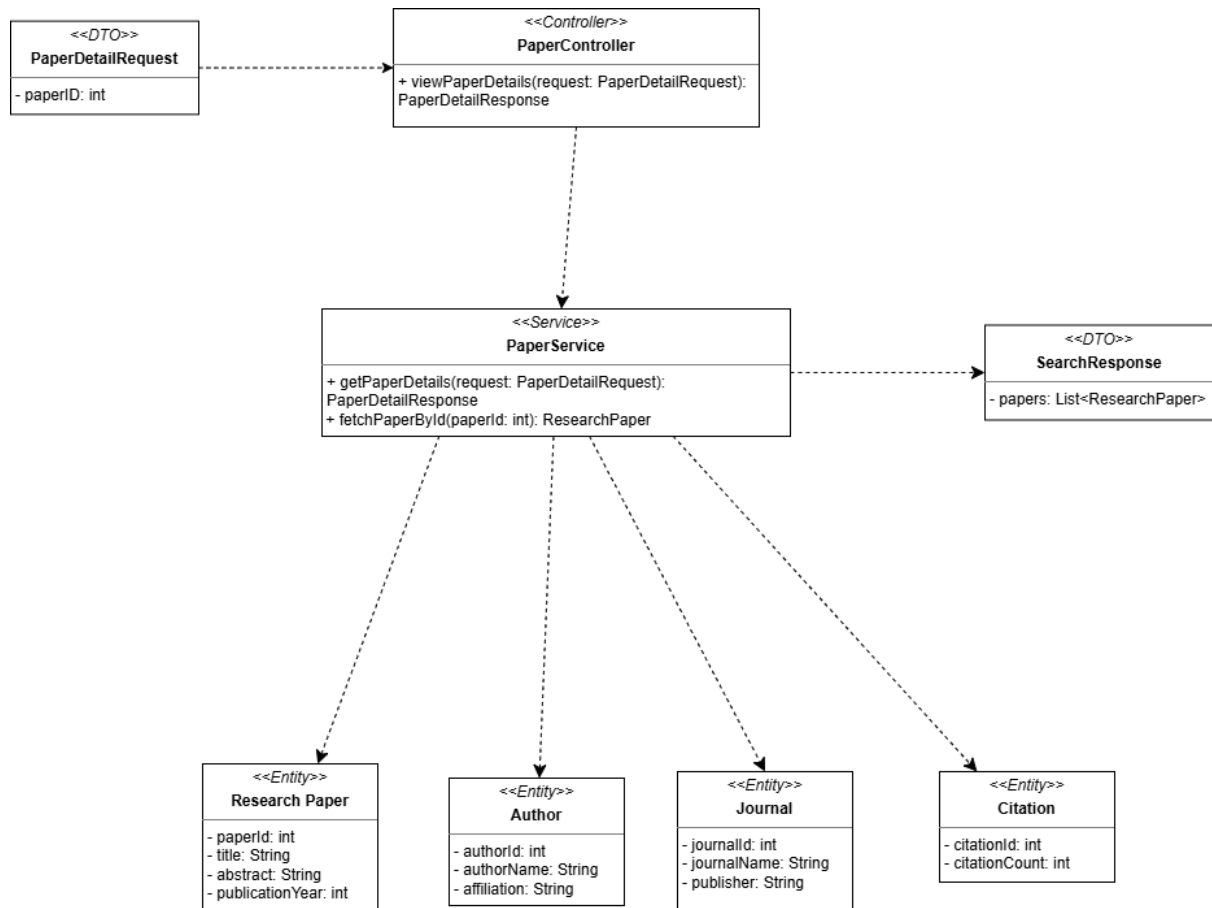
Hình 32: Sequence Diagram: View Paper Details

### 3.4.3 State Diagram



Hình 33: State Diagram: View Paper Details

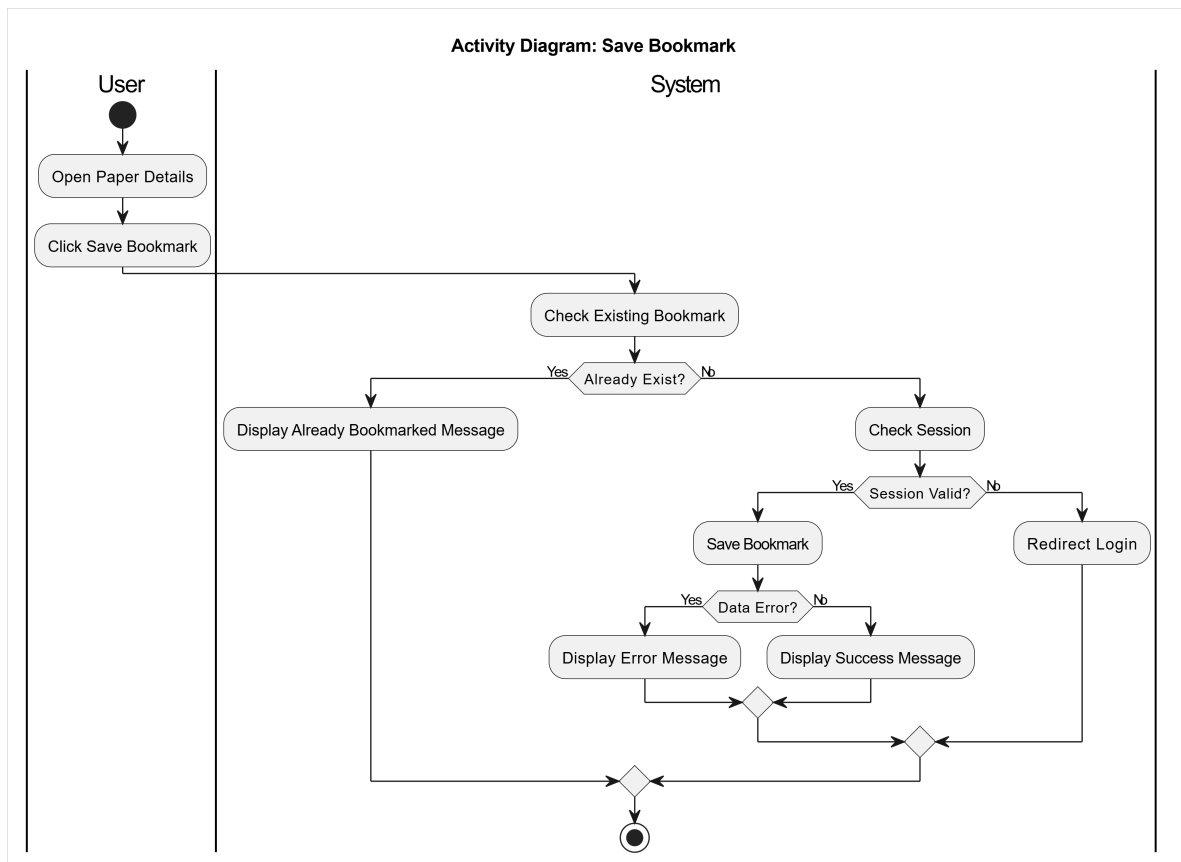
### 3.4.4 Class Diagram



Hình 34: Class Diagram: View Paper Detail

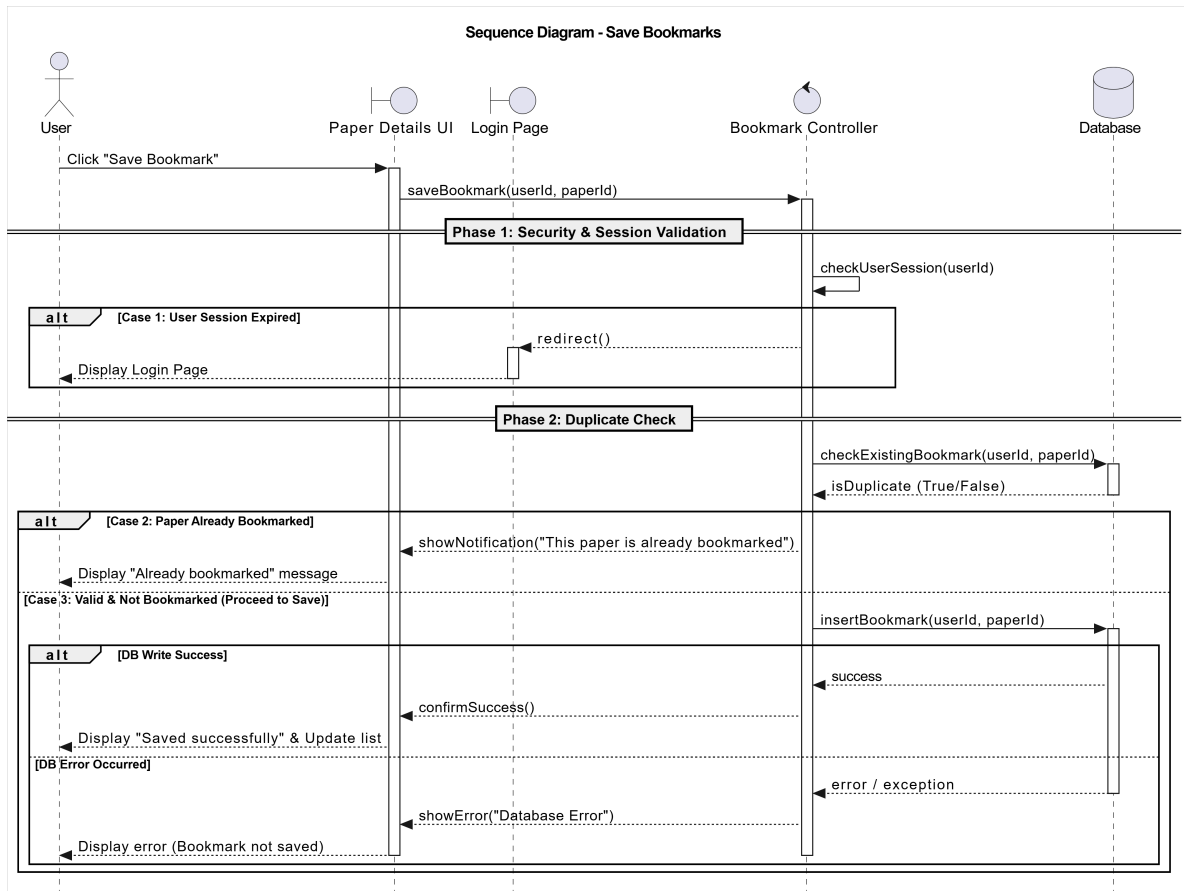
## 3.5 Save Bookmarks

### 3.5.1 Activity Diagram



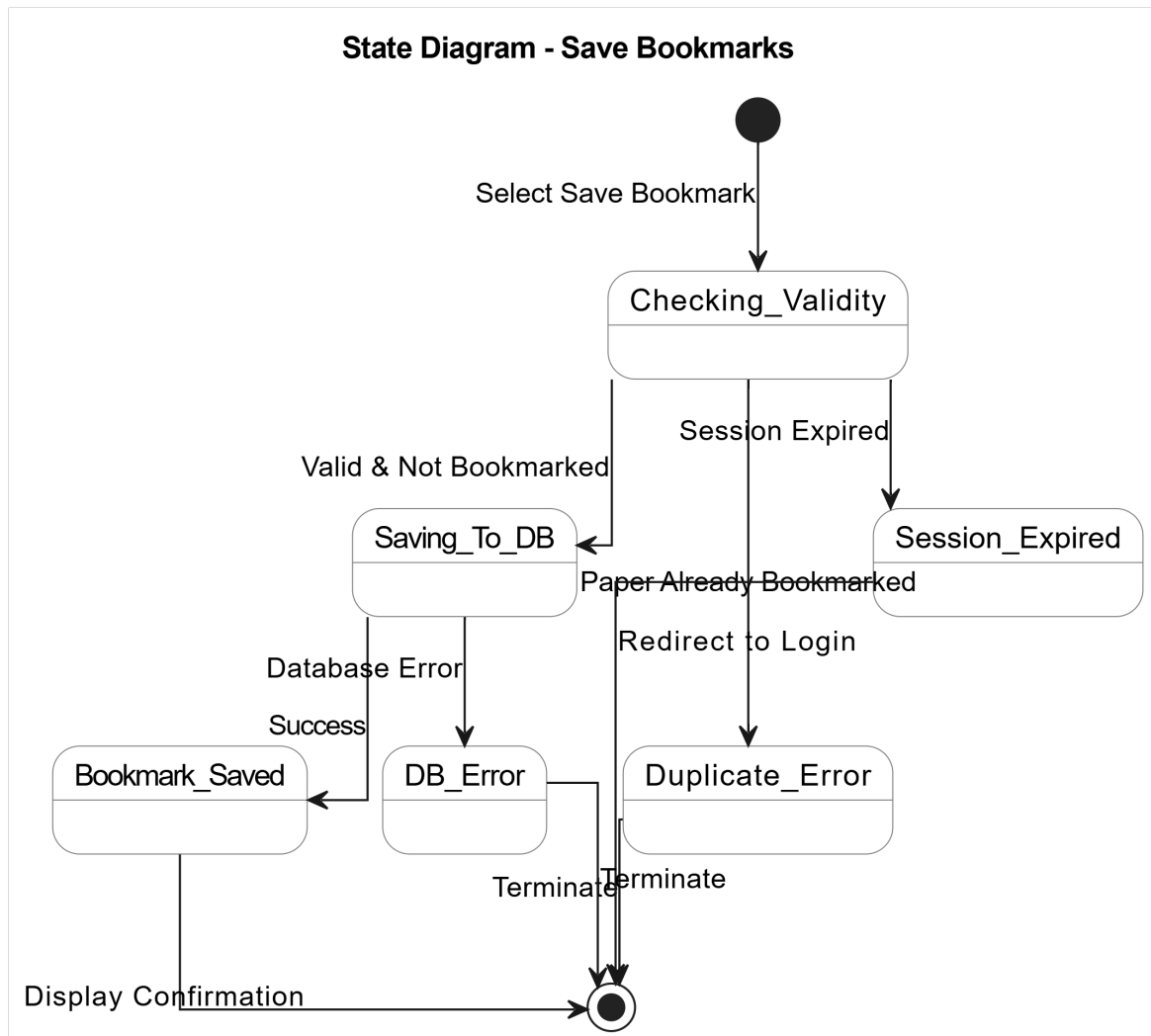
Hình 35: Activity Diagram: Save Bookmarks

### 3.5.2 Sequence Diagram



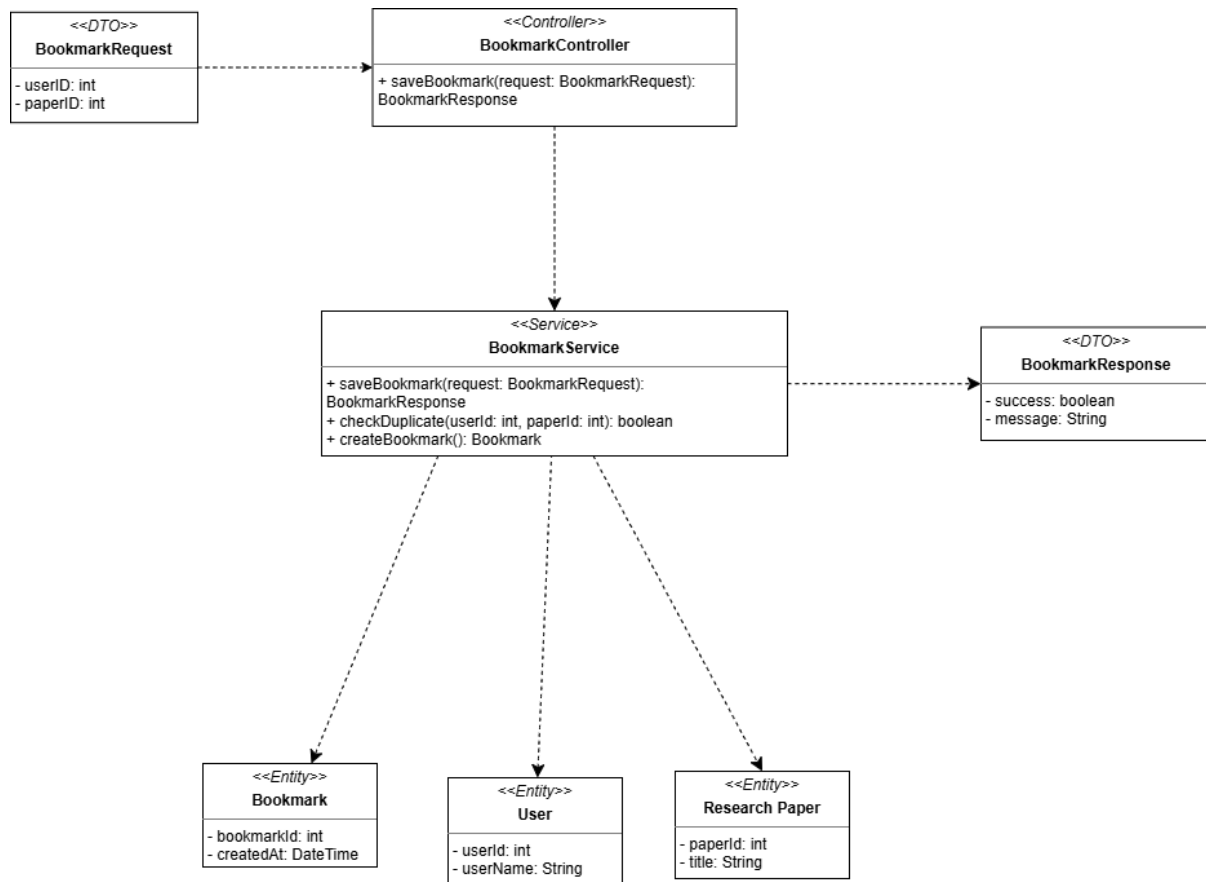
Hình 36: Sequence Diagram: Save Bookmarks

### 3.5.3 State Diagram



Hình 37: State Diagram: Save Bookmarks

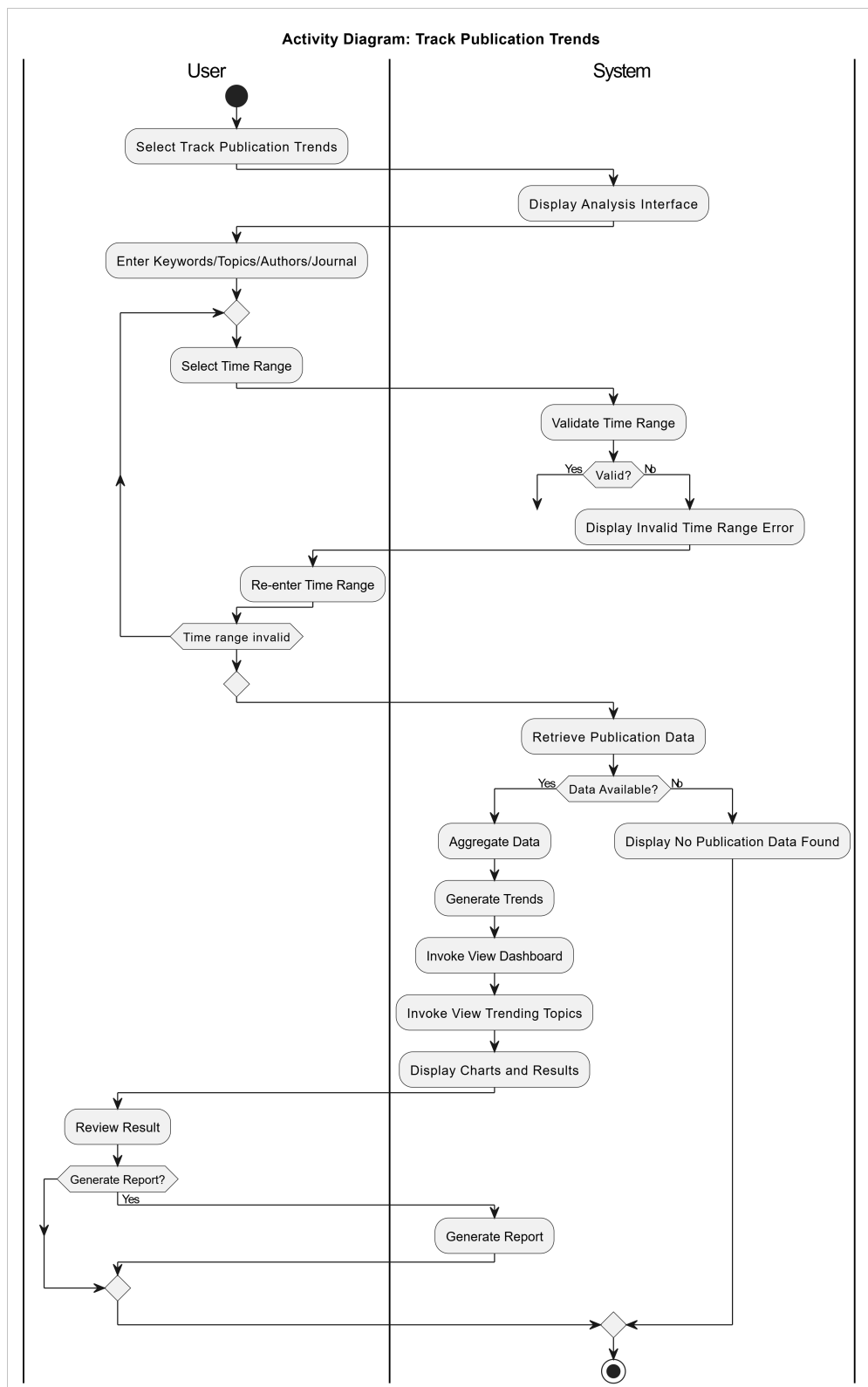
### 3.5.4 Class Diagram



Hình 38: Class Diagram: Save Bookmarks

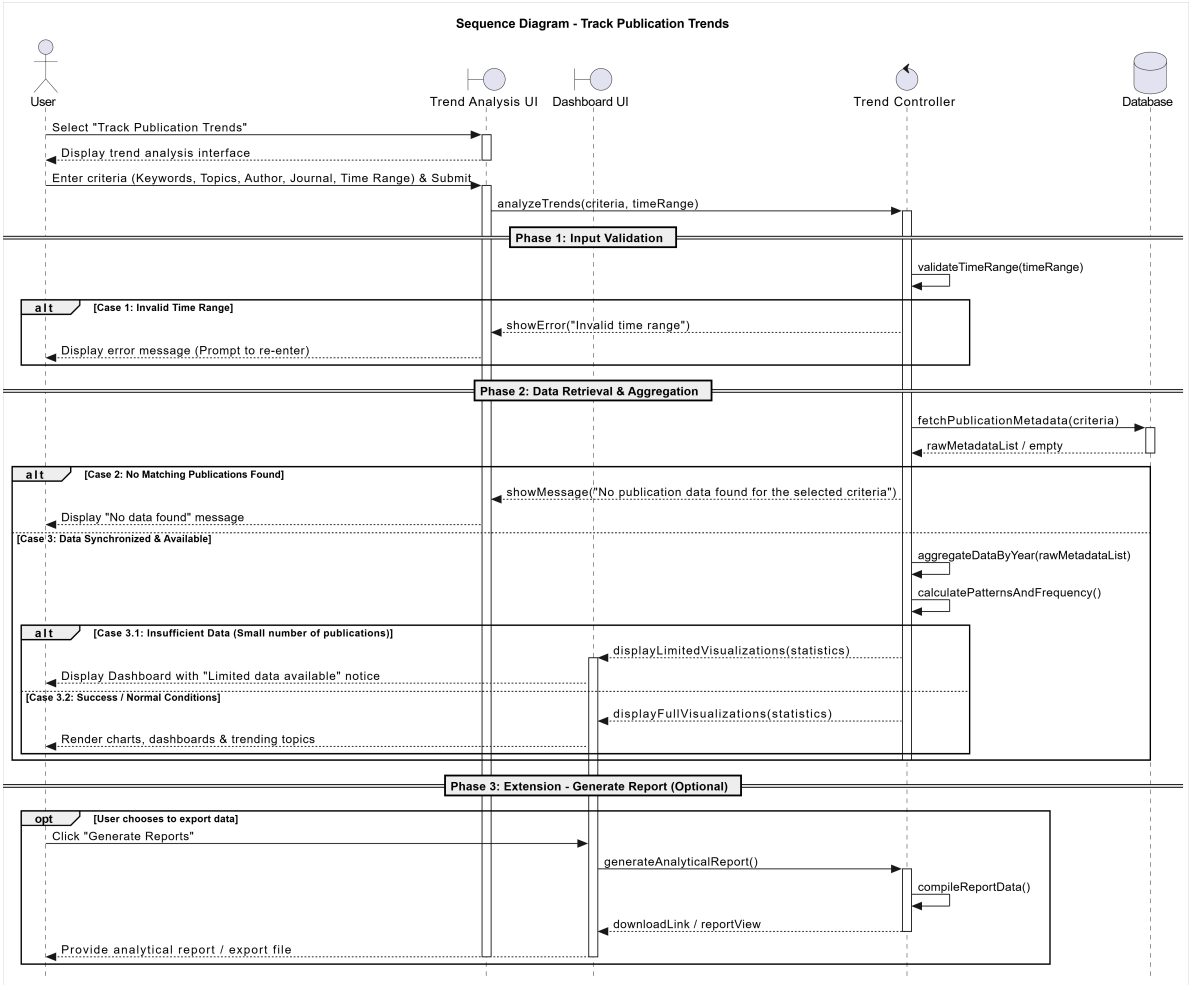
## 3.6 Track Publication Trends

### 3.6.1 Activity Diagram



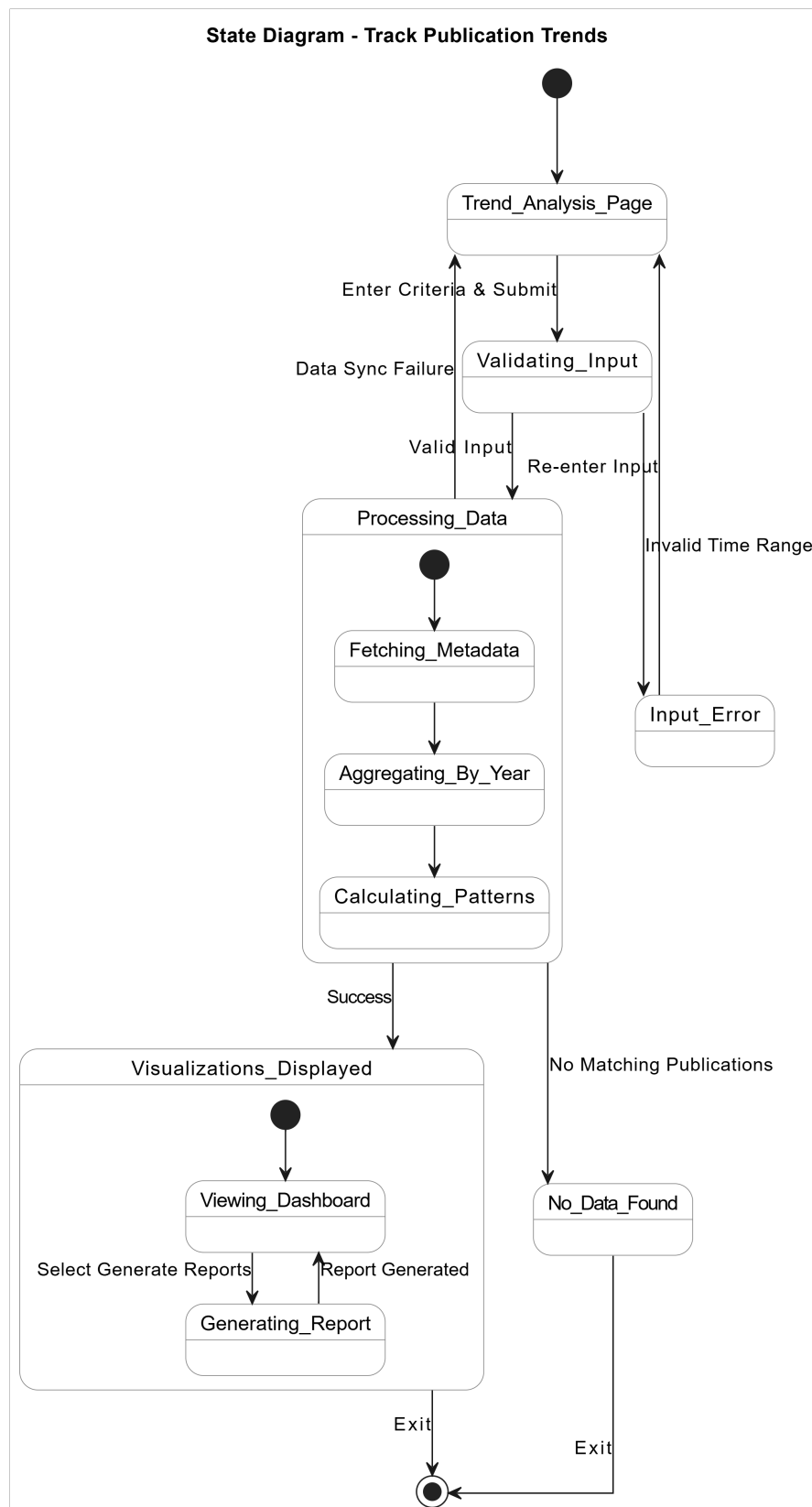
Hình 39: Activity Diagram: Track Publication Trends

### 3.6.2 Sequence Diagram



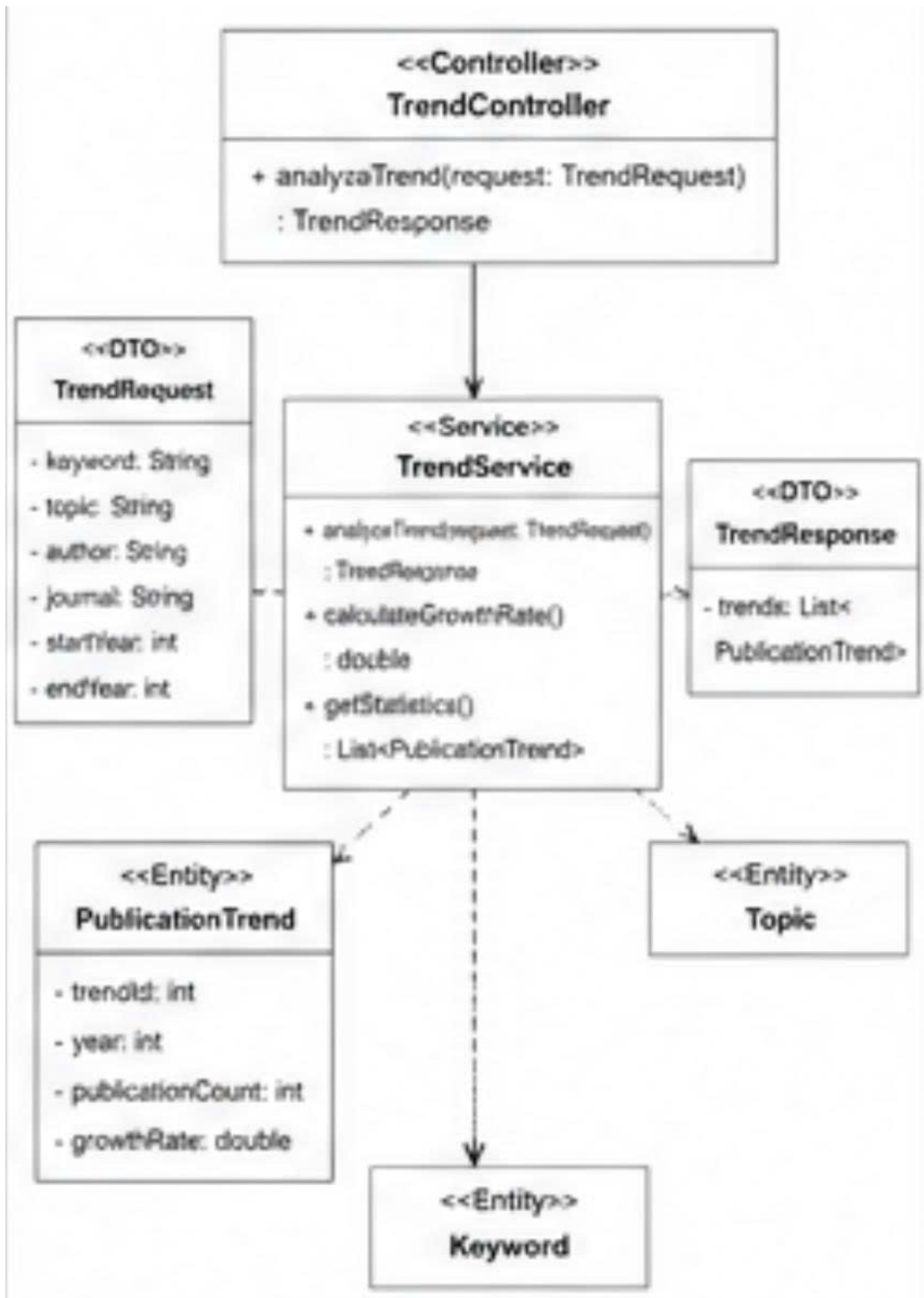
Hình 40: Sequence Diagram: Track Publication Trends

### 3.6.3 State Diagram



Hình 41: State Diagram: Track Publication Trends

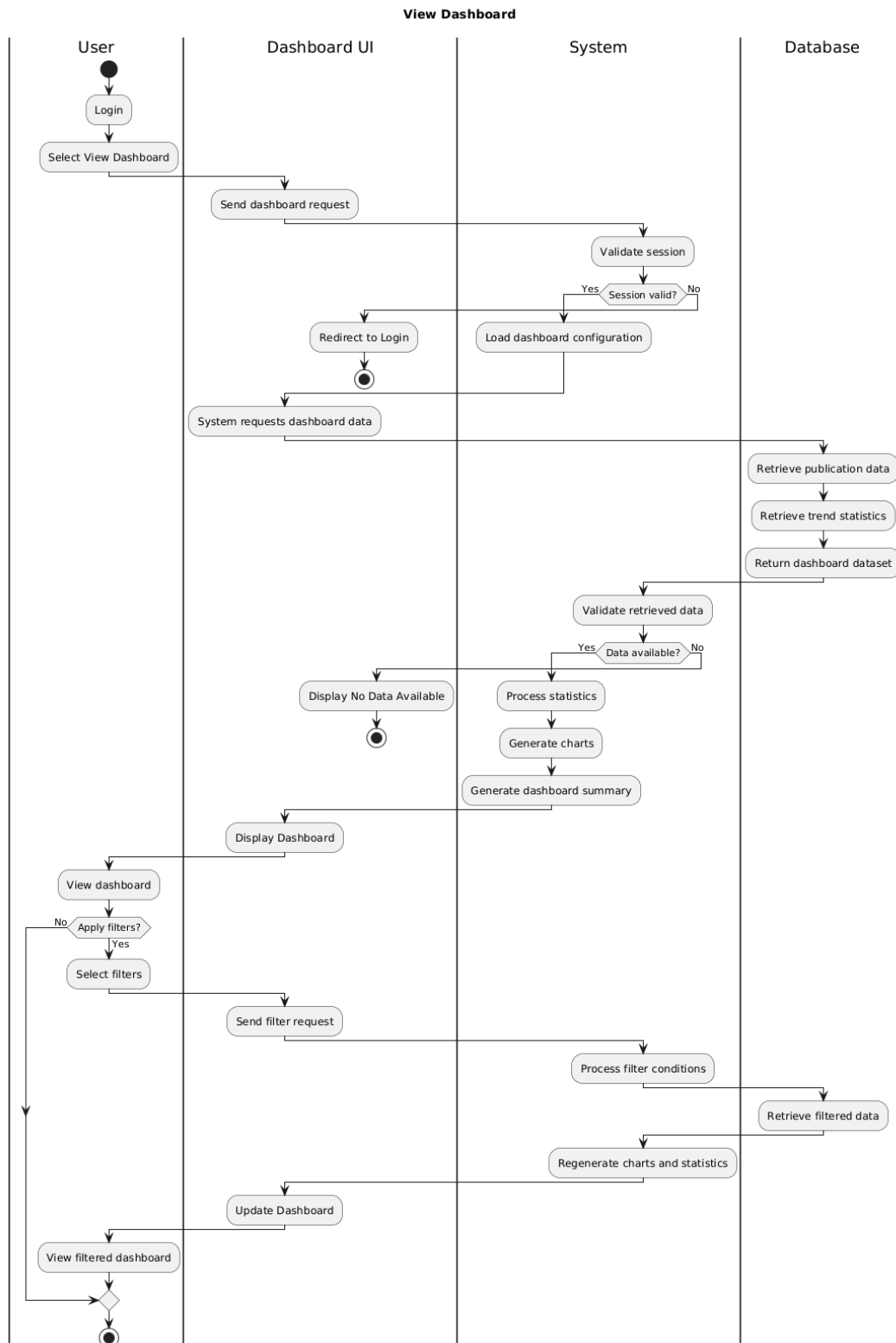
### 3.6.4 Class Diagram



Hình 42: Class Diagram: Track Publication Trends

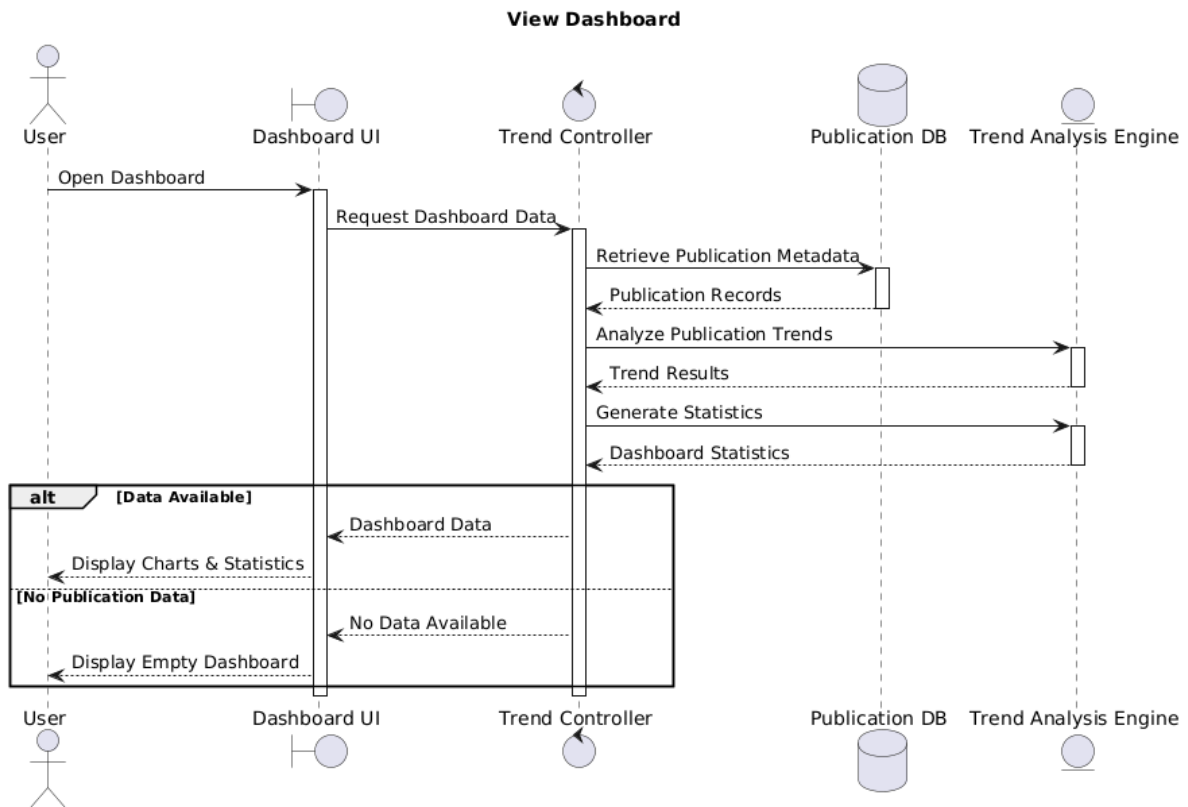
## 3.7 View Dashboard

### 3.7.1 Activity Diagram



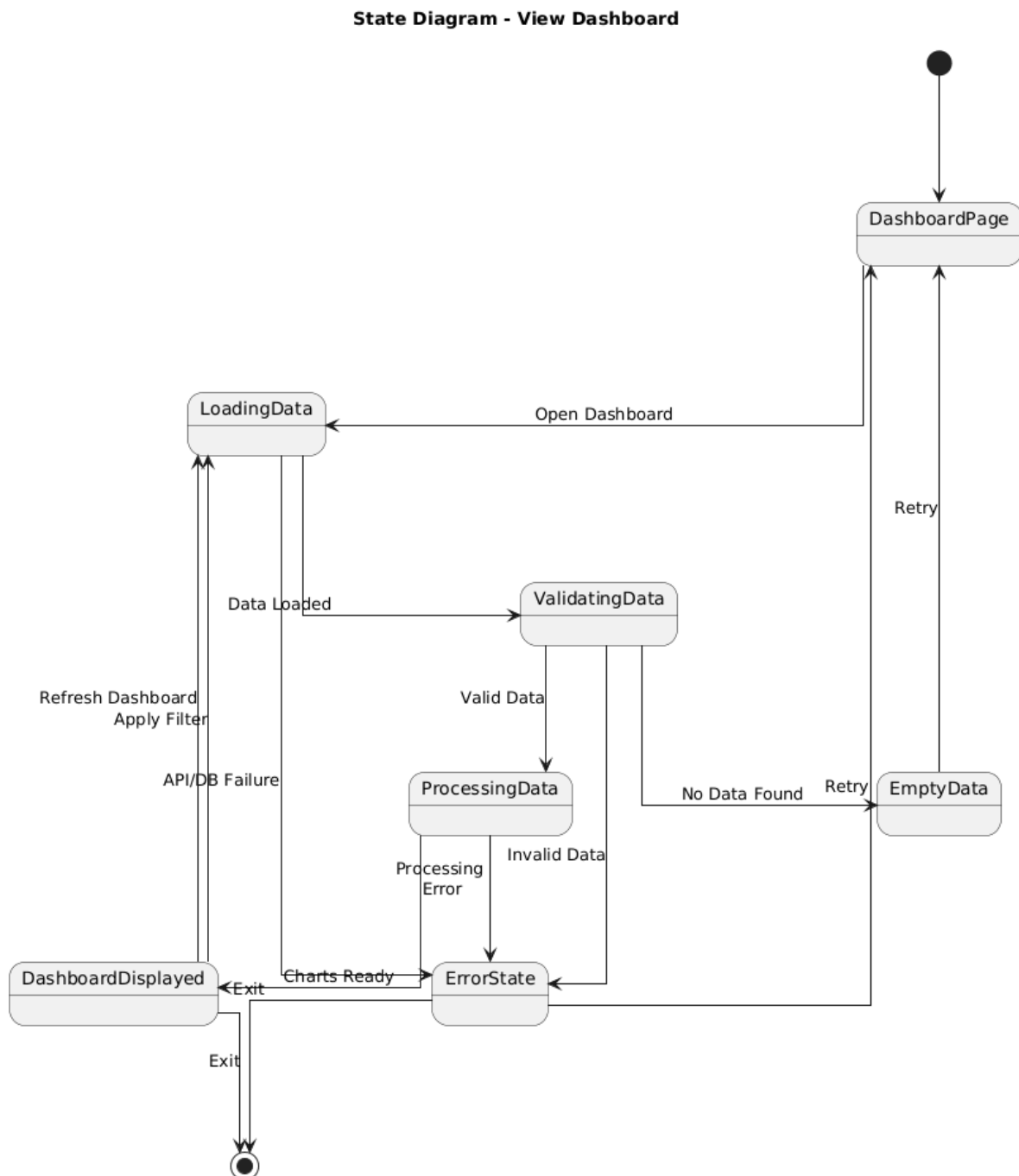
Hình 43: Activity Diagram: View Dashboard

### 3.7.2 Sequence Diagram



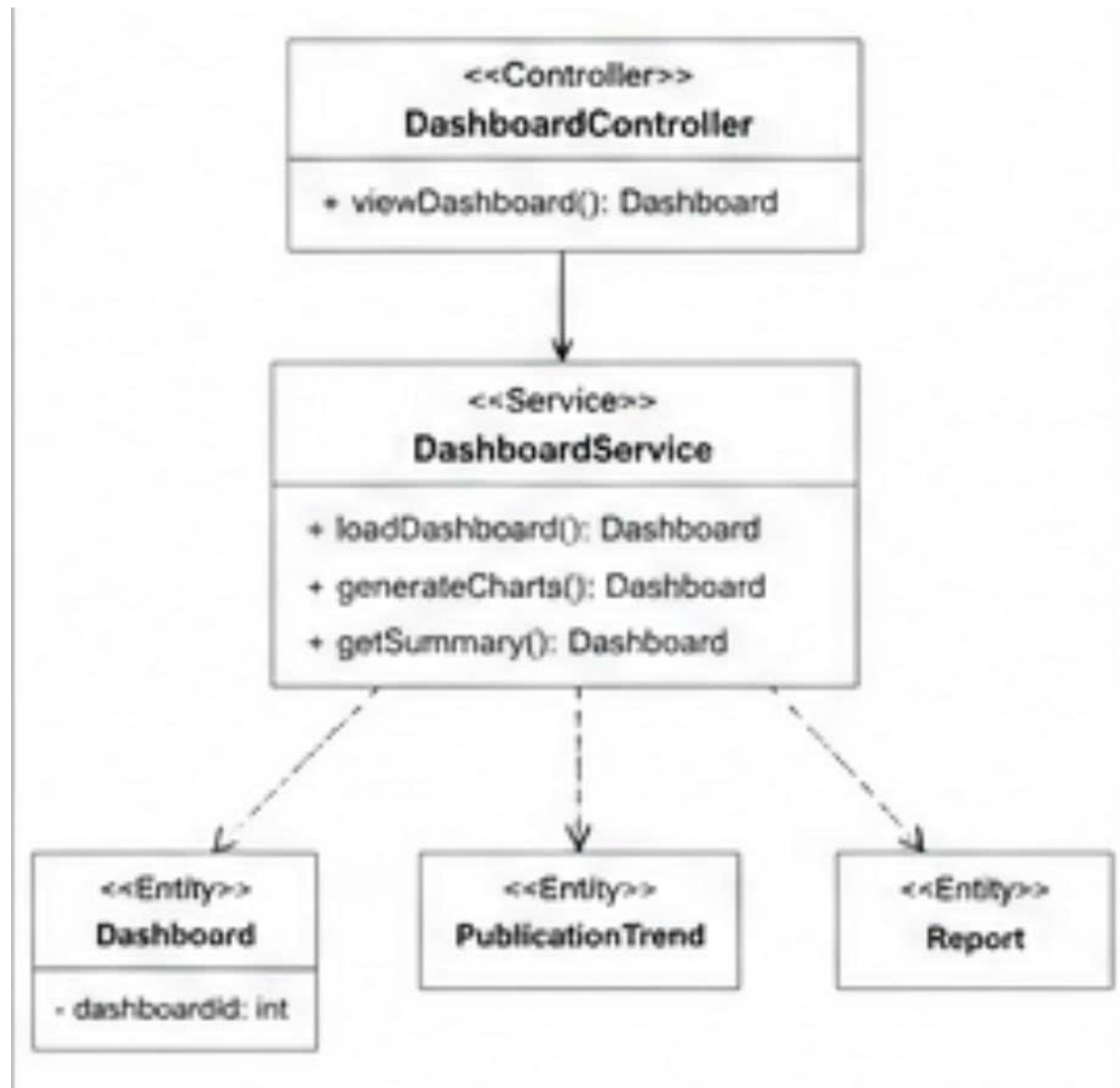
Hình 44: Sequence Diagram: View Dashboard

### 3.7.3 State Diagram



Hình 45: State Diagram: View Dashboard

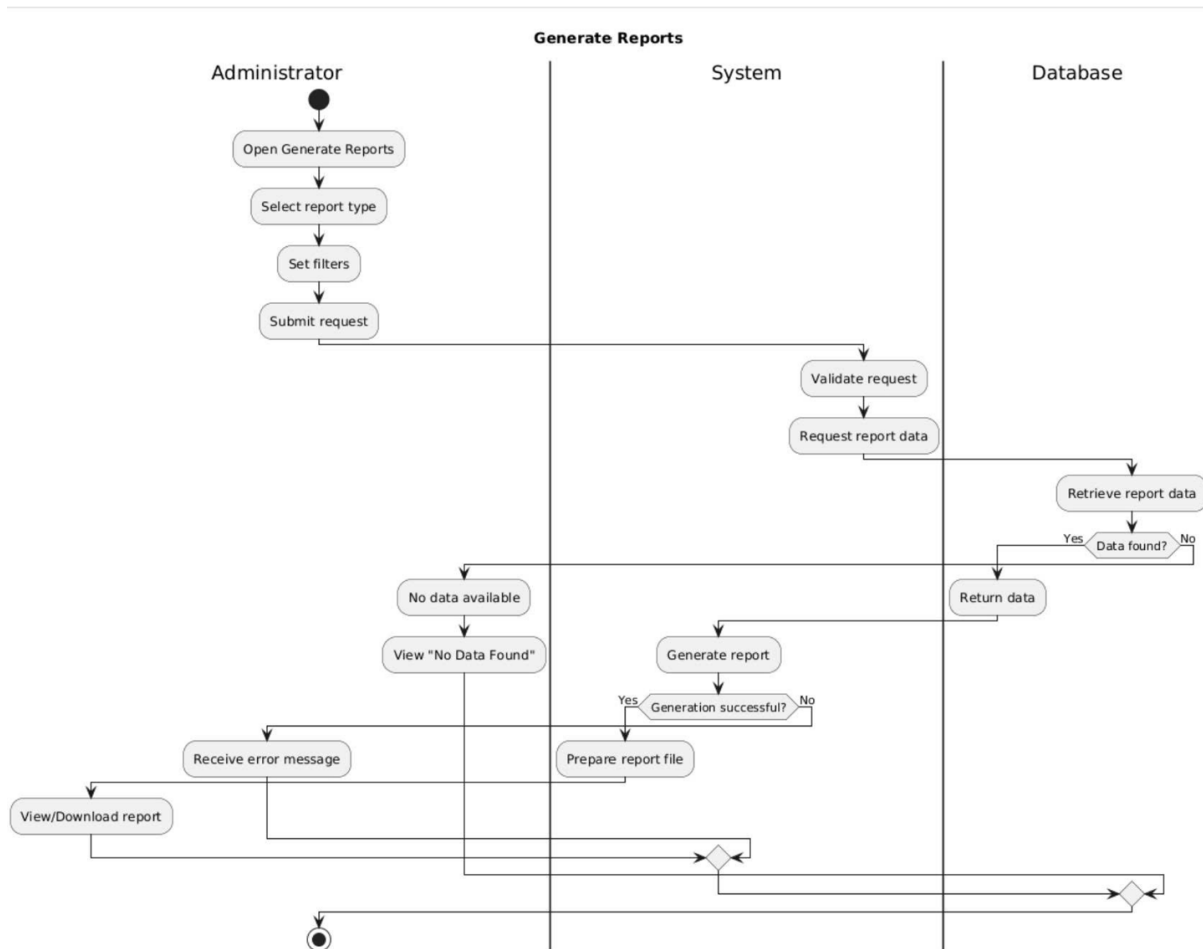
### 3.7.4 Class Diagram



Hình 46: Class Diagram: View Dashboard

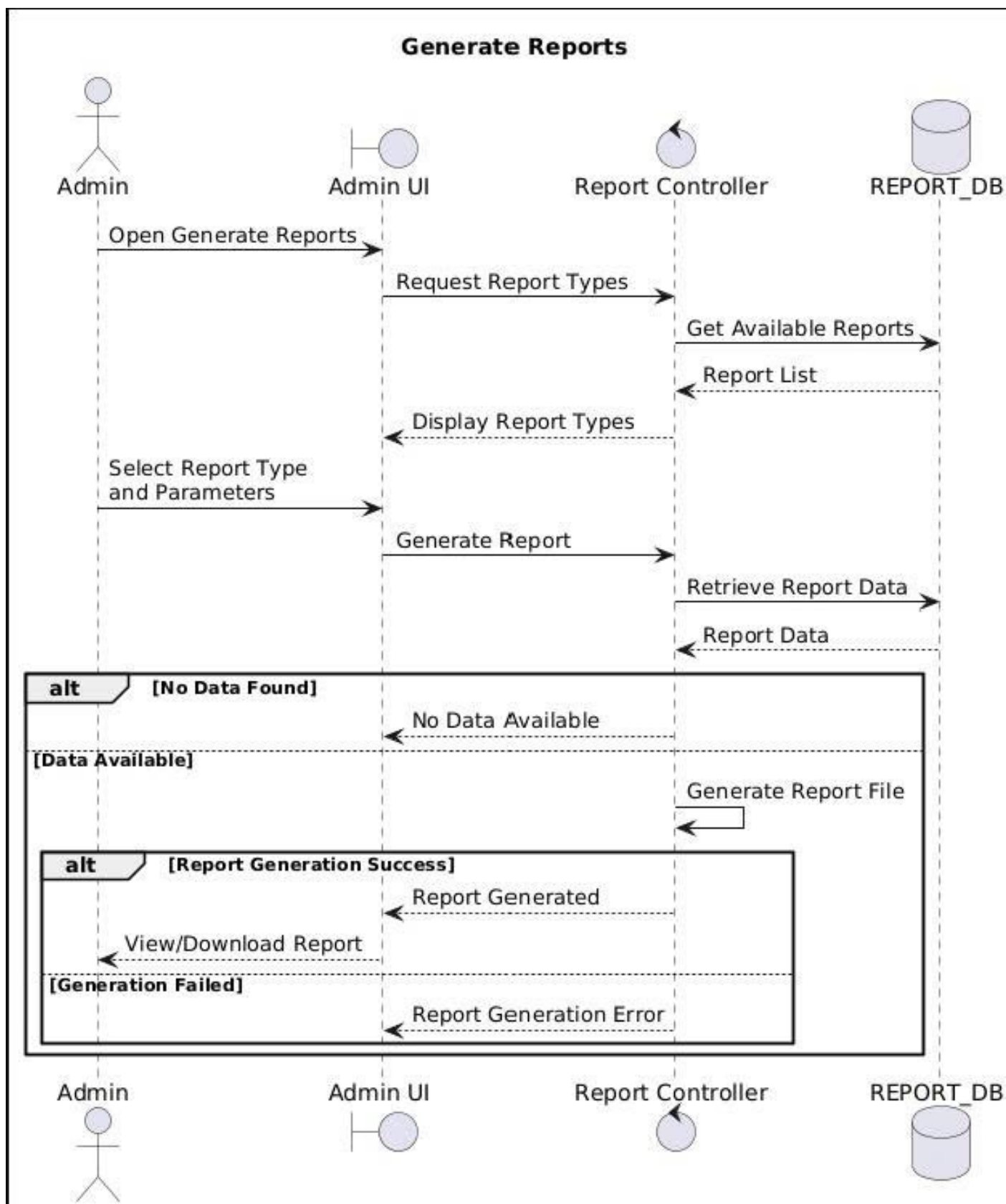
## 3.8 Generate Reports

### 3.8.1 Activity Diagram



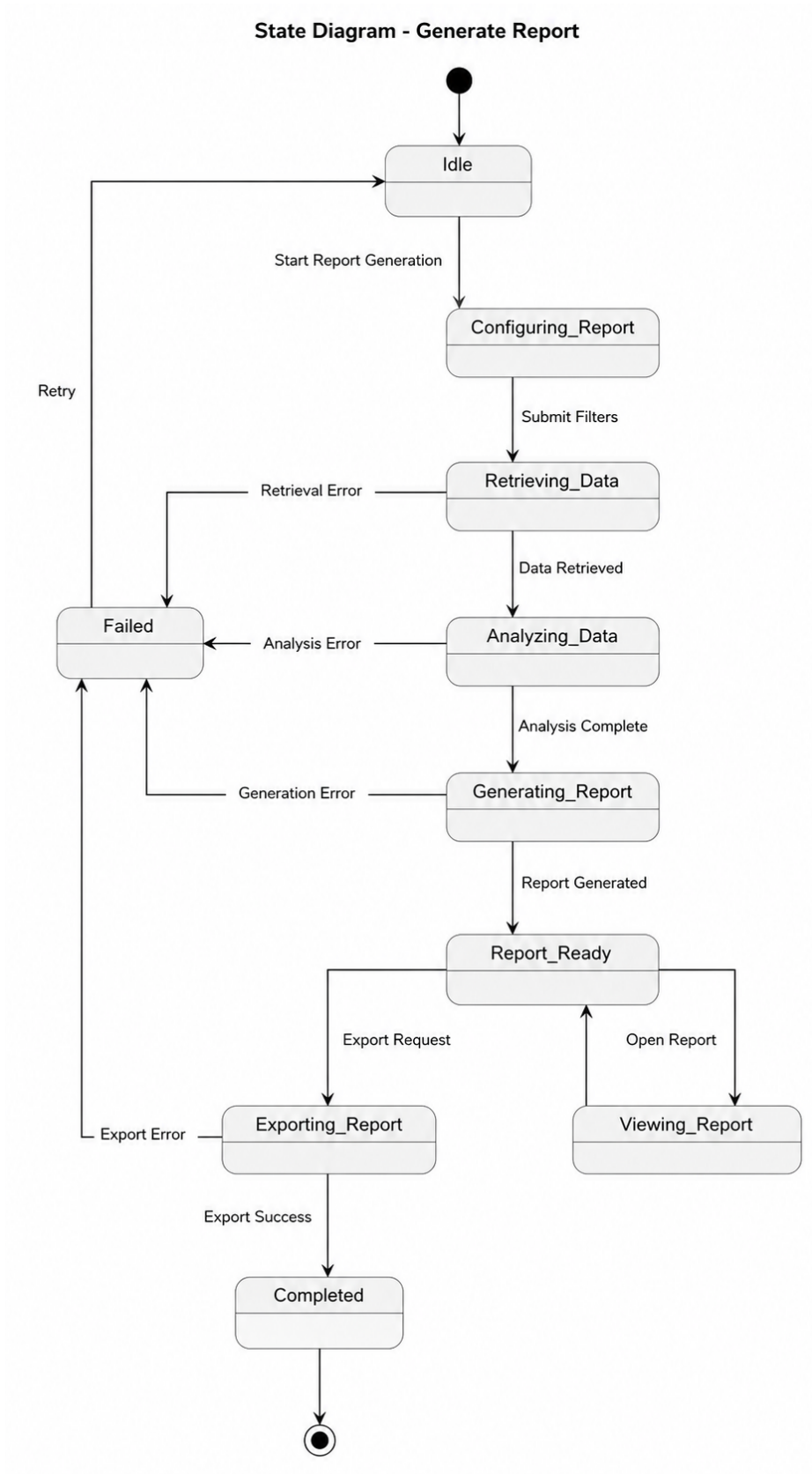
Hình 47: Activity Diagram - Generate reports

### 3.8.2 Sequence Diagram



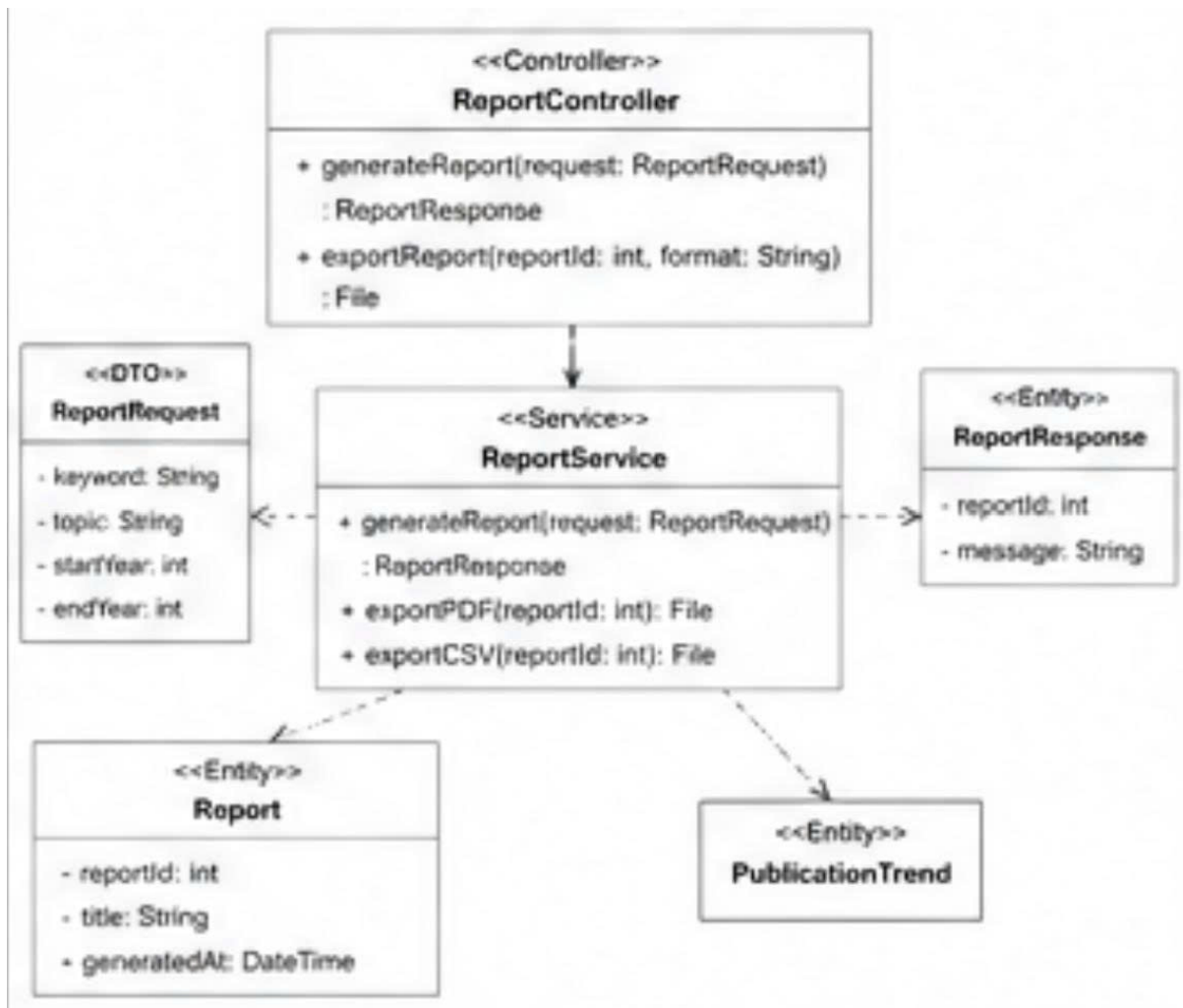
Hình 48: Sequence diagram - Generate reports

### 3.8.3 State Diagram



Hình 49: State diagram - Generate reports

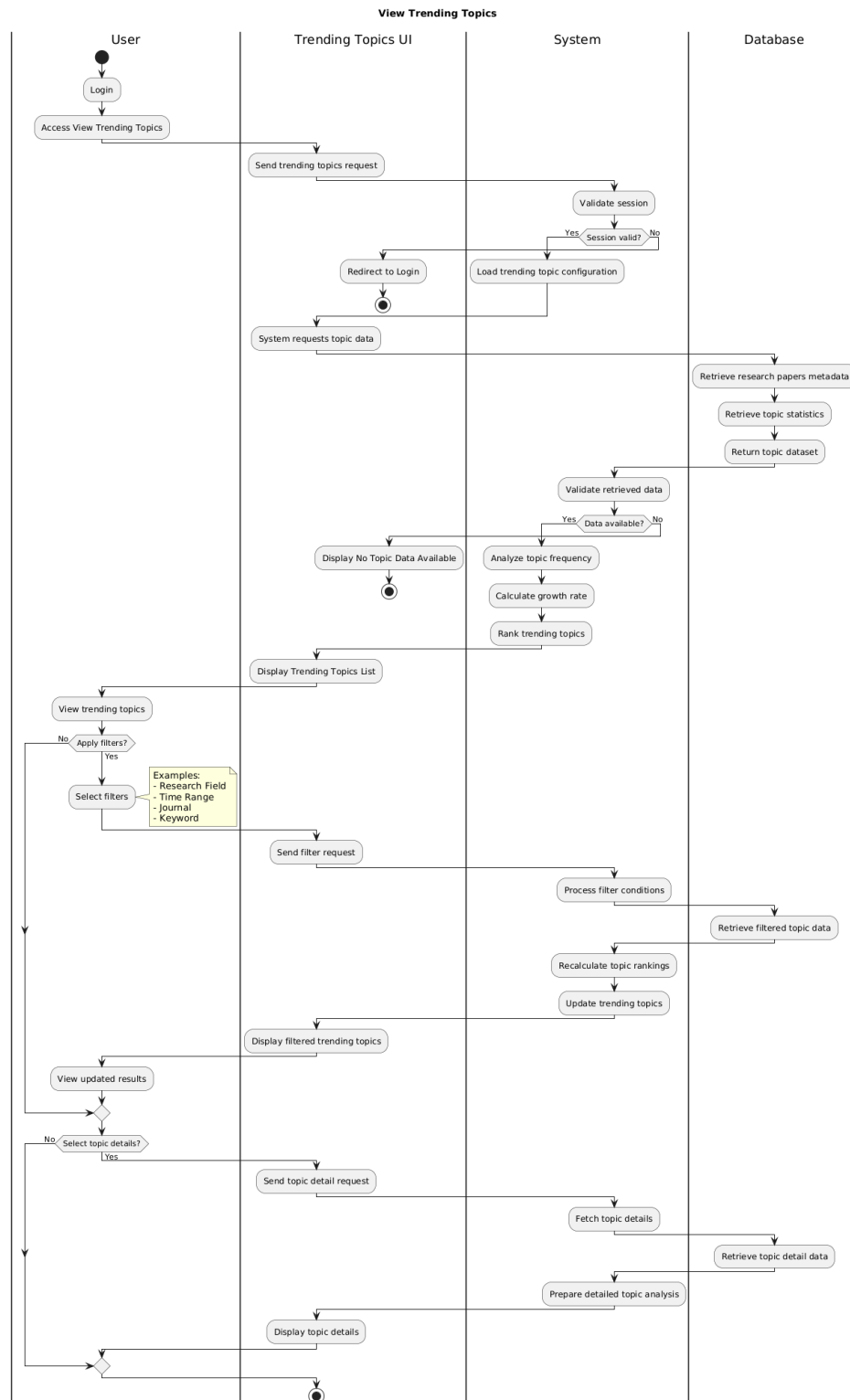
### 3.8.4 Class Diagram



Hình 50: Class Diagram - Generate Reports

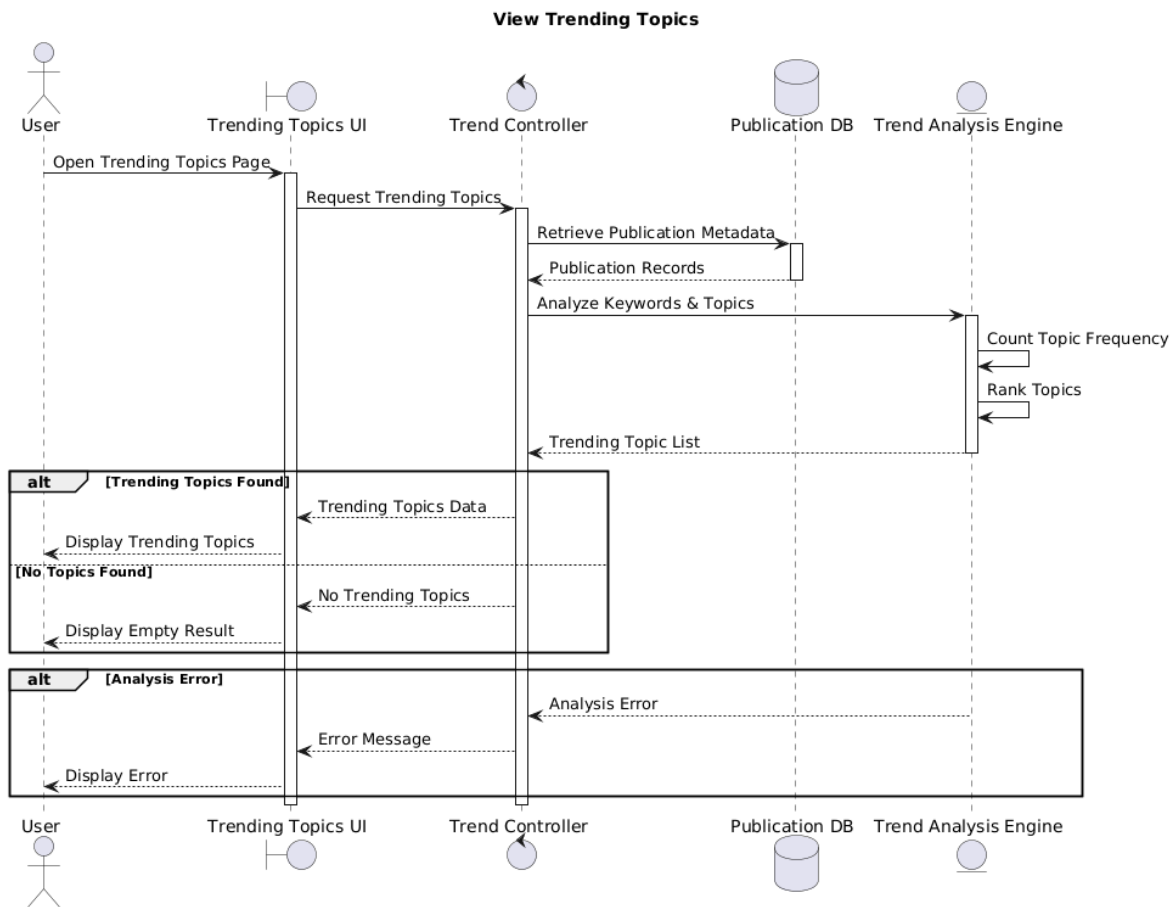
## 3.9 View Trending Topics

### 3.9.1 Activity Diagram



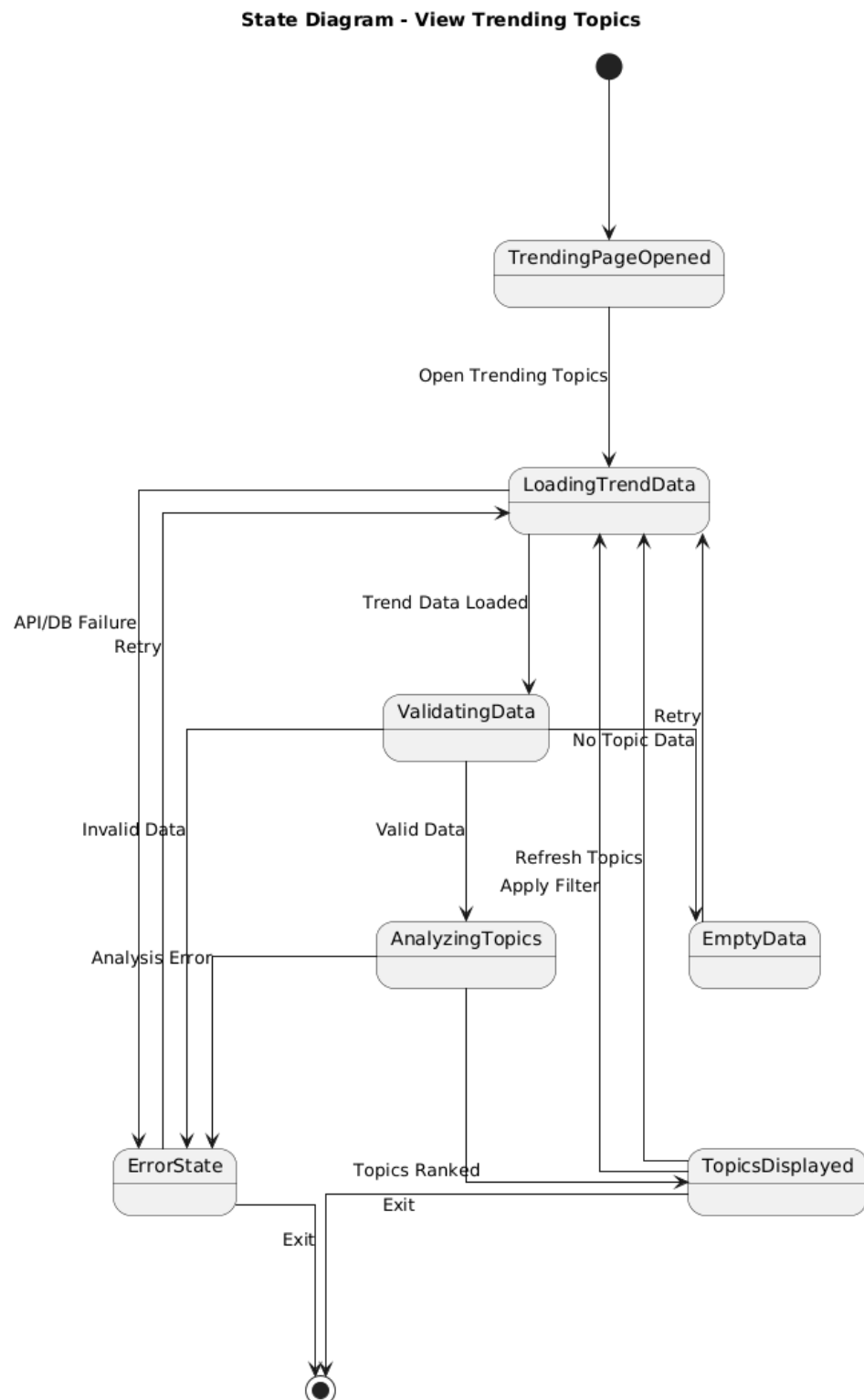
Hình 51: Activity Diagram - View Trending Topics

### 3.9.2 Sequence Diagram



Hình 52: Sequence diagram

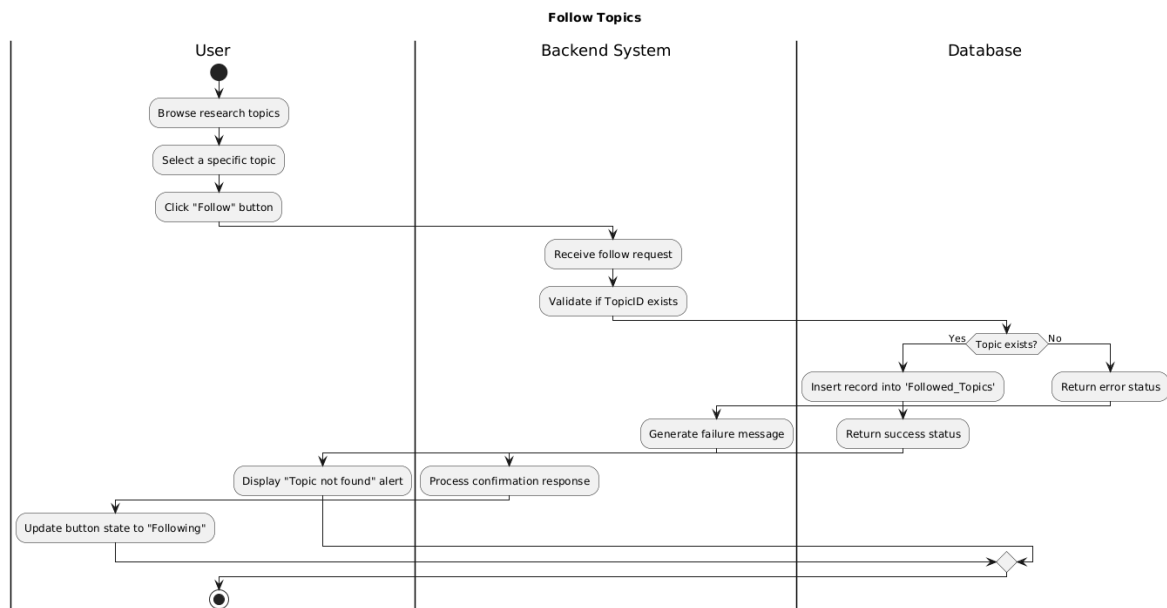
### 3.9.3 State Diagram



Hình 53: State diagram

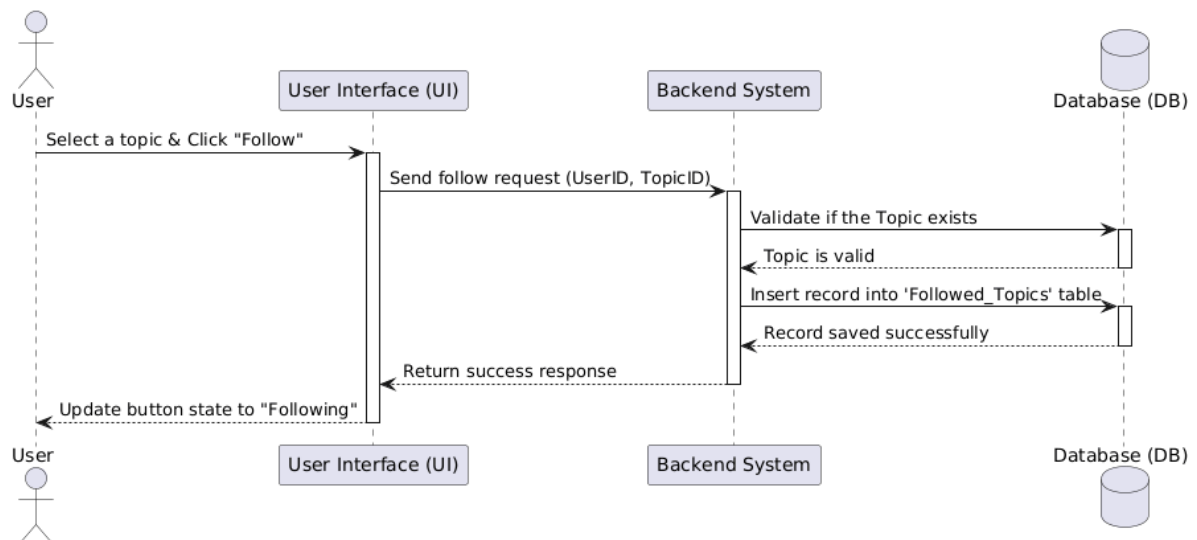
## 3.10 Follow Topics

### 3.10.1 Activity Diagram



Hình 54: Activity diagram

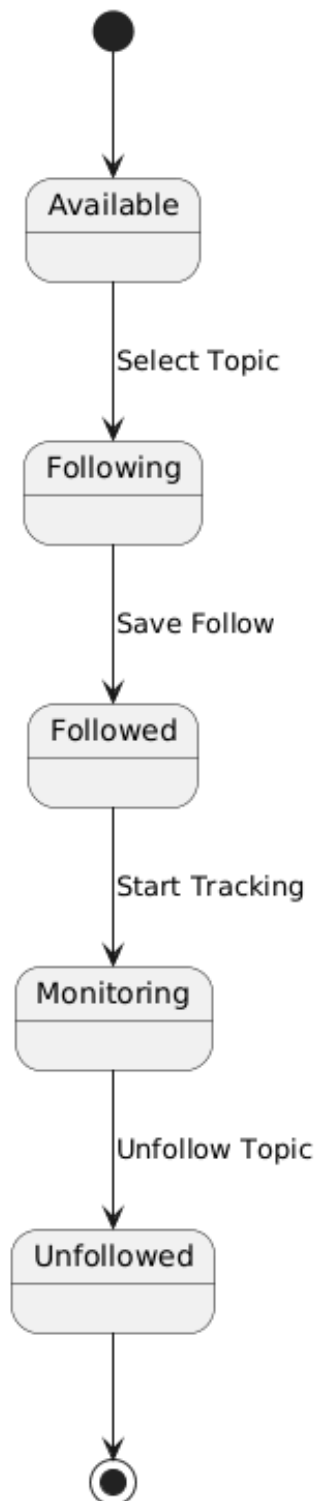
### 3.10.2 Sequence Diagram



Hình 55: Sequence diagram

### 3.10.3 State Diagram

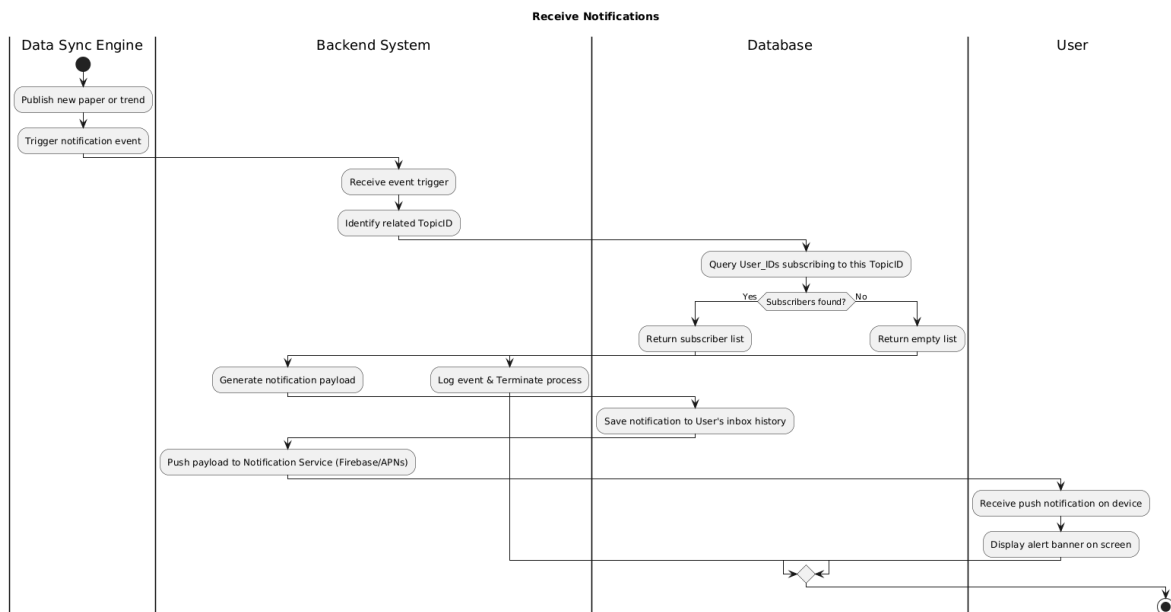
**State Diagram - Follow Topics**



Hình 56: State diagram

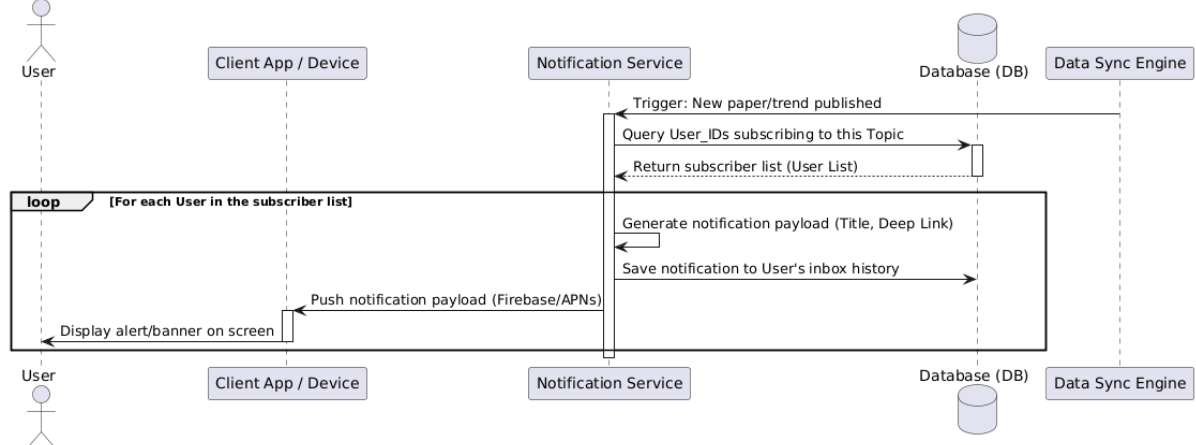
## 3.11 Receive Notifications

### 3.11.1 Activity Diagram



Hình 57: Activity diagram

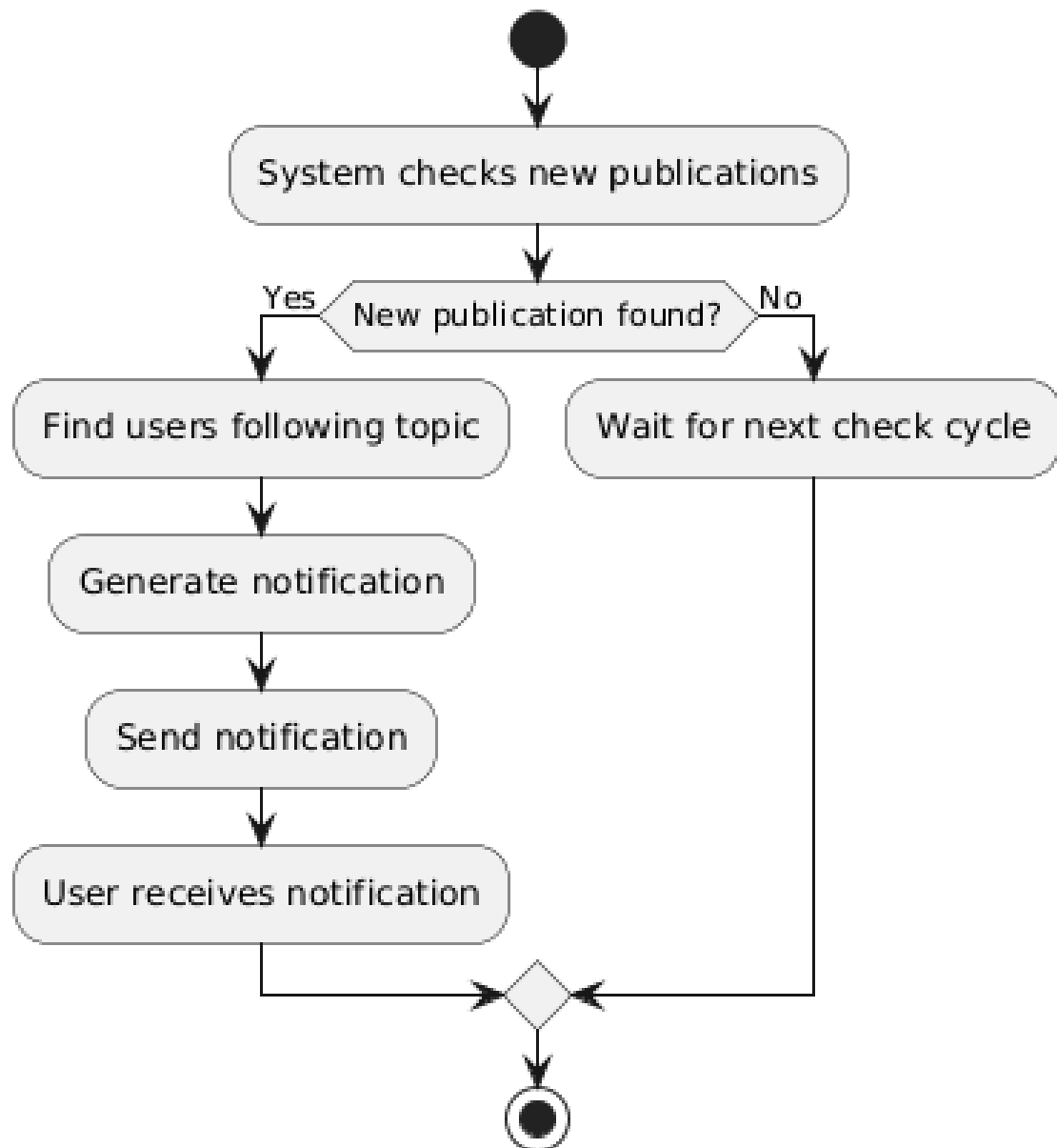
### 3.11.2 Sequence Diagram



Hình 58: Sequence diagram

### 3.11.3 State Diagram

#### State Diagram: 10. Receive Notifications

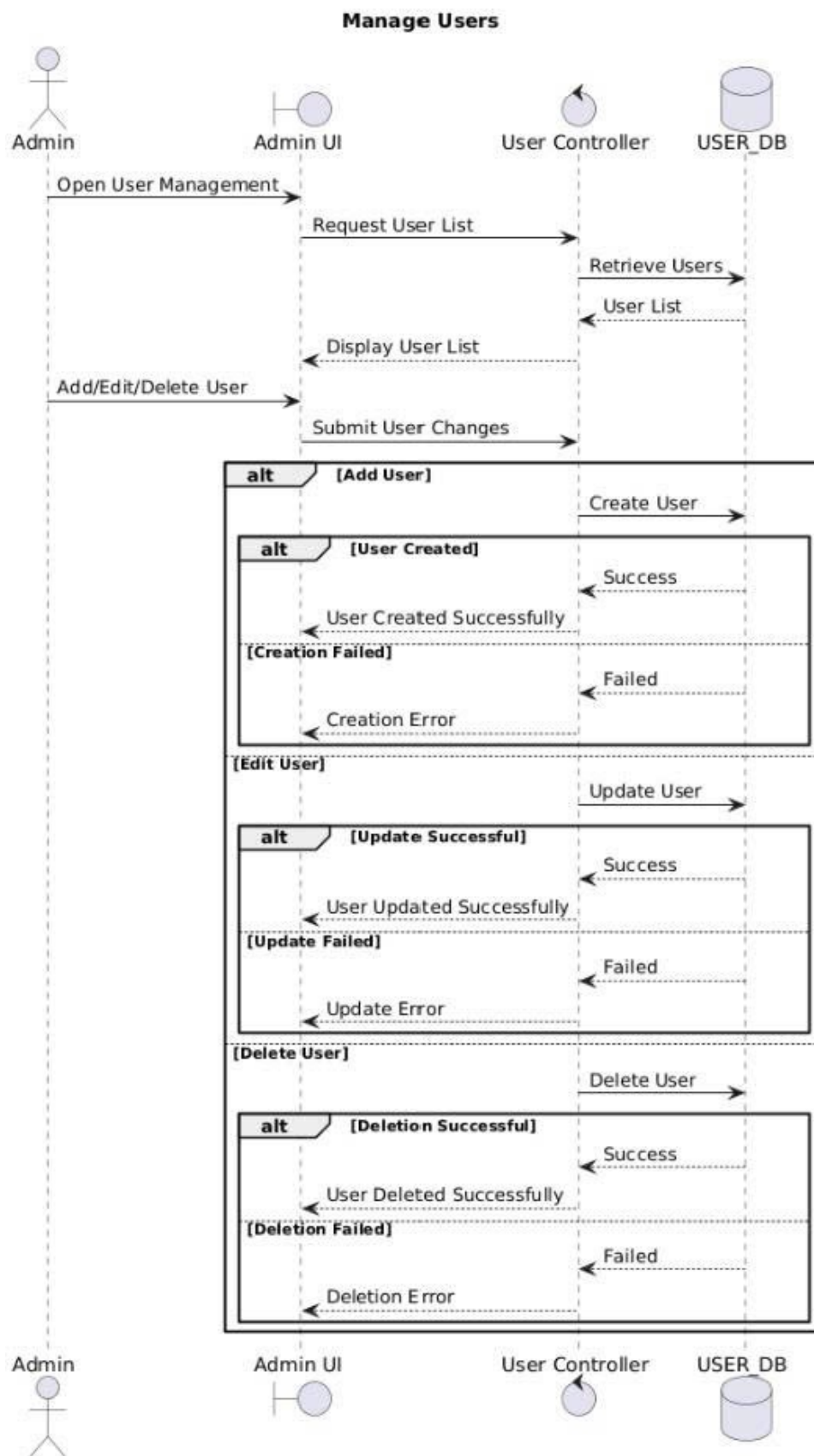


Hình 59: State diagram



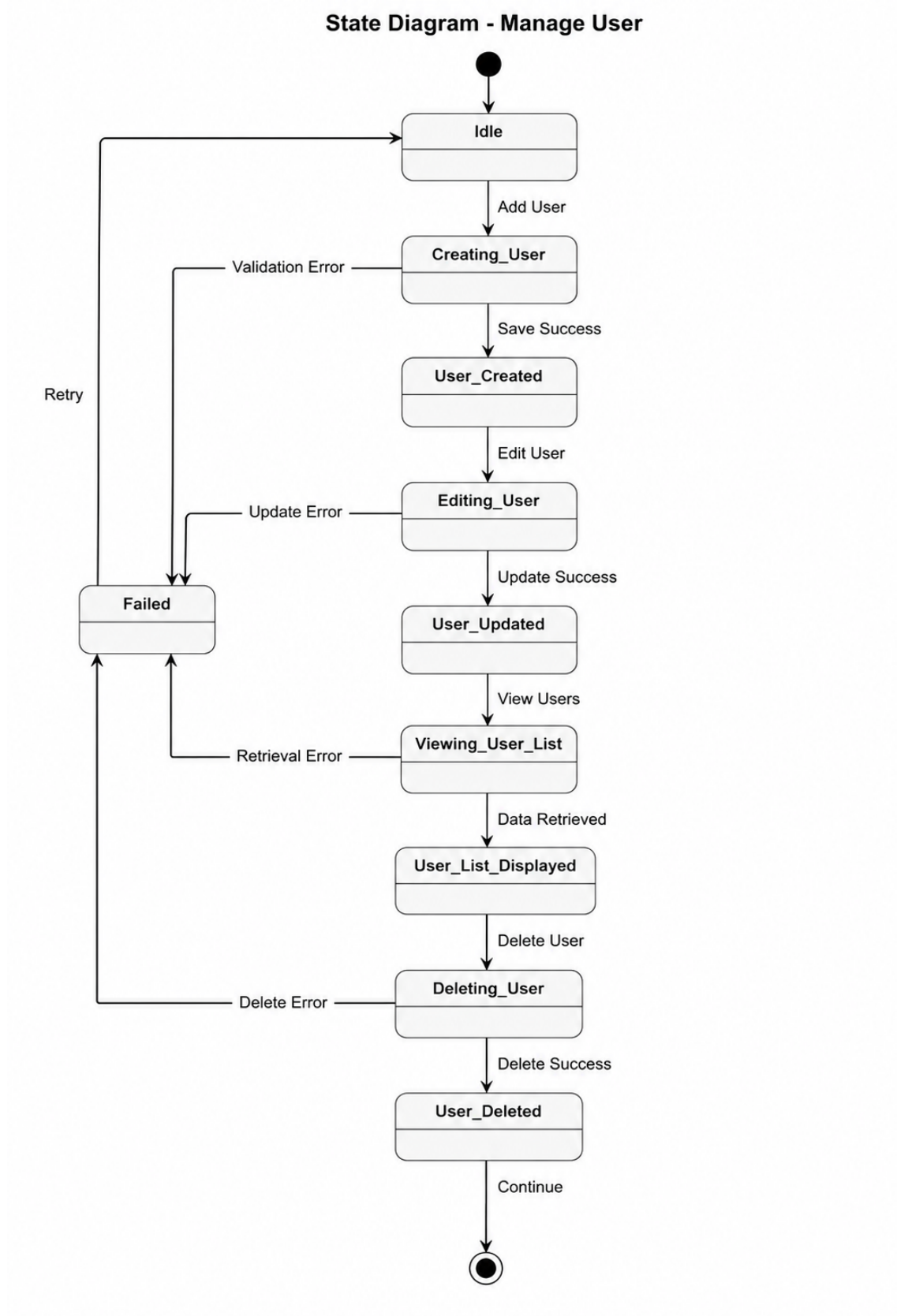


### 3.13.2 Sequence Diagram



Hình 61: Sequence diagram - Manage Users

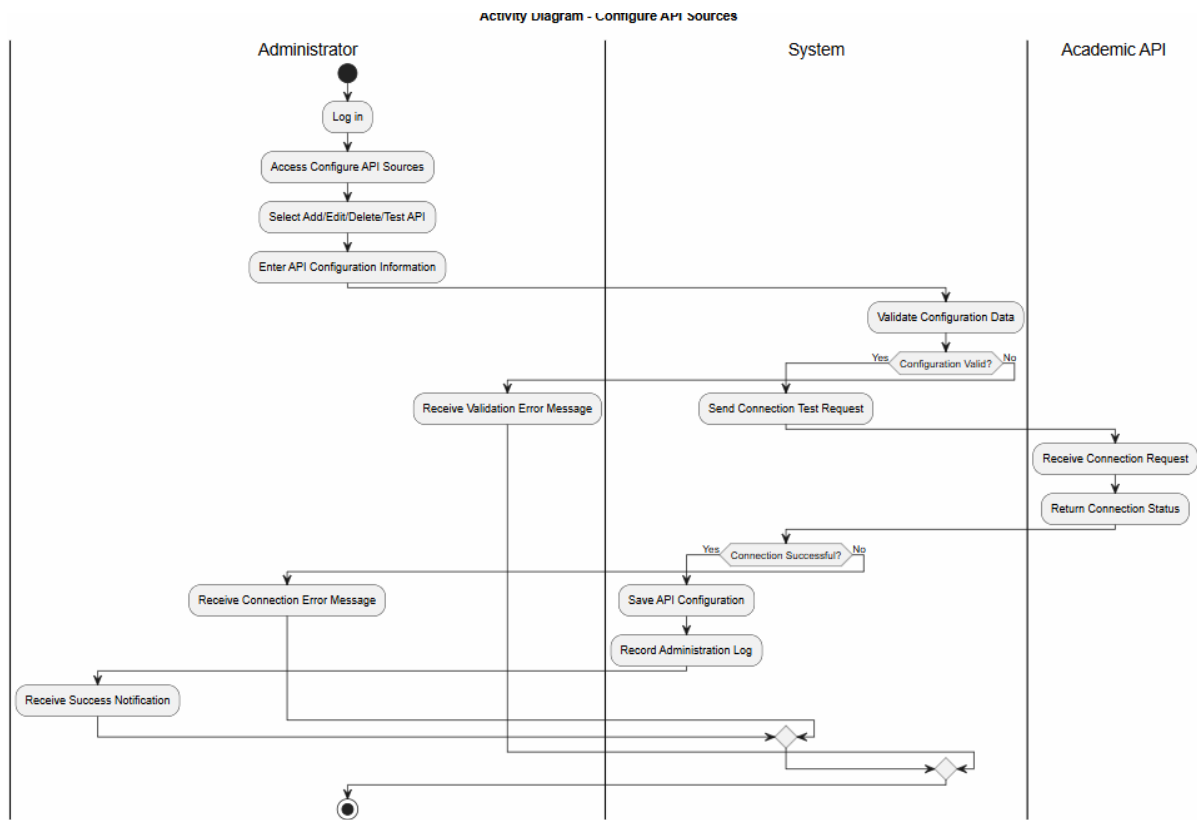
### 3.13.3 State Diagram



Hình 62: State diagram - Manage Users

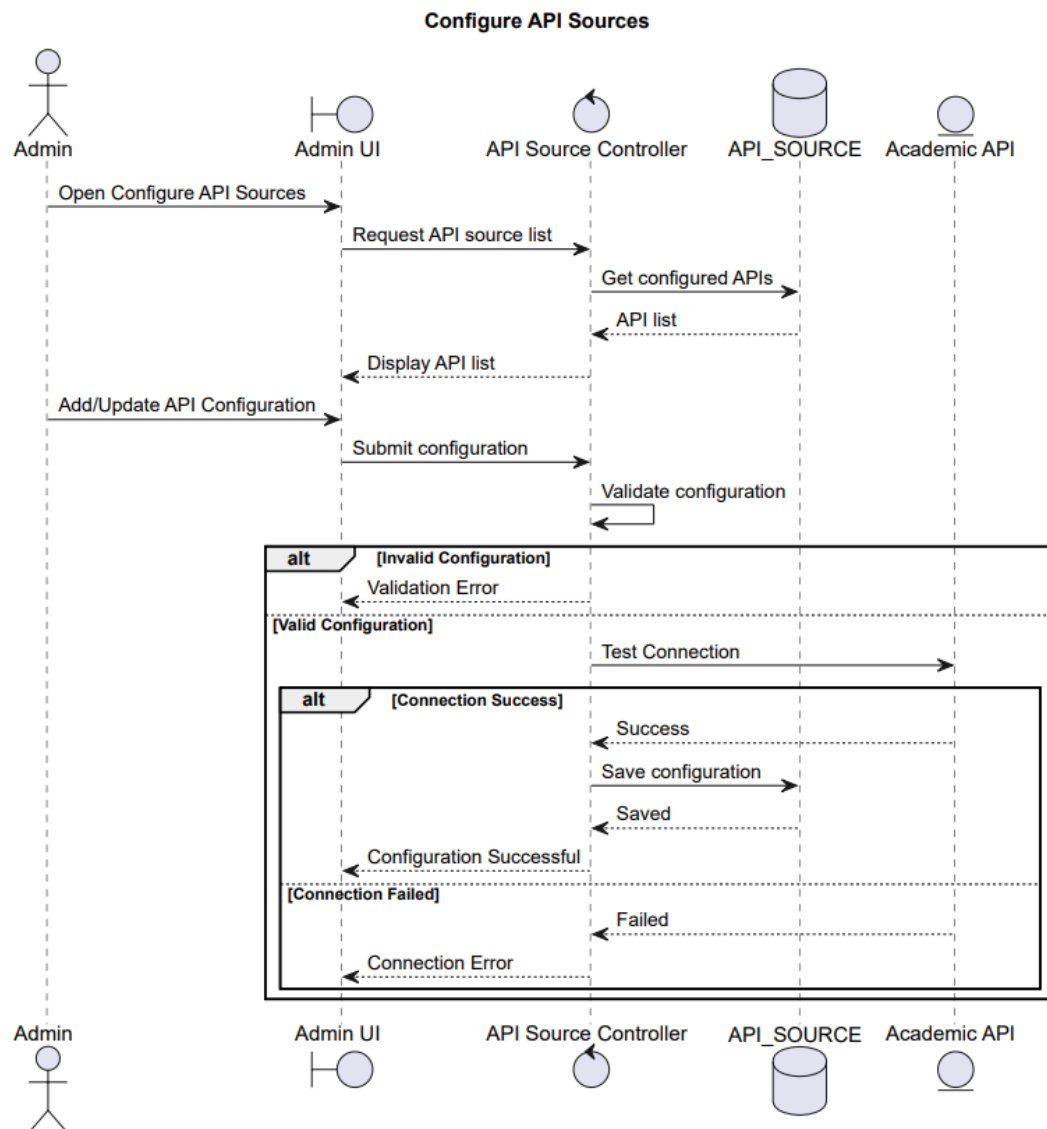
## 3.14 Configure API Sources

### 3.14.1 Activity Diagram



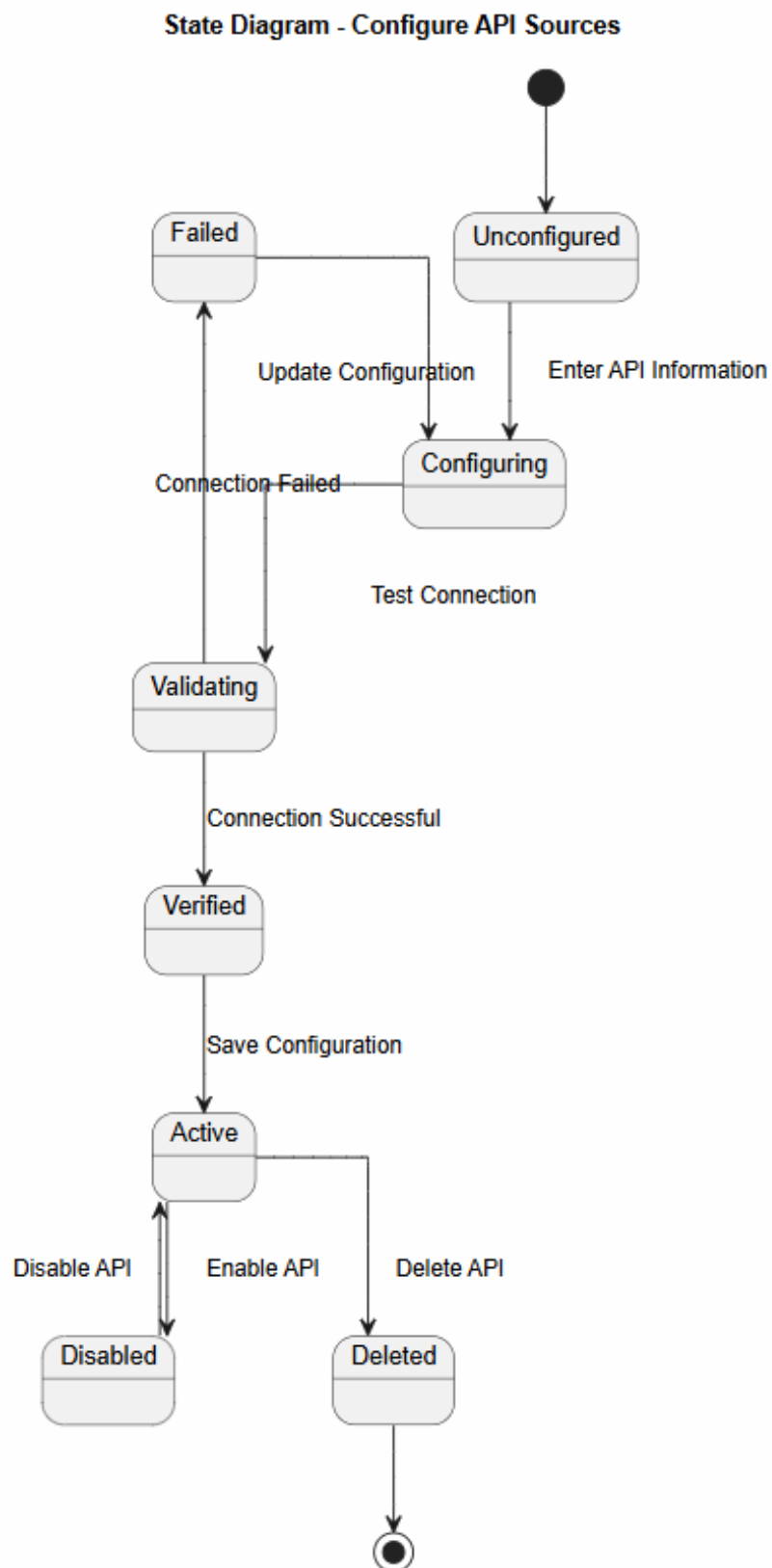
Hình 63: Configure API Sources

### 3.14.2 Sequence Diagram



Hình 64: Configure API Sources

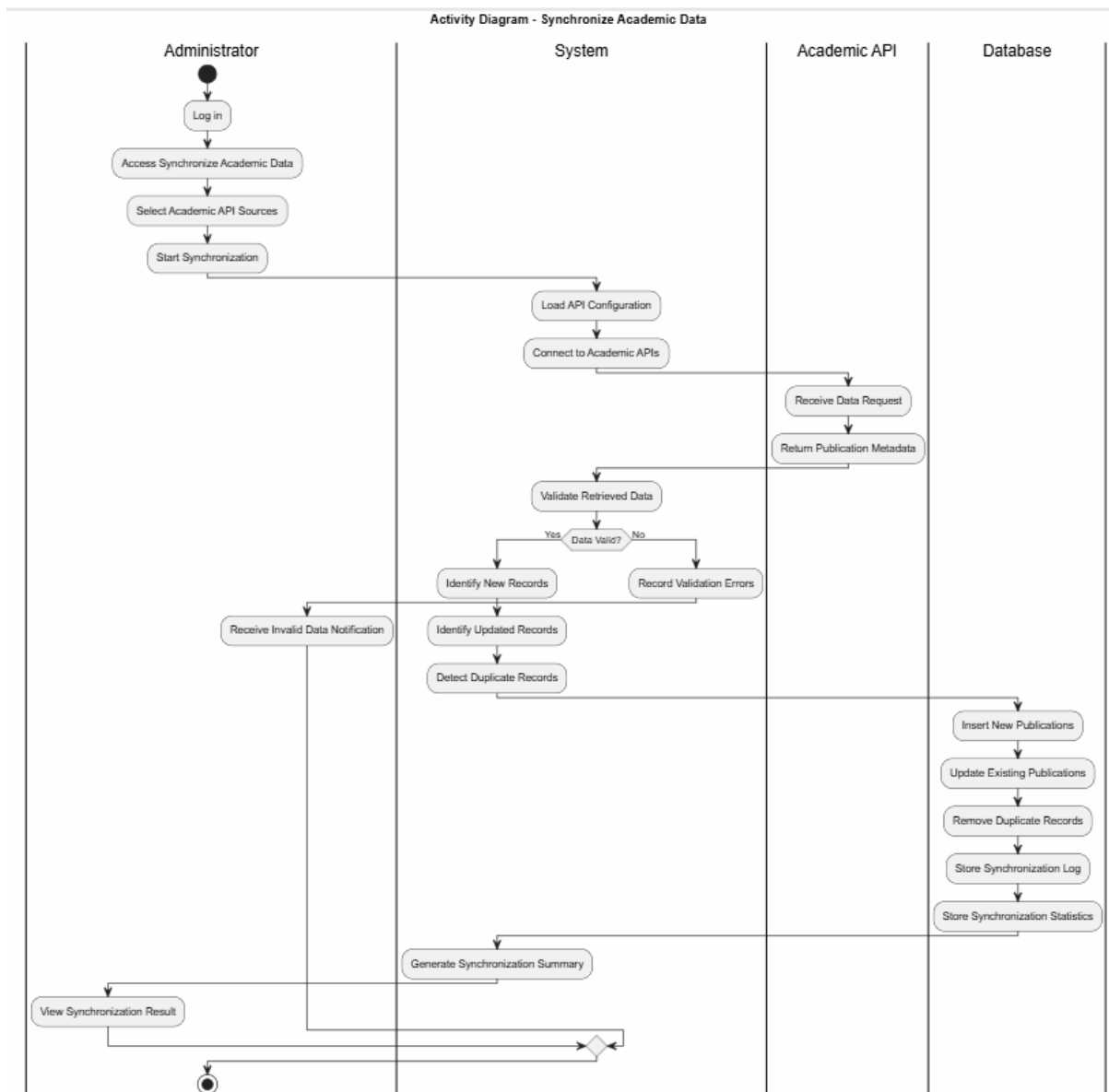
### 3.14.3 State Diagram



Hình 65: State diagram

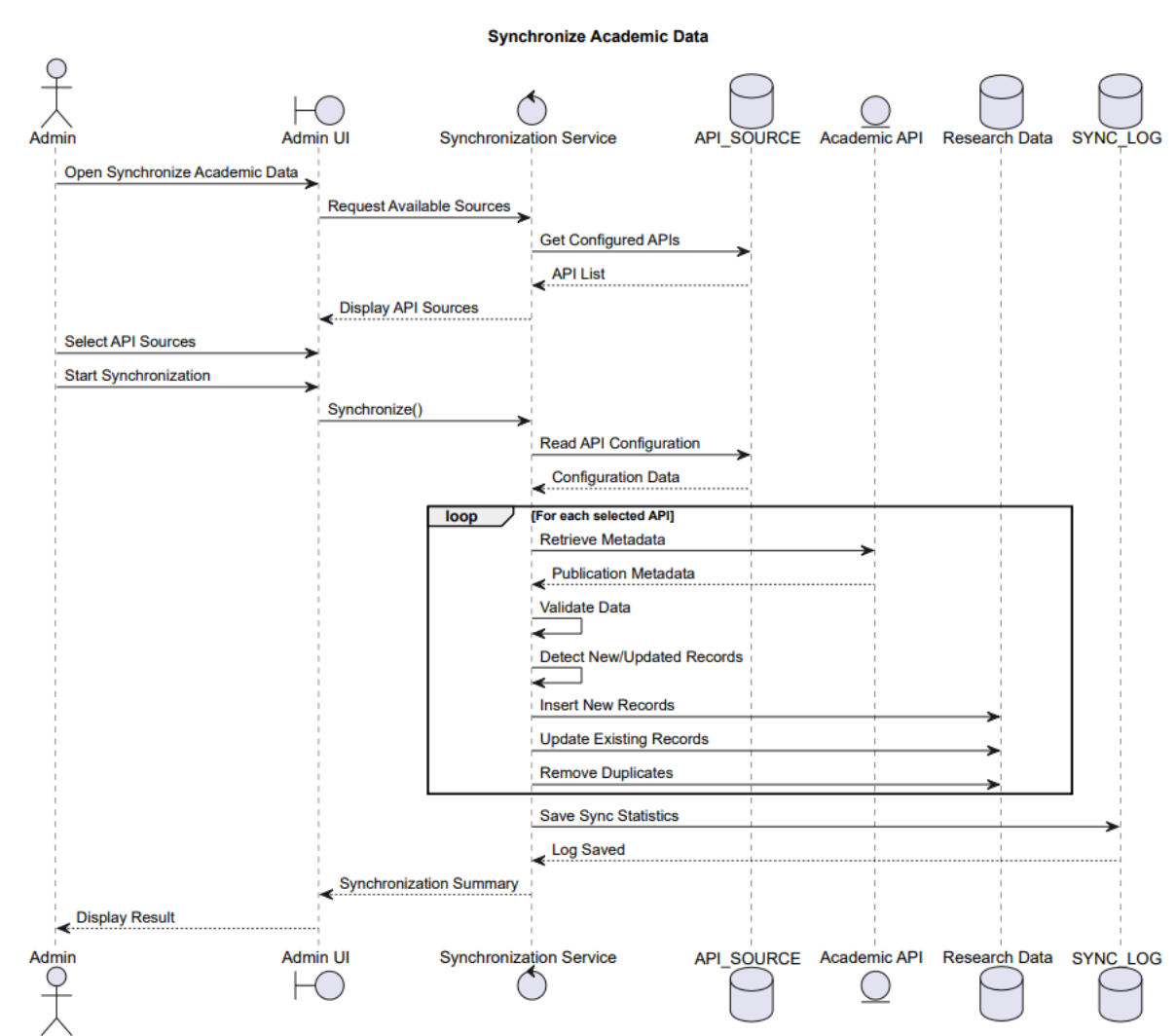
## 3.15 Sync Academic Data

### 3.15.1 Activity Diagram



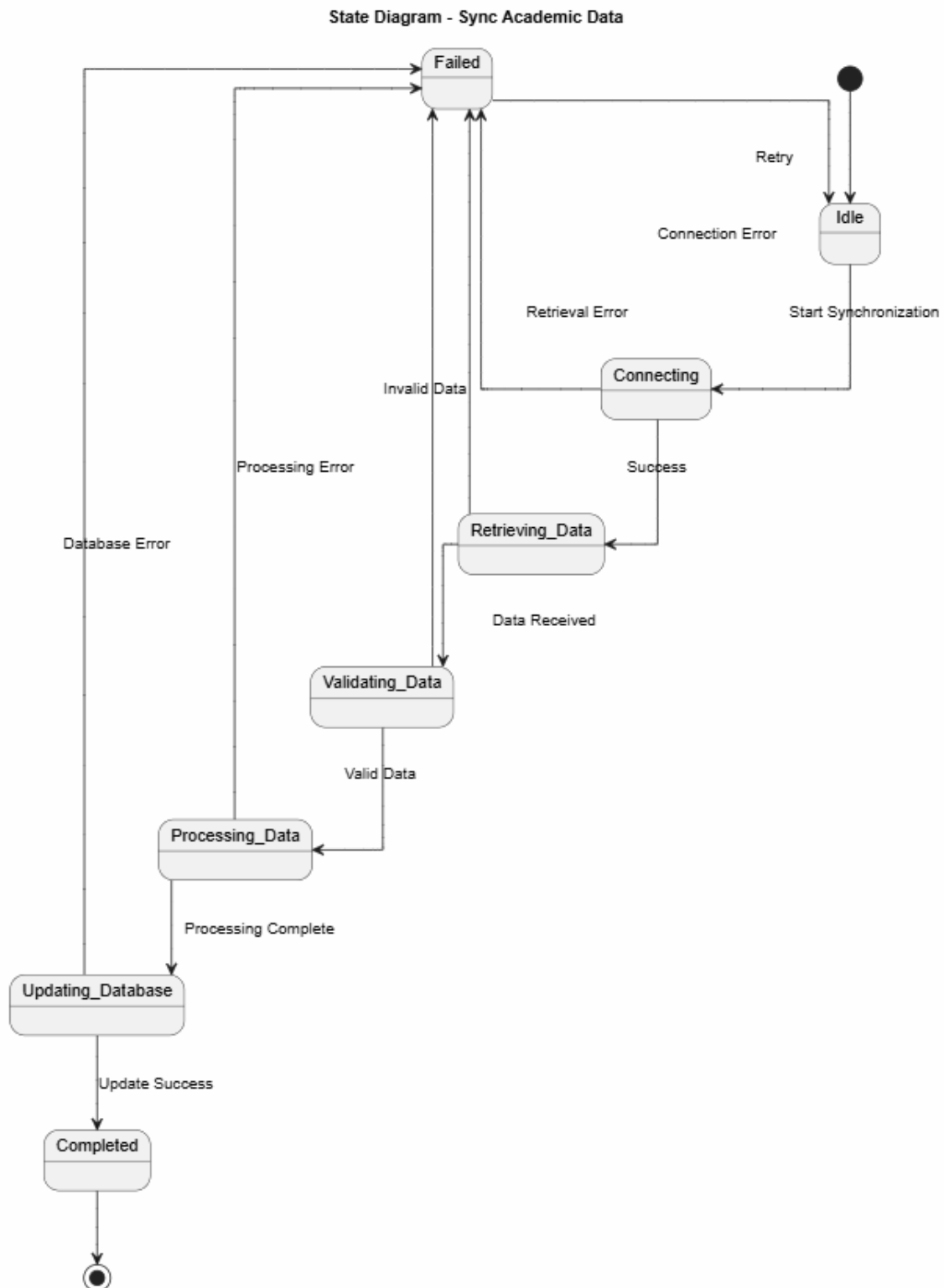
Hình 66: Sync Academic Data

### 3.15.2 Sequence Diagram



Hình 67: Sync Academic Data

### 3.15.3 State Diagram



Hình 68: Sync Academic Data